

CALIFORNIA CDL · WRITTEN TESTS

General Knowledge + Combination Vehicles

The study guide — most-asked and most-confusing topics first, hard topics explained with worked examples.

GENERAL KNOWLEDGE TEST

50 questions · pass with **40** correct (80%)

COMBINATION VEHICLES TEST

20 questions · pass with **16** correct (80%)

3 answer choices per question · 3 tries per application · no time limit. Answer the **California handbook way** (DL 650), even if another book or website says something else.

Built from your 18 GK/CV lessons and the California Commercial Driver Handbook DL 650 (REV. 12/2019); page numbers like "p. 2-15" point to that handbook. Every fact here was checked against it. Not affiliated with the California DMV. Rules change — confirm current rules with DMV before test day.

Made with the CDL Workshop app (same facts as the app's lessons).

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How to use this guide

1. **Part 1 — the 50 most-missed facts.** Learn these first. They come from your course's most-missed list (built from large California CDL question banks, each answer checked against the handbook). Tick each one when you can say it without looking.
2. **Part 2 — easy to confuse.** Look-alike numbers and "opposite" answers side by side. Most wrong answers on the test are one of these swaps.
3. **Part 3 — hard topics, explained.** Plain-language walk-throughs with diagrams and worked examples.
4. **Part 4 — everything, lesson by lesson.** Key points, every number to memorize, and every exam trap for all 18 lessons. Lessons are ordered by test weight (High first), then by how many most-missed facts they hold.
5. **Part 5 — practice tests.** A 50-question General Knowledge test and a 20-question Combination test, with an answer key that explains each answer. Aim for 90%+ before test day.

HOW TO READ THE BOXES

Worked example step-by-step numbers · **Exam trap** the wrong answer the test hopes you pick · **Memory hook** a short way to remember · **CA** a California rule that beats the federal answer.

A 7-day plan using this guide

Day	Do
1	Part 1 (all 50). Part 3: license classes, stopping distance, following distance.
2	Part 2 (all). Part 3: triangles, railroad, mountain driving.
3	Part 3: emergencies, ABS, penalties, California rules, cargo, crashes/fires. Part 4: first 5 GK lessons.
4	Part 4: remaining GK lessons. Re-read every red Exam trap box.
5	Part 3 combination topics (rollover, air brakes, coupling, inspection). Part 4: CV lessons.
6	Part 5: both practice tests without looking. Re-study every miss.
7	Part 1 again + the "test-day checklist". Retake Part 5; score 90%+.

The 50 most-missed facts

If you only have one hour, learn this page set. General Knowledge first (30), then Combination (20).

General Knowledge – 30 facts

1. An **empty** truck needs a **longer** stopping distance than a loaded one — less weight, less traction. [GK-07]
2. Total stopping distance at 55 mph = **419 ft** (142 perception + 61 reaction + 216 braking). [GK-07]
3. Double your speed → about **4×** the braking distance. [GK-07]
4. Following distance: **1 second per 10 ft** of length, **+1 second above 40 mph**. A 30-ft truck at 55 mph → **4 s**. [GK-08]
5. Someone tailgates you → **increase your following distance**. Don't speed up, and no brake-light tricks. [GK-08]
6. Tight right turn → turn wide **as you complete** it (button hook), never swing left first. [GK-08]
7. Convex mirrors make things look **smaller and farther away**. [GK-06]
8. Warning triangles within **10 minutes**. Divided highway: **10, 100, 200 ft behind**. Two-lane road: within 10 ft of a corner, **100 ft behind, 100 ft ahead**. Hill or curve: the rear one goes **100–500 ft** back. [GK-06]
9. Retarders: turn them **off** on wet, icy or snowy roads — they can skid the drive wheels. [GK-05]
10. Back toward the **driver's side**, with a helper. With a trailer, turn the wheel the **opposite** way. [GK-05]
11. Pick your downhill gear **before** the grade, usually **lower** than your climbing gear. [GK-05]
12. Long downgrade: brake to about **5 mph below** your safe speed, release, repeat. [GK-11]
13. Railroad crossing: stop **15–50 ft** from the rail, **never shift** on the tracks. Clearing a single track takes **at least 14 s**, a double track **more than 15 s**. [GK-11]
14. Use **high beams whenever safe and legal**. Dim within **500 ft**. [GK-10]
15. Hot tires: **never** let air out. Radiator cap: only when cool enough to **touch bare-handed**. [GK-10]
16. Tire blowout: hold the wheel firmly and **stay off the brake** until you slow down. [GK-12]
17. Something in your path: steering is usually faster than stopping — go **right**. Off the road: avoid braking until about **20 mph**. [GK-12]
18. ABS: **brake normally**. It does not shorten stopping distance. No stab braking with ABS. [GK-12]
19. Drive-wheel braking skid: **stop braking**, countersteer. [GK-12]
20. At a crash, **protect the area first**. [GK-13]
21. Fire: never pull into a service station, don't open the hood, keep cargo doors shut, aim at the **base**. A B:C extinguisher is for **electrical and burning liquids**. [GK-13]
22. Alcohol in a CMV: **.04** is DUI. **Any** detectable alcohol = **24 hours** out of service. Only time sobers you up. [GK-02]
23. Check cargo within the first **50 miles**, then every **3 hours or 150 miles**, and after every break. [GK-14]
24. Tie-downs: at least **1 per 10 ft**, never fewer than **2** (a 20-ft load needs 2). [GK-14]
25. The bridge formula allows **less** weight on axles that are **closer** together. [GK-14]

26. Tread depth: **4/32 in** front tires, **2/32 in** others. Carry **3** red reflective triangles. [GK-04]
27. **CA** Trucks with 3+ axles and anything towing: **55 mph** max. SR-1 to DMV within **10 days** (injury, death, or over \$1,000 damage). [GK-03]
28. **CA** Intrastate hours: **12 hours driving, 16 hours on duty** (11/14 is federal). [GK-03]
29. Serious violations: **2 in 3 years = 60 days**, 3 = 120 days. Excessive speeding = **15+ mph** over. [GK-02]
30. You may haul hazmat without an H endorsement **only if the load does not need placards**. [GK-13]

Combination Vehicles – 20 facts

1. A **fully loaded** rig is **10x** more likely to roll over than an empty one. Keep cargo low and take turns slowly. [CV-01]
2. In a sudden move, the **last trailer** is most likely to tip. A 2.0 rating means it tips **2x** as easily as the tractor. [CV-01]
3. Empty rigs and **bobtail** tractors take **longer** to stop. [CV-01]
4. Trailer skid (jackknife): **release the brakes**. Never use the hand valve to straighten it. You usually spot it first in your **mirrors**. [CV-01]
5. Trailer hand valve: **only for testing** trailer brakes. Never for parking, never while driving. [CV-02]
6. Emergency line loses air → the tractor protection valve **closes** and the trailer emergency brakes come on. [CV-02]
7. **Red 8-sided** knob = trailer air supply: push in to supply air, pull out for the trailer emergency brakes. **Yellow diamond** = parking brakes. [CV-02]
8. **Red** = emergency (supply) line, which fills the trailer tanks. **Blue** = service (control) line, which carries the brake signal. [CV-02]
9. The service line comes apart → nothing happens **until you brake**. [CV-02]
10. Crossed air lines: a trailer **with** spring brakes won't release. An old trailer **without** spring brakes can be driven with **no trailer brakes**. [CV-02]
11. Shut-off valves: **open** everywhere except the rear of the **last trailer (closed)**. [CV-02]
12. Dummy couplers keep **dirt and water** out of the lines. [CV-02]
13. Trailer ABS lamp: **yellow, left side**. Required on trailers and dollies built **on or after March 1, 1998**. [CV-02]
14. Coupling starts: **inspect the fifth wheel** → inspect the area and chock → position the tractor. Lock the trailer brakes **before** backing under, then do the **tug test**. [CV-03]
15. Trailer height: low enough to be **raised slightly**. Too high = may not couple. Too low = you hit the trailer nose. [CV-03]
16. After coupling: **no space** between the upper and lower fifth wheel, with the jaws around the **shank** of the kingpin. [CV-03]
17. Landing gear: raised **all the way**, crank handle secured. [CV-03]
18. Uncoupling: stop with the **tractor frame still under the trailer**. [CV-03]
19. Tractor protection valve test: **engine off**, pump the pedal — the knob pops out at **20–45 psi**. [CV-04]
20. Trailer emergency brakes: pull the knob out, tug gently. Trailer service brakes: move slowly and apply the **hand valve**. [CV-04]

PART 2

Easy to confuse

Test writers put a real handbook number from a *different* rule next to the right one. Learn these side by side.

The handbook way vs. what other sources say

Topic	Answer on the CA test	What others say (wrong here)
1st violation of an out-of-service order	90 days (2nd in 10 years: 1 year · 3rd: 3 years)	180 days (current federal law)
Dim your high beams	within 500 ft of an oncoming vehicle and when following within 500 ft	300 ft when following (car handbook)
Hazmat placard size	at least 9.84 in (250 mm) square, on point	10¾ in (older manuals)
Tell your employer your license was suspended	by the end of the next business day (p. 1-4) — or 2 business days (p. 1-16). The handbook says both; pick whichever is offered	30 days (that is for convictions)
Tie-down strength	total working load limit at least ½ the cargo weight	1½×
CDL needed for passengers CA	more than 10 people, including the driver	16 or more (federal)
Fifth wheel before coupling	tilted down toward the rear of the tractor	"level"
Tractor protection valve / red knob pops out	20–45 psi	25–40 psi
Trailer hand valve	GK (p. 2-9): may hold a combination from rolling back when starting. Combination test (6.2): only for testing trailer brakes — never to park, never while driving, never to straighten a jackknife	—

Look-alike numbers

Weight lines (who needs a CDL)

Rule	Right answer	Watch out
Single vehicle needs a CDL	26,001 lb GVWR or more	26,000 is the trap
Combination needs a CDL	26,001 lb GCWR or more	
Towed unit that makes it Class A	10,001 lb GVWR or more	10,000 is the trap
Passengers CA	more than 10 people including the driver	federal/other books: 16+

Rule	Right answer	Watch out
3-axle vehicle CA	over 6,000 lb → CDL (Class B)	
Tank endorsement (N)	each tank over 119 gal and total 1,000 gal or more	

Feet

Rule	Right answer	Watch out
Dim high beams	within 500 ft of oncoming AND when following	300 ft is the car-book trap
Low beams / high beams show	about 250 ft / 350-500 ft	
Railroad stop (no stop line)	15-50 ft from nearest rail	
Triangles, divided highway	10, 100, 200 ft behind	
Triangles, two-lane road	≤10 ft from corner, 100 ft behind, 100 ft ahead	
Hill/curve hides you within	500 ft → rear triangle 100-500 ft back	
Mirror must show behind you	at least 200 ft	
Following listed vehicles (CA)	at least 300 ft ; caravan gap 100 ft	
Headlights on when you can't see	a person/vehicle at 1,000 ft	

Seconds

Rule	Right answer	Watch out
Look ahead	12-15 s (≈1 block in town, ≈¼ mile on highway)	
Perception / reaction	1¾ s / ¾-1 s	
Following distance	1 s per 10 ft of length, +1 s over 40 mph	
Clear railroad tracks	single: at least 14 s · double: more than 15 s	
Snub braking application	about 3 s	
Stab braking: wheels restart in	up to 1 s	

Speeds

Rule	Right answer	Watch out
CA max for 3+ axle trucks & anything towing	55 mph	
Following distance gets +1 s	over 40 mph	
Off the road: don't brake until	about 20 mph	
Downgrade snub braking	brake to 5 mph below safe speed, release, repeat	
Excessive speeding (serious violation)	15 mph or more over the limit	
Hydroplaning can start at	as low as 30 mph	

Rule	Right answer	Watch out
Wet road	slow by about $\frac{1}{3}$ (55 → about 35); packed snow: by half or more	

Time, miles and deadlines

Rule	Right answer	Watch out
Place warning triangles	within 10 minutes	
First cargo check	within the first 50 miles	
Later cargo checks	every 3 hours or 150 miles , and after every break	
Hot-weather tire checks	every 2 hours or 100 miles	same numbers as rest breaks
No alcohol before duty	4 hours	
Any alcohol under .04	out of service 24 hours	
SR 1 crash report to DMV CA	within 10 days (injury, death, or over \$1,000)	
Tell employer about a crash in their vehicle	within 5 days	
Report a conviction (DMV / employer)	within 30 days	
Tell employer about a suspension	next business day (p. 1-4) — or 2 business days (p. 1-16)	handbook says both

Hours of service

Rule	Right answer	Watch out
California intrastate CA	12 h driving, 16 h on duty, 80 h in 8 days	
Federal	11 h driving, 14th hour, 60/7 or 70/8	the trap on a CA question
Off-duty rest	10 h in a row	

Air pressure (psi)

Rule	Right answer	Watch out
Tractor protection valve closes / red knob pops out	20–45 psi	25–40 is the trap
Pressure build-up check (GK inspection)	50 → 90 psi within 3 min	
Governor cut-out	usually 120–140 psi	
Leakage per minute, brakes released	3 psi (2 vehicles) · 5 psi (3+)	
Leakage per minute, foot brake held	4 psi (2 vehicles) · 6 psi (3+)	

Tires, cargo, other

Rule	Right answer	Watch out
Tread depth	front 4/32 in · all others 2/32 in	swapping them is the trap
Tie-downs	at least 1 per 10 ft , never fewer than 2	
Tie-down strength (total WLL)	at least ½ the cargo weight	1½× is the trap
Steering play	more than 10° (≈2 in on a 20-in wheel) = problem	
Missing spring leaves	¼ or more = out of service	
Warning triangles to carry	3 red reflective	
Rollover: fully loaded vs empty	10× more likely	5× is the website trap
Placard size	at least 9.84 in (250 mm)	10¾ in is the old-book trap

Same situation, opposite answers

Topic	Handbook answer	Tempting wrong answer
Empty vs loaded: which stops longer?	Empty truck/rig needs a longer stopping distance (less weight = less traction; stiff springs, wheels lock). A bobtail tractor also takes longer than a fully loaded rig.	Loaded is heavier, so it takes longer.
Right turn: button hook vs jug handle	Button hook : go straight into the intersection and turn wide as you finish ; rear stays near the curb.	Swing left first (jug handle) — cars squeeze in on your right.
Drive-wheel skid vs front-wheel skid vs trailer jackknife	Drive-wheel braking skid → stop braking , then countersteer. Front-wheel skid → let it slow ; stop turning/braking so hard. Trailer skid → release the brakes .	Brake harder / use the trailer hand valve to straighten.
Controlled vs stab braking vs ABS	Controlled: as hard as possible without locking . Stab: full → release when wheels lock → reapply (only without ABS). With ABS: brake normally ; full braking only with working ABS on all axles in an emergency.	Pump the brakes with ABS; ABS shortens stops.
Red vs blue vs yellow	Red line = emergency (supply); red 8-sided knob = trailer air supply. Blue line = service (control). Yellow diamond knob = parking brakes.	Swap colours or knobs.
Emergency line break vs service line break	Emergency (red) line loses air → trailer emergency brakes come on , TPV closes. Service (blue) line breaks → nothing until you brake ; then air escapes fast.	Service line break sets the brakes at once.
Crossed air lines	Trailer with spring brakes → brakes won't release . Old trailer without spring brakes → you could drive off with no trailer brakes .	
Shut-off valves	All open — except the rear of the last trailer: closed . No air at the back? Check they are open.	All closed / all open.

Topic	Handbook answer	Tempting wrong answer
Trailer hand valve (Johnson bar)	Combination test: only for testing trailer brakes . Never to park, never while driving, never to fix a skid. (GK p. 2-9 also mentions holding a rig from rolling back when starting — on the Combination test answer "testing only".)	Use it to park or to straighten a jackknife.
Engine braking vs retarder vs service brakes	Downhill: engine braking in a low gear is the main control; brakes only help. Retarder off on wet, icy or snowy roads (it can skid the drive wheels).	Rely on the brakes; retarder on ice.
Backing	Back toward the driver's (left) side , lowest reverse gear, with a helper ("STOP" signal). Trailer: turn the wheel opposite , then follow it.	Turn the wheel the way you want the trailer to go.
Convex (fisheye/spot) mirror	Wider view, but things look smaller and farther away than they are.	Bigger / closer.
Tailgater behind you	Increase your following distance ahead; avoid quick changes. No speeding up, no brake-light tricks.	Speed up or flash brake lights.
Fire extinguishers	B:C = electrical + burning liquids. A:B:C adds wood, paper, cloth. Never water on electrical or gasoline fires; a burning tire needs lots of water. Aim at the base , stand upwind.	Aim at the flames; open the hood.
HazMat without an H	Allowed only if the load does not need placards . Placards needed → H. Tank + placarded HazMat → X .	
Hot tires / radiator	Hot tire: never let air out. Radiator cap: only when cool enough to touch bare-handed , turn slowly to the first stop.	Let air out of hot tires.

Hard topics, explained

General Knowledge topics first, then Combination. Read the text, then check the worked example.

1 Which license class do I need?

Ask these questions **in order**. Stop at the first "yes".

- Combination** (towing) with GCWR **26,001 lb or more**, and the towed unit's GVWR is **10,001 lb or more**? → **Class A**.
CA Towed unit 10,001+ lb but GCWR under 26,001 lb → Class A with **Restriction 88**.
- Power unit GVWR **26,001 lb or more** (alone, or towing something under 10,001 lb)? → **Class B**. **CA** So is a 3-axle vehicle over 6,000 lb.
- Built or used for **more than 10 people including the driver** for pay, profit or a nonprofit (or a farm labor vehicle for 10+)? → **Class B** in the handbook's chart.
- None of the above but hauling **placarded HazMat**? → **Class C**. Otherwise no CDL.

WORKED EXAMPLE

Vehicle	Answer	Why
Dump truck, GVWR 33,000 lb, no trailer	B	single vehicle $\geq 26,001$
Same truck towing a 12,000-lb GVWR trailer	A	combination $\geq 26,001$ and towed $\geq 10,001$
Same truck towing an 8,000-lb GVWR trailer	B	towed unit under 10,001
Pickup (GVWR 12,000) + 11,000-lb trailer = GCWR 23,000	A with Restriction 88 CA	towed $\geq 10,001$ but GCWR under 26,001
Nonprofit church van built for 12 incl. driver (not a vanpool)	CDL needed	more than 10 people incl. driver, nonprofit use
Pickup carrying placarded HazMat	C + H	only reason for a CDL is the placards

EXAM TRAP

Every line is **one pound above** a round number: 26,001 and 10,001. "26,000" and "10,000" are the wrong answers.

2 Stopping distance

Total stopping distance = **perception distance + reaction distance + braking distance**.

You are still moving at full speed during the first two boxes

Perception 142 ft 1¼ s: you see it and realise	Reaction 61 ft foot to pedal	Braking about 216 ft brakes working (dry road, good brakes)
--	--	---

Total at 55 mph: at least 419 ft (longer than a football field at 60 mph)

- **Speed hurts fast.** Double your speed (20 → 40 mph) → about **4×** the braking distance and impact. Triple it → **9×**. Four times → **16×**.
- **Empty trucks need MORE distance** than loaded ones: less weight = less traction, so wheels lock more easily.
- Wet road can **double** stopping distance → slow by about ⅓ (55 → about 35 mph). Packed snow → slow by half or more. Ice → crawl, and stop driving as soon as it's safe.
- Bridges freeze before the road. Black ice looks like a wet road.
- Rule of thumb: you must be able to **stop within the distance you can see** ahead.

WORKED EXAMPLE

Call your braking distance at 20 mph "one unit". At 40 mph (2× the speed) it is not 2 units but about **4 units**. At 60 mph (3×) it is about **9 units**; at 80 mph (4×) about **16**.

Add the perception and reaction distance on top, and a truck at 55 mph needs at least **419 ft** — more than a football field at 60 mph.

EXAM TRAP

"A loaded truck takes longer to stop." **False on the test** — the empty one does.

3 Following distance (the seconds rule)

At least 1 second for every 10 ft of vehicle length below 40 mph. **Over 40 mph, add 1 more second.**

How to count: when the car ahead passes a landmark (a sign, a shadow), count "one thousand-and-one, one thousand-and-two..." until you reach it.

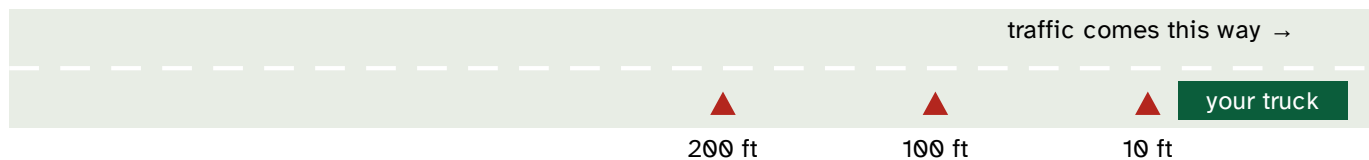
WORKED EXAMPLE

Your vehicle	Speed	Math	Seconds
40-ft truck	35 mph	$40 \div 10 = 4$	4 s
40-ft truck	55 mph	$4 + 1$ (over 40)	5 s
30-ft truck	55 mph	$3 + 1$	4 s
50-ft truck	60 mph	$5 + 1$	6 s
60-ft rig	30 mph	$60 \div 10 = 6$	6 s
60-ft rig	55 mph	$6 + 1$	7 s

- Slippery road → leave **more** than the rule.
- Someone tailgates you → **increase** your space ahead (so you can brake gently), avoid quick changes, don't speed up, no brake-light tricks.
- The space **ahead** is the most important space.

4 Warning triangles after a stop

1. Divided or one-way highway: 10 ft, 100 ft, 200 ft behind you (toward traffic)



2. Two-lane road with traffic both ways (or undivided): 1 within 10 ft of a corner, 100 ft behind, 100 ft ahead



3. Hill or curve hides you within 500 ft: move the rear triangle back 100–500 ft so drivers see it in time

rear triangle: before the hill crest / curve (100–500 ft)



Place all triangles within 10 minutes of stopping. Carry them held in front of you, between you and oncoming traffic.

MEMORY HOOK

10 minutes to place them. Divided highway counts up: **10 – 100 – 200**, all behind. Two-lane road: one at the truck, **100 behind, 100 ahead** (traffic comes from both sides).

- Stopped on the roadside: turn on your **4-way flashers**; don't rely on taillights.

- You must carry **3** red reflective triangles (plus a charged fire extinguisher and spare fuses unless you have circuit breakers).

5 Railroad crossings

Closer than 15 ft = the train may hit you. Farther than 50 ft = you can't see well enough.



Clear a single track: at least 14 s · a double track: more than 15 s · never shift gears on the tracks

- Passive** crossing = no signals — you decide if it's safe. **Active** crossing = flashing lights, bells and/or gates.
- Round black-on-yellow sign = advance warning: slow down, look, listen, be ready to stop. Crossbuck = yield to trains.
- Vehicles carrying **passengers or HazMat** must stop at crossings.
- Gates: stay stopped until the gates are **up AND the lights stop flashing**.
- Never shift gears** while crossing.
- Low units (lowboy, car carrier, moving van, possum-belly livestock trailer) can hang up on raised crossings.
- Stuck on the tracks → **get out and away first**, then call 9-1-1 and give the crossing's **DOT number**.

EXAM TRAP

The "15 ft" and "50 ft" are distances to the **nearest rail**, and only when there is no stop line. Clearing times: single track **at least 14 s**, double **more than 15 s**.

6 Mountain driving and long downgrades

- Pick your gear before** you start down — usually a **lower** gear than you used to climb. (You can't safely downshift once you're gaining speed.)
- Let **engine braking** do the main work (strongest near governed rpm in lower gears). The brakes only help.
- Use **snub braking**: when you reach your safe speed, brake firmly enough to feel a real slowdown until you are **5 mph below** it, then release. Each application lasts about **3 seconds**. Repeat.

WORKED EXAMPLE

Your safe speed down the grade is **40 mph**. When you reach 40 → brake to **35** (about 3 s) → release. When you are back at 40 → do it again.

- Brake fade**: overheated brakes need harder and harder pressure for the same stop. Brakes out of adjustment stop working first, so the others overheat.
- Brakes fail on a downgrade → use an **escape ramp** (usually a few miles below the top). No ramp → least dangerous route (open field, uphill side road).
- Safe speed depends on: **weight, length and steepness of the grade, road conditions, weather**.

EXAM TRAP

"Ride the brakes lightly all the way down" is wrong — it overheats them.

7 Emergencies: steer, stop, or leave the road?

- You can almost always **steer around** an obstacle faster than you can stop a heavy truck.
- Oncoming driver drifts into your lane → move **right**. Right is usually best (you won't hit oncoming traffic; the shoulder is often clear).
- Both hands on the wheel. **Don't brake while turning** (wheels can lock). Turn only as much as needed. Be ready to **countersteer**.
- Leaving the road: **don't brake until about 20 mph**, then brake gently; keep one set of wheels on the pavement if you can. To come back, **turn sharply**, and countersteer once both front tires are on the pavement.
- Tire blowout** (bang, vibration, heavy steering = front, fishtailing = rear): **hold the wheel firmly, stay off the brake** until the truck slows, then brake gently.
- Hydraulic brake failure** (pedal spongy or to the floor): downshift, pump the brakes, use the parking brake while holding its release.

Situation	Do this
Drive-wheel braking skid (most common)	Stop braking → countersteer quickly as it straightens
Drive-wheel acceleration skid on ice/snow	Foot off the accelerator (push in the clutch if very slippery)
Front-wheel skid (front goes straight)	Let it slow down: stop turning so sharply / braking so hard
Trailer jackknife (combination)	Release the brakes — never the trailer hand valve; you see it first in your mirrors
Hydroplaning	Release the accelerator, push in the clutch, don't brake
Emergency stop, no ABS	Controlled braking, or stab braking (full → release when locked → reapply; up to 1 s to roll)
Emergency stop, ABS	Brake normally; full brakes only if ABS works on all axles

4 causes of skids: **over-braking, over-steering, over-acceleration, driving too fast** (the cause of most serious skids).

8 ABS in plain words

- ABS is a computer that stops wheels from **locking** in hard braking. Its job is **control** (you can still steer), **not a shorter stop**.
- It only switches on when a wheel is about to lock. Otherwise brakes work normally.
- Yellow** ABS lamp: on the dash for tractors/trucks/buses; on trailers on the **left side, front or rear corner**. It lights at start-up then goes out (older systems: may stay on until over **5 mph**). Stays on = ABS problem; you still have normal brakes.

- Required on: air-brake **tractors built on/after March 1, 1997**; other air-brake vehicles **and trailers/dollies on/after March 1, 1998**; hydraulic trucks/buses $\geq 10,000$ lb on/after **March 1, 1999**.
- ABS only on the tractor → watch the trailer, ease off if it swings. ABS only on the trailer → ease off if the tractor starts to jackknife.

EXAM TRAP

ABS is **no** reason to drive faster or follow closer. It does not stop power skids or turning skids.

9 Losing your CDL: the penalty ladders

Offense type	1st	2nd	3rd	Window
Major (DUI .04+, refusing a test, leaving a crash, felony with a CMV...)	1 year (3 years if placarded HazMat)	life	—	lifetime
Serious (excessive speeding 15+ mph over, reckless driving, improper lane change, following too closely, violation tied to a deadly crash...)	—	60 days	120 days	3 years
Railroad crossing (6 offenses)	60 days	120 days	1 year	3 years
Out-of-service order violation	90 days	1 year	3 years	10 years
Hands-free / texting (CMV)	—	60 days + 1 point	120 days + 1 point	3 years

- Alcohol: **.04** or more in a CMV = DUI. **Any** detectable alcohol below that = out of service **24 hours**. Driving a CMV = you already agreed to be tested (implied consent).
- Using a CMV in a controlled-substance felony = disqualified **for life**.
- Points: a violation in a CMV counts **1½ times**. Negligent operator = **4 points in 12 months, 6 in 24, 8 in 36**.
- No hardship license for a CMV.

MEMORY HOOK

Serious = "**2 → 60, 3 → 120**". Out-of-service = "**90 days, 1, 3**" over 10 years. Railroad = "**60, 120, 1 year**".

10 California-only rules (they beat federal answers)

Topic	California answer	Don't pick
Speed limit: 3+ axle trucks, anything towing, school bus with pupils, farm labor vehicle with passengers, explosives	55 mph	65
Intrastate hours of service	12 h driving, 16 h on duty ; 80 h in 8 days; 10 h off in a row	11/14 (federal)
Passengers needing a CDL	more than 10 incl. driver	16+

Topic	California answer	Don't pick
SR 1 crash report	to DMV within 10 days (injury, death, damage over \$1,000)	
Diesel idling (GVWR over 10,001 lb)	5 minutes max; none at a school	
Turnouts	use when 5 or more vehicles are behind you	
Lanes on 4+ lanes each way	right lane or the lane just left of it	
Length	single vehicle 40 ft ; tractor + trailer 65 ft ; tractor + semi + trailer 75 ft (each trailer 28 ft 6 in)	
Width / height	102 in body or load / 14 ft	
Triple trailers	not legal in California	

11 Cargo: weight, balance and tie-downs

- Check cargo securement within the first **50 miles**, then every **3 hours or 150 miles** (whichever first), and after every break.
- Keep the load **low**; put heavier cargo **under** lighter cargo.
- Too much weight on the steering axle → hard steering. Too little on the front → can't steer safely. Too little on drive axles → wheels spin.
- **Bridge formula**: axles that are **closer together** may carry **less** weight.
- Legal maximum weight may still be **unsafe** in bad weather or mountains.

WORKED EXAMPLE

Cargo	Minimum tie-downs	Why
8-ft crate	2	never fewer than 2
20-ft load	2	$20 \div 10 = 2$ (not 3!)
30-ft load	3	$30 \div 10 = 3$
25-ft load	3	handbook doesn't say how to count the extra 5 ft; adding one is the safe choice

Strength: total working load limit of all tie-downs $\geq \frac{1}{2}$ the cargo weight. A 10,000-lb machine needs at least **5,000 lb** total WLL: four 1,500-lb straps = 6,000 lb → OK.

- Header board ("headache rack") protects you from cargo sliding forward.
- Sealed load you can't inspect → still check gross and axle weights.

12 Crashes, fires and alcohol

At a crash, in this order: 1) **Protect the area** (stop a second crash: move your truck to the side, 4-way flashers, triangles) → 2) notify authorities → 3) care for the injured → 4) collect information → 5) report.

- Stopping to help: park **away** from the wreck. Have a phone/CB → call before you get out.
- Don't move a badly hurt person unless fire or traffic makes it necessary. Heavy bleeding → direct pressure.

Fires:

- Pull off in an open area — **never** into a service station.
- Engine fire → engine off, **don't open the hood**; spray through louvers, radiator or from underneath. Van/box cargo fire → **keep the doors shut**.
- Stand upwind, as far away as possible; aim at the **base**.
- Tire fires come from under-inflated tires and duals that touch; check tires for heat at every stop.

Alcohol: only **time** sobers you — the liver handles about $\frac{1}{3}$ oz of alcohol per hour. One drink = 12 oz beer (5%) = 5 oz wine (12%) = $1\frac{1}{2}$ oz 80-proof liquor. Alcohol hits **judgment and self-control** first.

13 Combination vehicles: rollovers, crack-the-whip, off-tracking

- Rollovers cause **more than half** of truck-driver crash deaths. A **fully loaded** rig is **10x** more likely to roll over than an empty one.
- Two ways to prevent rollover: keep cargo **as low as possible**, and go **slowly around turns** (and on ramps).
- **Crack-the-whip:** a quick lane change whips the rear trailer much harder than the tractor. The **last** trailer is the most likely to tip.

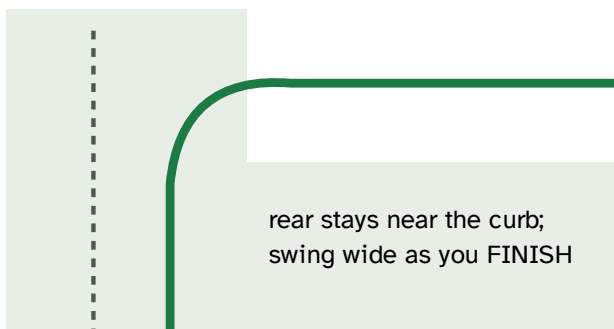
Rearward amplification: how much harder the LAST trailer is thrown than the tractor

5-axle tractor-semi, 45-ft	1.0 (lowest)
A rig rated 2.0	2.0 → last trailer tips 2x as easily
Triples (27-ft trailers)	3.5 (highest)

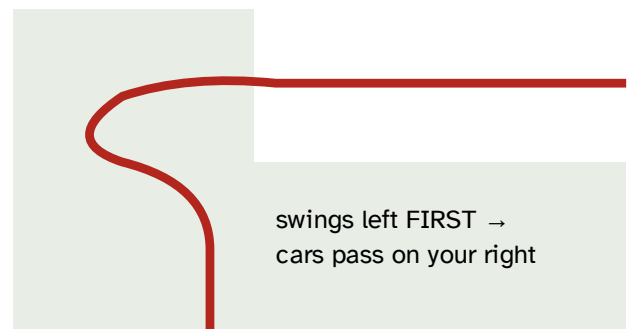
Triples aren't legal in California, but the 3.5 number can still be on the test.

- **Off-tracking** ("cheating"): in a turn, rear wheels cut **inside** the path of the front wheels. Longer rigs off-track more; the last trailer cuts in most.

✓ **BUTTON HOOK (correct)**

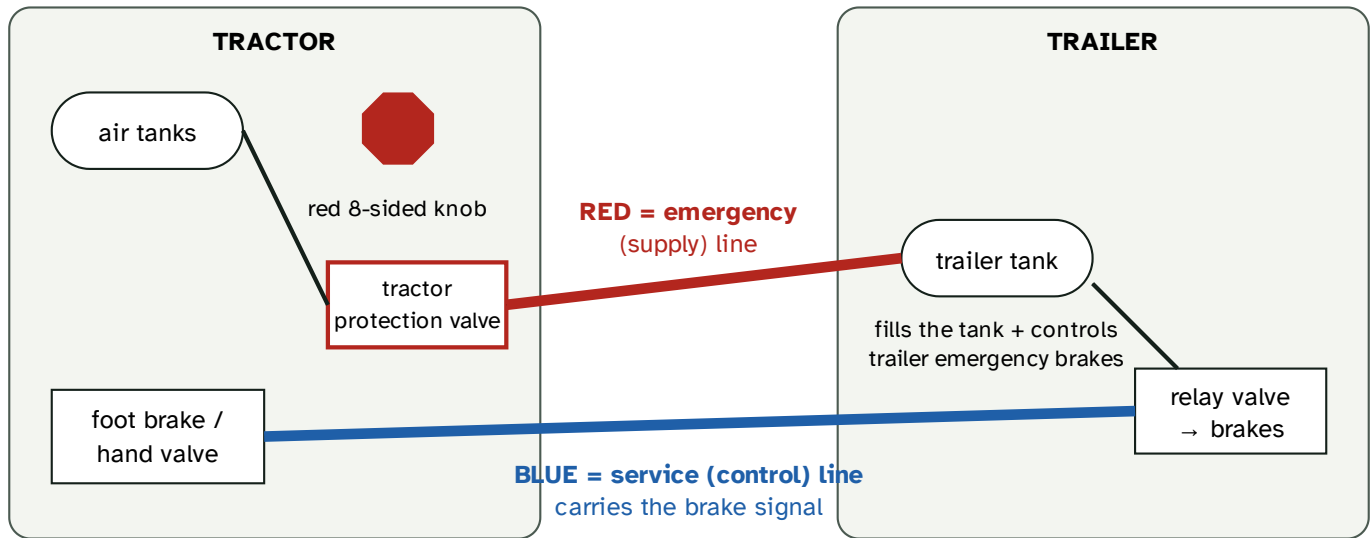


✗ **JUG HANDLE (wrong)**



- Backing a trailer: turn the wheel the **opposite** way (top of wheel left → trailer goes right). Back straight; if you must curve, curve toward the **driver's side**.
- Stopping: an **empty** rig and a **bobtail** tractor both take **longer** to stop than a loaded rig.

14 Combination air brakes: what happens when...

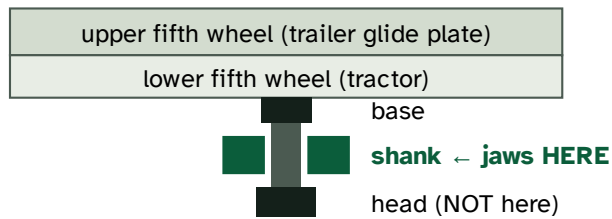


Glad hands join the lines (held at 90°). Blue to blue, red to red.

If this happens...	...then
Air pressure falls to 20–45 psi	Tractor protection valve closes ; red knob pops out ; trailer emergency brakes come on; tractor keeps its air
You pull the red knob out	Trailer air shut off → trailer emergency brakes on
You push the red knob in	Air goes to the trailer (fills its tanks, releases its brakes)
Emergency (red) line breaks or the trailer breaks away	Line loses pressure → trailer emergency brakes come on ; TPV closes
Service (blue) line comes apart	Nothing until you brake; then air escapes, pressure drops fast, trailer emergency brakes may set
Lines crossed , trailer has spring brakes	Trailer brakes won't release
Lines crossed, old trailer without spring brakes	You could drive away with no trailer brakes
Trailer has no spring brakes and you park it	Emergency brakes hold only while tank air lasts → chock the wheels

- Trailer tanks are filled through the **emergency (supply)** line. Drain all tanks **every day**.
- Glad hands: hold at **90°**, push the seals together, turn to lock. Unused lines go on **dummy couplers** (keep out dirt and water).
- Shut-off valves: **open** except at the rear of the **last** trailer (closed).
- Spring brakes aren't required on converter dollies and trailers built **before 1975**.

Coupling (16 steps) and uncoupling (10 steps)



After coupling, check:

- NO space between upper and lower fifth wheel
- jaws closed around the SHANK of the kingpin
- locking lever in "locked"; safety latch over it
- landing gear ALL the way up, crank secured
- trailer height: low enough to be raised slightly

Before backing under: fifth wheel tilted DOWN toward the rear, jaws open.

Coupling

1. Inspect the **fifth wheel** (greased, tilted down toward the rear, jaws open, safety handle on automatic lock)
2. Inspect the area; **chock** the wheels
3. Position the tractor straight in front of the trailer
4. Back slowly until the fifth wheel **just touches** the trailer
5. Secure the tractor (parking brake, neutral)
6. Check **trailer height** (low enough to be raised slightly)
7. **Connect the air lines**
8. Supply air; check for crossed lines (engine off, apply/release trailer brakes, listen, watch the gauge)
9. **Lock the trailer brakes** (pull the knob)
10. Back under the trailer in the **lowest reverse gear**
11. **Tug test:** pull forward gently with trailer brakes locked
12. Secure the vehicle (take the key)
13. Inspect: **no gap**, jaws around the **shank**, lever locked
14. Connect the electrical cord; check lines
15. Raise landing gear **all the way**; secure the crank; check clearances
16. **Remove the chocks** (last step)

Uncoupling

1. Park on firm ground, tractor **in line** with the trailer
2. Lock trailer brakes, back up gently to ease jaw pressure, set parking brakes while pushing
3. Chock trailer wheels if no spring brakes (or not sure)
4. Lower the landing gear (loaded: a few extra turns, but don't lift off the fifth wheel)
5. Disconnect air lines (to dummy couplers) and cable (hang **plug down**)
6. Unlock the fifth wheel; keep legs away from the rear tractor wheels
7. Pull partly clear; stop with the **tractor frame still under the trailer**
8. Secure the tractor
9. Inspect the ground and landing gear
10. Pull clear

MEMORY HOOK

Coupling order: **Look → Line up → Air → Lock → Back under → Tug → Check → Legs up → Chocks out.**
Air lines and trailer-brake lock come **before** backing under.

EXAM TRAP

Trailer too **high** → may not couple correctly. Too **low** → the tractor hits and damages the trailer nose. "Fifth wheel level" is wrong — it tilts **down toward the rear**.

Combination inspection and brake check

Use the same **7-step** walkaround as General Knowledge, plus:

- Lower fifth wheel mounted, greased, **no visible space** between plates, jaws around the **shank** (not the head); release arm seated, safety latch on.
- Air lines and electrical cord secure, not leaking, with **enough slack** for turns.
- Sliding fifth wheel: all locking pins present and locked; not too far forward.
- Landing gear fully raised, crank handle secured.

Test	How	Pass
Air flow to all trailers	Push red knob in; apply trailer hand brake; open the last trailer's emergency then service shut-off valves	Air comes out of both; then close them
Tractor protection valve	Engine off , pump the brake pedal to drop pressure	Knob pops out at maker's range, usually 20–45 psi
Trailer emergency brakes	Pull the knob out, tug gently	Trailer holds
Trailer service brakes	Move slowly, apply the hand valve	You feel the trailer brakes grab
Leakage (engine off)	Brakes released / foot brake held for 1 minute	$\leq 3 / 4$ psi (2 vehicles) · $\leq 5 / 6$ psi (3+)

General vehicle inspection (7 steps)

Walk up to the vehicle noticing its general condition, then:

1. Vehicle overview (review the last inspection report)
 2. Check the engine compartment (parking brakes on and/or wheels chocked **first**)
 3. Start the engine and inspect inside the cab
 4. Engine off, check lights (take the key; low beams + 4-way flashers on)
 5. Walkaround inspection
 6. Check signal lights
 7. Start the engine and check brake system
- Air pressure must build from **50 to 90 psi within 3 minutes**; governor cut-out usually **120–140 psi**. (The 85–100 psi in 45 s figure is the Air Brakes dual-system check — for a General Knowledge question, answer 50–90 in 3 minutes.)
 - Tread: front tires **4/32 in** in every major groove; all others **2/32 in**.
 - Steering play more than **10°** (about 2 in on a 20-in wheel) → hard to steer. Missing **¼** or more of spring leaves → out of service. Rust around lug nuts → may be loose. Welded wheel/rim repairs are unsafe.
 - Front clearance lights/reflectors **amber**, rear **red**.
 - Hydraulic brakes: pump **3** times, then press and hold **5 seconds** — the pedal shouldn't move (trap: pump 5, hold 3). Test service brakes at about **5 mph**.
 - ABS light should go out after start; staying on = fault.

- Emergency equipment: fire extinguisher, spare fuses (unless breakers), **3** red reflective triangles.
- Most important reason to inspect: **safety**.

18 Seeing: mirrors, night, fog, winter, heat

- Look **12–15 seconds** ahead. Convex mirrors: things look **smaller and farther**. Check mirrors 4 times in a lane change.
- Use **high beams whenever safe and legal**; dim within **500 ft** of oncoming traffic **and** when following within 500 ft. Never sunglasses at night.
- Headlights on from ½ hour after sunset to ½ hour before sunrise, when wipers are needed, or when you can't see 1,000 ft.
- Fog: low beams + fog lights + 4-way flashers (even in daytime).
- Fatigue: 15–20 min nap or sleep; the only cure is **sleep**. Crash-heavy hours midnight–6 a.m. Break every ~100 miles or 2 hours.
- Winter: melting ice is **more** slippery; no spray from other vehicles = ice. Retarder **off**.
- Heat: check tires every **2 h or 100 mi**; never let air out of hot tires; radiator cap only when cool to the bare hand. Bleeding tar is very slippery.

Everything, lesson by lesson

For each lesson: key points by topic, every number to memorize, and every exam trap. General Knowledge lessons come first, ordered by test weight and how often they are missed; then Combination.

Order	Lesson	Test weight	Most-missed facts
1	GK-12 Driving emergencies, ABS, and skid control	High	4
2	GK-07 Controlling speed and stopping distance	High	3
3	GK-08 Managing space: following distance, turns, clearance	High	3
4	GK-13 Accidents, fires, alcohol and drugs, hazmat rules for all drivers	High	3
5	GK-14 Transporting cargo safely	High	3
6	GK-02 Disqualifications, violation points, hands-free rules	High	2
7	GK-06 Seeing, mirrors, signals, warning devices	High	2
8	GK-10 Night, fog, winter and hot-weather driving	High	2
9	GK-11 Railroad crossings and mountain driving	High	2
10	GK-04 Vehicle inspection	High	1
11	GK-05 Basic control, backing, shifting, retarders	Medium	3
12	GK-03 Other CDL rules and California laws	Medium	2
13	GK-01 Getting your CDL: classes, tests, medical rules	Medium	0
14	GK-09 Seeing hazards, distracted and aggressive drivers	Medium	0
15	CV-02 Combination air brakes and trailer ABS	High	9
16	CV-03 Coupling and uncoupling	High	5
17	CV-01 Driving combination vehicles safely	High	4
18	CV-04 Inspecting a combination vehicle	High	2

GK-12 · Driving emergencies, ABS, and skid control

General Knowledge · test weight **High** · handbook Section 2.17–2.19

You will be able to answer

- When steering around a hazard is better than stopping, and why the right side is usually the best escape.
- How to leave the road safely and how to get back on (the "steer-countersteer" move).
- The difference between controlled braking and stab braking, and why stab braking is only for trucks without ABS.
- What to do when hydraulic brakes fail, when brakes fail on a downgrade, and when a tire fails.
- What ABS does and does NOT do, which vehicles must have it, how to tell if a vehicle has it, and how to brake with it.
- The four causes of skids and how to correct drive-wheel and front-wheel skids.

Key points

Driving emergencies p. 2-38

- A **traffic emergency** is two vehicles about to collide; a **vehicle emergency** is a tire, brake or other critical part failing.
- Safe driving helps prevent emergencies, and your chance of avoiding a crash depends on how well you act.

Steering to avoid an accident pp. 2-38, 2-39

- **Stopping is not always safest:** turning away almost always takes less time than stopping a heavy truck.
- Keep **both hands on the wheel** at all times, **don't brake while turning**, turn only as much as needed, and be ready to countersteer.
- If an oncoming driver drifts into your lane, **move right**; right is often best when the shoulder is clear or both sides are blocked.
- Leaving the road, avoid braking until you slow to about **20 mph**, and keep one set of wheels on the pavement if you can.
- To return before stopping, turn sharply (don't edge back), then **countersteer immediately** once both front tires are on pavement.

How to stop quickly and safely p. 2-39

- **Controlled braking** means braking as hard as possible without locking the wheels, with very small steering moves.
- **Stab braking** means brake fully, release when the wheels lock, and brake again once they roll; wheels may need **up to 1 second**.
- Stab braking is **only for vehicles WITHOUT ABS**.
- **Don't jam on the brakes**, because locked wheels mean a skid you cannot control.

Brake failure p. 2-40

- Most hydraulic brake failures have **2 causes**: loss of hydraulic pressure and brake fade on long hills.
- If the pedal is spongy or goes to the floor, **downshift, pump the brakes**, and use the parking brake while holding its release.
- If brakes fail on a downgrade, **use an escape ramp**; if none, take the least dangerous route, such as an open field or uphill road.
- **Act as soon as you know** the brakes don't work, because every second the truck gets faster.

Tire failure p. 2-40

- Signs of tire failure are a loud **bang** (assume it was yours), heavy vibration, **heavy steering** (front tire), or fishtailing (rear tire).
- **Hold the steering wheel firmly**, because a failed front tire can twist it out of your hands.
- **Stay off the brake** and let the truck slow first, then brake very gently, pull off, and check all tires.

Anti-lock braking systems — ABS p. 2-41

- **ABS** uses a computer to **prevent wheel lockup** during hard braking, working on top of your regular brakes.
- It switches on only when a wheel is about to lock, and its job is **control, not a shorter stop**.

How ABS works p. 2-41

- **Sensors** detect a wheel about to lock, and the **ECU** lowers brake pressure on that wheel.
- It gives the **most braking possible without lockup**, reacting far faster than any driver; otherwise the brakes work normally.

Vehicles required to have ABS p. 2-41

- ABS is required on **truck tractors with air brakes** built on or after **March 1, 1997**.
- ABS is required on **all other air-brake vehicles** (buses, trucks, trailers, converter dollies) built on or after **March 1, 1998**.
- ABS is required on hydraulic-brake trucks and buses of **10,000 lb or more** GVWR built on or after **March 1, 1999**.

How to know if your vehicle has ABS p. 2-41

- Tractors, trucks and buses have a **yellow** ABS malfunction lamp on the instrument panel; trailers have one on the **left side**, front or rear corner.
- The lamp lights at start-up then goes out; on older systems it may stay on until you exceed **5 mph**.
- A lamp that **stays on** after the bulb check or lights while driving means you may have **lost ABS control**.
- On older towed units, look underneath for an **ECU** box and wheel-speed sensor wires.

How ABS helps you p. 2-41

- Without ABS, locked **steering wheels** mean lost steering; other locked wheels can cause a **skid, jackknife, or spin**.
- With ABS you **may or may not stop faster**, but you should be able to **steer around an obstacle while braking**.

ABS on only the tractor or only the trailer p. 2-42

- Even ABS on only the tractor, only the trailer, or a single axle improves control, so **brake normally**.
- **Tractor-only ABS**: watch the trailer and **ease off the brakes** if it starts to swing out.
- **Trailer-only ABS**: if you lose steering or the tractor starts to jackknife, **ease off the brakes** until back in control.

Braking with ABS p. 2-42

- Brake the way you always have, using **only as much force as you need** for a safe, controlled stop.
- Brake the same way whether ABS is on the bus, tractor, trailer or both, and back off if needed to stay in control.
- Exception: with **working ABS on ALL axles**, you may **brake fully** in an emergency stop.

Braking if ABS is not working p. 2-42

- Without ABS you **still have your normal brakes**, so drive and brake as usual.
- A lamp that stays on or lights while driving means lost ABS on **1 or more wheels**; **get it repaired soon**.

Safety reminders — what ABS will NOT do p. 2-42

- ABS is **no excuse** for more speed, shorter following distance, or careless driving.
- ABS does **not** stop power skids or turning skids; it should stop skids and jackknifes **caused by braking**.
- ABS will **not necessarily shorten** stopping distance and doesn't change your ultimate stopping power.
- Worn or poorly maintained brakes are still bad brakes, and a careful **driver** remains the most important safety feature.

Skid control and recovery p. 2-42

- A **skid** happens when the tires stop gripping the road.
- The **4 causes** are over-braking (including a speed retarder on slippery roads), over-steering, over-acceleration, and driving too fast.
- **Driving too fast** causes most serious skids.

Drive-wheel skids p. 2-43

- **Drive-wheel skids** are by far the most common, when rear drive wheels lose grip from hard braking or too much throttle.
- For an acceleration skid on ice or snow, **take your foot off the accelerator**; if very slippery, also **push in the clutch**.
- A locked wheel grips less than a rolling one, so a bus or straight truck may **spin out** and a tractor-trailer may **jackknife**.

Correcting a drive-wheel braking skid p. 2-43

- First **stop braking** so the rear wheels roll again and stop sliding.
- Then **countersteer** quickly the other way as the vehicle straightens, or you will skid in the **opposite direction**.
- These skills need practice, and a large driving range or **skid pad** is the best place to learn them.

Front-wheel skids p. 2-43

- Front-wheel skids are most often caused by **driving too fast for conditions**, plus worn front tread or too little front-axle weight.
- The front end keeps going **straight** no matter how much you turn the wheel.
- **The only fix is to let the vehicle slow down:** stop turning so sharply and/or braking so hard, without starting a new skid.

Numbers and terms to memorize

Item	Value	Page
Traffic emergency	Two vehicles about to collide	2-38
Vehicle emergency	Tires, brakes, or other critical parts fail	2-38
Turn vs. stop	You can almost always turn to miss an object faster than you can stop	2-38
Countersteer	Turning the wheel back the other way after the emergency turn	2-38
Oncoming driver in your lane	Steer right	2-39
Leaving the road	Avoid braking until about 20 mph , then brake very gently	2-39
Back on the road	Turn sharply; countersteer when both front tires are on pavement	2-39
Controlled braking	Brake as hard as possible without locking wheels; tiny steering moves	2-39
Stab braking	Brake fully → release when wheels lock → reapply when wheels roll	2-39
Wheel restart time (stab braking)	Up to 1 second	2-39
Stab braking allowed	Only without ABS	2-39
Hydraulic brake failure causes	2: loss of hydraulic pressure; brake fade on long hills	2-40
Loss of pressure sign	Pedal spongy or goes to the floor	2-40
Escape ramp location	Usually a few miles below the top of the downgrade	2-40
Heavy steering	Probably a front tire failure	2-40
Tire failure response	Hold wheel firmly; stay off brake until slowed	2-40
ABS required: air-brake tractors	Built on/after March 1, 1997	2-41
ABS required: other air-brake vehicles	Built on/after March 1, 1998	2-41
ABS required: hydraulic trucks/buses ≥ 10,000 lb GVWR	Built on/after March 1, 1999	2-41
ABS lamp color	Yellow	2-41
Trailer ABS lamp location	Left side , front or rear corner	2-41
Older ABS lamp may stay on until	Over 5 mph	2-41

Item	Value	Page
ECU	Electronic control unit (ABS computer)	2-41
Full brake application allowed	Working ABS on all axles , emergency stop	2-42
Causes of skids	4: over-braking, over-steering, over-acceleration, driving too fast	2-42
Most common skid	Drive-wheel (rear-wheel) skid	2-43
Most serious skids	Driving too fast for conditions	2-42
Drive-wheel braking skid fix	Stop braking , then countersteer	2-43
Front-wheel skid fix	Let the vehicle slow down ; stop turning/braking so hard	2-43

Exam traps

Tempting wrong idea	Handbook answer
In an emergency, always stop.	stopping is not always safest; turning is usually quicker than stopping. The instinct to brake is strong, so this feels right.
Steer left around an oncoming car in your lane	steer right . Left looks like the "passing lane," but the other driver will likely swerve back to that side.
Brake hard as soon as you leave the road	avoid braking until about 20 mph , then brake gently. Braking feels safe, but loose shoulders make locked wheels skid.
Ease back onto the road slowly and gradually	turn sharply back on, then countersteer when both front tires are on pavement. "Gentle" sounds safe, but tires can grab the edge.
Emergency braking means pushing the pedal as hard as you can	that locks the wheels and causes a skid; use controlled or stab braking.
Use stab braking in a truck with ABS	stab braking is only for vehicles without ABS .
In stab braking, reapply the instant you release	wait until the wheels roll again (up to 1 second), or the truck won't straighten.
With a hydraulic failure, pull the parking brake hard and hold it	hold the release button/lever at the same time so you can control it and prevent lockup.
Brakes fail on a hill — wait to see if they come back	take the escape route as soon as you know; speed only grows.
Brake hard after a blowout	stay off the brake unless about to hit something; hold the wheel firmly. The instinct to brake is the danger.
ABS shortens stopping distance / adds braking power	ABS helps you keep control ; it does not necessarily shorten stopping distance or change stopping power.
ABS prevents all skids	it prevents braking skids, not power skids or turning skids.
With ABS, brake differently (pump or press hard every time)	brake as you always have ; full application only with working ABS on all axles in an emergency.
ABS lamp is red / on the right side of the trailer	yellow , trailer lamp on the left side.

Tempting wrong idea	Handbook answer
1998 is the tractor date	air-brake tractors 1997 ; other air-brake vehicles 1998 ; hydraulic $\geq 10,000$ lb 1999 .
The most common skid is a front-wheel skid	drive-wheel skids are by far the most common.
Drive-wheel braking skid — keep braking	stop braking , then countersteer .
Acceleration skid — hit the brakes	take your foot off the accelerator (push in the clutch if very slippery).
Front-wheel skid — turn the wheel more	the front goes straight no matter what; let the vehicle slow down .
Retarders help on slippery roads	using the retarder on a slippery road can cause a skid (over-braking).

GK-07 · Controlling speed and stopping distance

General Knowledge · test weight **High** · handbook Section 2.6–2.6.8

You will be able to answer

- Name the three parts of total stopping distance and give each one's value at 55 mph ($142 + 61 + 216 = 419$ ft).
- Say how much braking distance and impact grow when you double, triple or quadruple your speed (4×, 9×, 16×).
- Explain why an empty truck needs **MORE** room to stop than a loaded one.
- Pick a safe speed for wet roads, packed snow, ice, curves, fog or low beams, heavy traffic, downgrades and work zones.
- Recognize slippery surfaces (black ice, bridges, shade, first rain) and say what to do when hydroplaning.
- **CA** Recall the brake-performance distances from 20 mph, the rule for passing slow vehicles on a grade, the 300-foot following rule and the 100-foot caravan gap.

Key points

Why speed matters p. 2-15

- Driving too fast is a **major cause of fatal crashes**.
- Your speed must fit **five things**: traction (tire grip), curves, visibility, traffic, and hills.

Stopping distance pp. 2-15, 2-16

- Total stopping distance = perception + reaction + braking distance: at 55 mph $\approx 142 + 61 + 216 = 419$ ft, a minimum under ideal conditions.
- Perception takes about **1¾ seconds** and reaction ¾ to 1 second; during both you are still rolling at full speed.
- Double the speed → impact and braking distance **4 times** greater (triple → 9, quadruple → 16); empty trucks need **longer** stopping distances because they have less traction.
- **CA** Service brakes must hold the vehicle still on any grade, loaded or empty, and stop it from **20 mph** within a maximum stopping distance.
- **CA** Limits from 20 mph: car 25 ft; single under 10,000 lb 30 ft; single 10,000 lb or more, any bus, light vehicle towing 40 ft; other combinations **50 ft**.

Matching speed to the road surface pp. 2-16, 2-17

- Traction is the friction between tires and road; without it you **cannot steer or brake**.
- Wet roads can double stopping distance → slow about **one-third** (55 → about 35 mph); packed snow → slow **half or more**; ice → crawl, stop when safe.
- Watch for shaded areas, bridges (freeze **before the road**; take care near 32°F), melting ice, **black ice** (road just looks wet), and the first rain mixing with oil.
- Hydroplaning can happen as low as **30 mph**; release the accelerator and push in the clutch, and **do not brake**.

Speed and curves p. 2-17

- Too fast in a curve → the tires lose traction and you **skid off the road**, or they keep traction and the vehicle **rolls over**.
- **Slow down before** entering the curve, because braking inside it can lock the wheels and start a skid.
- Never exceed the posted curve speed, and be in a gear that lets you **accelerate slightly** through the curve for control.

Speed and distance ahead p. 2-17

- Your stopping distance must always fit **inside the distance you can see ahead**.
- Fog, rain or other conditions that cut visibility → slow down.
- At night, **low beams** show less road than high beams, so slow down when using them.

Speed and traffic flow p. 2-17

- In heavy traffic, the safest speed is **the same speed as the vehicles around you**, if legal and safe.
- **CA** Truck and bus speed limits are lower than car limits, by **as much as 15 mph**, so take extra care changing lanes or passing.
- Speeding to save time saves very little; constant passing raises **crash risk** and fatigue.

Speed on downgrades p. 2-18

- Pick a speed that fits total weight, grade length, grade steepness, road conditions, and weather; never exceed a **"Maximum Safe Speed"** or posted limit.
- **Engine braking** is your main speed control, strongest near governed rpm in a lower gear; save the service brakes for slowing or stopping.
- **Shift to a low gear before** starting down the grade, so you have full engine braking from the top.

Roadway work zones p. 2-18

- **Speeding traffic** is the number one cause of injuries and deaths in work zones.
- Obey the posted work-zone speed limit and watch your speedometer so speed doesn't **creep up** in long construction areas.
- Slow down for bad weather or road conditions, and **even more** when a worker is close to the road.

CA Overtaking or following another vehicle p. 2-18

- **CA** Outside business or residential districts, pass a vehicle going under **20 mph** on a grade only if at least **10 mph faster** and finishing within ¼ mile.
- **CA** Stay at least **300 feet** behind trucks with 3+ axles, vehicles towing, school buses with pupils, farm labor vehicles, explosives carriers, and trailer buses.
- **CA** The 300-foot rule does not apply while passing, where there are **2 or more lanes** each way, or in business or residential districts.
- **CA** Large vehicles traveling in a caravan on open highway must leave at least **100 feet** between them so others can pass.

Numbers and terms to memorize

Item	Value	Page
Total stopping distance formula	Perception + reaction + braking distance	2-15
Perception time (alert driver)	1¾ seconds	2-15
Perception distance at 55 mph	142 ft	2-15
Reaction time (average driver)	¾ second to 1 second	2-15
Reaction distance at 55 mph	61 ft	2-15
Braking distance at 55 mph (dry, good brakes)	about 216 ft	2-16
Total stopping distance at 55 mph	at least 419 ft	2-16
Stopping distance at 60 mph	longer than an American football field	2-16
Speed $\times 2$ / $\times 3$ / $\times 4$ (from 20 mph)	impact and braking distance $\times 4$ / $\times 9$ / $\times 16$	2-16
Empty truck	needs longer stopping distance (less traction)	2-16
CA Service brake hold	stationary on any grade, loaded or empty (CVC §26454)	2-16
CA MSD from 20 mph	car 25 ft · single vehicle $<10,000$ lb 30 ft · single $\geq 10,000$ lb or any bus 40 ft · light vehicle towing 40 ft · all other combinations 50 ft	2-16
Traction	friction between tires and road	2-16
Wet road	can double stopping distance; slow by about $\frac{1}{3}$ (55 $\rightarrow$ about 35)	2-16
Packed snow	slow by half or more	2-16
Ice	crawl; stop driving as soon as safe	2-16
Bridges	freeze before the road; watch near 32°F	2-17
Black ice	thin clear ice; road looks wet	2-17
Ice check	feel front of mirror, mirror support or antenna	2-17
Hydroplaning	tires ride on water; can occur as low as 30 mph	2-17
Hydroplaning response	release accelerator, push in clutch, no brakes	2-17
Curve too fast	skid off road OR roll over	2-17
Sight rule	stop within the distance you can see	2-17
CA Truck/bus vs car speed limits	truck/bus lower, by up to 15 mph	2-17
Downgrade speed factors	weight, grade length, steepness, road conditions, weather	2-18
Main downgrade speed control	engine braking (low gear, near governed rpm)	2-18

Item	Value	Page
Work zones	speeding traffic = number one cause of injury and death	2-18
CA Passing on a grade	vehicle <20 mph: pass ≥10 mph faster, finish within ¼ mile	2-18
CA Following listed vehicles	at least 300 ft (not when passing, 2+ lanes each way, or business/residential district)	2-18
CA Caravan gap	at least 100 ft	2-18

Exam traps

Tempting wrong idea	Handbook answer
An empty truck stops in a shorter distance because it is lighter.	empty trucks need a longer stopping distance. Lighter <i>feels</i> easier to stop, but an empty truck has less traction.
About 300 ft for total stopping distance at 55 mph	419 ft. 300 ft sounds like "a football field" and the 300-ft following rule.
216 ft as the total at 55 mph	216 ft is only the braking part; the total is 419 ft.
Double the speed = double the braking distance.	4 times. Speed and distance feel like they should grow together, but braking distance grows with speed × speed.
Perception time = ¾ second	¾ to 1 second is reaction time; perception is 1¾ seconds.
Brake hard when hydroplaning	release the accelerator and push in the clutch; do not brake. Braking is instinct, but the tires have no grip.
Hydroplaning only happens at highway speeds.	it can happen as low as 30 mph with a lot of water.
Black ice is easy to see.	it is clear and makes the road look wet.
A road is most slippery after hours of rain.	it is very slippery just after rain begins (water + oil); steady rain washes the oil away.
Slowing for a curve while in the curve	slow down before entering; braking in the curve can lock wheels.
Packed snow → reduce speed by one-third	one-third is for wet roads; packed snow needs half or more.
Use the service brakes as the main way to control speed downhill.	engine braking in a low gear is the main way; save the brakes.
Driving faster than traffic saves a lot of time.	it saves little and raises crash risk and fatigue; match the flow of traffic.
40 ft as the MSD for a tractor-semitrailer	50 ft ("all other combinations"). 40 ft is for single heavy vehicles, buses and light vehicles towing.
Mixing the 300-ft following rule with the 100-ft caravan gap	300 ft behind listed vehicles; 100 ft between caravan vehicles.

GK-08 · Managing space: following distance, turns, clearance

General Knowledge · test weight **High** · handbook Section 2.7–2.7.7

You will be able to answer

- Work out the right following distance, in seconds, for any truck length at any speed.
- Measure the following distance you actually have, using a landmark and a count.
- Say what to do — and what NOT to do — when someone tailgates you.
- Explain why you avoid driving beside other vehicles, and what strong wind changes.
- Judge clearance above and below your vehicle, including why an empty van can hit a bridge that a loaded one cleared.
- Make right and left turns safely: button hook vs jug handle, oncoming traffic, and which lane to use when there are two turn lanes.

Key points

Why space matters p. 2-19

- Space around your vehicle is **time**: when something goes wrong, space gives you time to think and act.
- A truck needs more space than a car because it **takes up more room** and needs more room to stop and turn.
- Manage space on every side (ahead, behind, sides, above, below) plus space for turns and crossing traffic.

Space ahead p. 2-19

- The space **ahead** is the most important space, because a smaller vehicle ahead can usually stop faster than your truck.
- Below 40 mph, keep at least **1 second for every 10 feet** of vehicle length; above 40 mph, **add 1 second** (once, not per extra 10 mph).
- Example: a 40-ft truck needs **4 seconds** (5 above 40 mph); a 60-ft rig needs 6 seconds (7 above 40 mph).
- Measure by counting "one thousand-and-one..." from when the vehicle ahead passes a fixed landmark until you reach it; slippery roads need **more** space.

Space behind pp. 2-19, 2-20

- Trucks get tailgated when **traveling slowly** (often uphill) or in **bad weather**; with a heavy load, stay in the right lane where possible.
- If tailgated, **avoid quick changes**: signal early and slow down very gradually, because a sudden slowdown is how the tailgater hits you.
- **Increase your following distance** ahead, so you won't need to brake or swerve suddenly and the tailgater can pass more easily.
- **Do not speed up**, since being tailgated at a low speed is safer, and avoid tricks like turning on taillights or blinking brake lights.

Space to the sides p. 2-20

- Stay **centered in your lane** to keep some clearance on both sides.
- Avoid traveling next to other vehicles: a driver may **swerve into your lane**, or you may be boxed in when you need to change lanes.
- **Strong winds** can push you out of your lane, especially lighter vehicles and when coming out of a tunnel.

Space overhead p. 2-20

- **Do not trust posted heights** at bridges and overpasses, because repaving or packed snow may have lowered the clearance.
- An **empty** van sits higher than a loaded one, so a bridge you cleared loaded may not clear empty.
- Not sure you will fit → **take another route**; where the road tilts, move a little toward the center.
- Before backing, get out and check for overhanging objects such as **tree branches or electric wires**.

Space below p. 2-20

- Space **under** the vehicle can be very small when it is heavily loaded.
- Watch for **dirt roads and unpaved yards**, and cross drainage channels carefully so the vehicle's ends don't drag.
- Don't risk getting hung up on **railroad tracks**, especially with low-clearance trailers.

Space for turns pp. 2-20, 2-21

- Right turns: turn slowly and **turn wide as you complete the turn** (button hook), keeping the back near the curb so no one passes on the right.
- **Do not swing left** before a right turn (jug handle); a driver behind may pull up on your right and get hit.
- If the turn puts you in the oncoming lane, leave room or stop and wait, but **do not back up**.
- Left turns: start only at the **center of the intersection**; with 2 left-turn lanes, use the **right-hand** (outside) lane.

Space to cross or enter traffic p. 2-21

- Trucks accelerate slowly and take up a lot of room, so joining traffic needs a **much larger gap** than a car.
- Acceleration depends on the load; when heavily loaded, allow **even more room**.
- Start across a road only when sure you can get **all the way across** before any traffic reaches you.

Numbers and terms to memorize

Item	Value	Page
Most important space	The space ahead	2-19
Following distance, below 40 mph	1 second per 10 ft of vehicle length	2-19
Following distance, above 40 mph	Same rule + 1 second	2-19
40-ft truck	4 s below 40 mph · 5 s above	2-19
60-ft rig	6 s below 40 mph · 7 s above	2-19
50-ft truck above 40 mph	6 s	2-19

Item	Value	Page
30-ft vehicle at 55 mph	4 s	2-19
How to measure	Fixed landmark + count "one thousand-and-one..."	2-19
Slippery road	Needs more space than the rule	2-19
Tailgated when	Traveling slowly · bad weather · can't keep up (often uphill)	2-19
Tailgated → do	Avoid quick changes · increase following distance · don't speed up · no light tricks	2-19, 2-20
Dangers of driving alongside	2: sudden lane change into you · being trapped	2-20
Strong wind	Worse for lighter vehicles; bad leaving tunnels	2-20
Posted bridge height	May be wrong (repaving, packed snow)	2-20
Empty vs loaded van	Empty is higher	2-20
Tilted road	Drive a little closer to the center	2-20
Space below — trouble spots	Dirt roads, unpaved yards, drainage channels, railroad tracks	2-20
Button hook	Correct right turn: wide at the end , rear near curb	2-20
Jug handle	Incorrect right turn: swinging left first	2-20
Oncoming car during a right turn	Give room or stop — never back up	2-20
Left turn starts	At the center of the intersection	2-20
Two left-turn lanes	Use the right-hand lane	2-21
Entering traffic	Much larger gap than a car; more when loaded	2-21

Exam traps

Tempting wrong idea	Handbook answer
When tailgated, slow down or flash your brake lights to warn the driver	Increase your following distance , signal early and slow very gradually. No taillight or brake-light tricks. Warning the driver feels natural, but sudden signals invite a rear-end crash.
Speed up to get away from the tailgater	Do not speed up — being tailgated at low speed is safer.
Decrease your following distance so the tailgater has room behind you.	False — you increase the space ahead.
Above 40 mph, add 1 second for every extra 10 mph	Add 1 second, once . A 40-ft truck at 65 mph still needs 5 s.
A 30-ft truck at 55 mph needs 3 seconds	4 seconds (3 + 1, because it is over 40 mph). People forget the extra second.
The height on the bridge sign is exact	Clearance may be lower than posted (repaving, packed snow).
If you fit under a bridge loaded, you will fit empty	An empty van rides higher .

Tempting wrong idea	Handbook answer
Swing left before a tight right turn to make room	No — that is the jug handle . Turn wide as you complete the turn (button hook).
Back up to let an oncoming car pass during your turn	Do not back up — give room or stop.
With two left-turn lanes, take the inside (left) one so you have more room	Take the right-hand lane.
Start a left turn early so your trailer clears the corner	Reach the center of the intersection first; turning early can make off-tracking swing the left side of your vehicle into another vehicle.
Strong wind is mainly a problem for heavy, loaded trucks	It is usually worse for lighter vehicles.
On a road that tilts you toward roadside signs, move toward the edge to stay out of traffic	Move a little toward the center of the road.

GK-13 · Accidents, fires, alcohol and drugs, hazmat rules for all drivers

General Knowledge · test weight **High** · handbook Section 2.20–2.23

You will be able to answer

- The five basic steps at any accident scene, what to do first, and what information to collect.
- The main causes of vehicle fires (including tire fires) and how to prevent them.
- How to fight a truck fire: where to stop, engine vs. cargo fires, where to stand, where to aim, and which extinguisher works on which fire.
- How alcohol works in the body, the CMV blood alcohol limit, the 4-hour rule, and why coffee doesn't sober you up.
- Which drugs and medicines are a problem, and what to do if you get sick while driving.
- What a HazMat placard is, why it is used, how many are needed and how big they are, and when you need an "H" endorsement.

Key points

Accident procedures p. 2-44

- If you crash but are not badly hurt, act to prevent more damage and injuries.
- The **5 basic steps** at any accident → protect the area, notify authorities, care for the injured, collect the required information, report the accident.
- Think in time order → make the scene safe, get help coming, help people, then do the paperwork.

Protect the area p. 2-44

- Your **first job** at any crash is to stop a second accident from happening at the same spot.
- Your truck is involved → move it to the **side of the road**, so it prevents another crash and keeps traffic moving.
- Stopping to help → park **away from** the wreck, because emergency vehicles need that space.
- Turn on your **4-way flashers** and place reflective triangles so other drivers see them in time.

Notify authorities p. 2-44

- Have a cell phone or CB → call for help **before you get out** of your truck.
- No phone or radio → first **protect the scene**, then phone the police or send someone to phone.
- Figure out **exactly where you are** so you can give the precise location.

Care for the injured p. 2-44

- If someone **trained** (like a medic) is already treating people, let them work and help only if asked.
- Do not move a badly injured person unless there is danger from **fire or passing traffic**; moving them can worsen injuries.
- Control heavy bleeding with **direct pressure** on the wound, and keep hurt people **warm**.

Gather information p. 2-44

- If you were in the accident, you must file an **accident report** and collect the information for it.
- Get each driver's name, address, and **DL number**, plus plate numbers and types of vehicles, and the owner's name and address if not the driver.
- Record the officer's **name, badge number, and agency**, plus names and addresses of injured people and witnesses.
- Note the damage, the **exact location**, and each vehicle's direction of travel.
- **CA** The SR 1 report to DMV is due **within 10 days** if there is injury, death, or damage over **\$1,000**.

Fires p. 2-44

- Truck fires come from five sources → after accidents, **tires**, the electrical system, fuel, and cargo.
- Prevent fires with pre-trip and en route inspections, safe procedures, and a **charged extinguisher**.
- Untrained drivers have turned small fires into **bigger ones**, so learn how your extinguisher works before you need it.

Causes of fire p. 2-44

- After accidents → **spilled fuel** and improper use of flares; electrical → short circuits from damaged insulation and loose connections.
- Tire fires come from **under-inflated tires** and **dual tires that touch**, which build heat through friction and flexing.
- Fuel → driver smoking, improper fueling, loose fuel connections; cargo → flammable cargo, cargo not properly sealed or loaded, **poor ventilation**.

Fire prevention pp. 2-44, 2-45

- Pre-trip, fully inspect the **electrical, fuel, and exhaust systems**, tires, and cargo, and make sure the extinguisher is **charged**.
- At **every stop** on a trip, check the tires, wheels, and truck body for any **sign of heat**.
- Check gauges often for overheating, use mirrors to look for **smoke**, and be careful with anything flammable.

Fire fighting p. 2-45

- Stop in an **open spot** away from buildings, trees, brush, and vehicles; **never pull into a service station**, which is full of fuel.
- Engine fire → shut the engine off and **avoid opening the hood**; van or box cargo fire → keep the **doors shut** so the fire gets no oxygen.
- Stay as far away as possible, **upwind**, aim at the **base** of the fire, and keep going until the burning material has cooled.
- **B:C** extinguisher → electrical fires and burning liquids; **A:B:C** adds wood, paper, and cloth.
- **Never use water** on electrical fires (shock) or gasoline fires (spreads); unsure which extinguisher, especially with HazMat → wait for firefighters.

Alcohol and driving pp. 2-46, 2-47

- Never drink on duty, and drink **no alcohol at all** within **4 hours** before going on duty.
- **CA** It is illegal to drive a CMV with a BAC of **0.04% or more**; below 0.04% is still not automatically safe or legal.
- The liver processes only about **1/3 ounce of alcohol per hour**, so coffee or cold showers do nothing; **only time** lowers your BAC.
- **One drink** = 12 oz 5% beer = 5 oz 12% wine = **1½ oz** 80-proof liquor; BAC depends on how much, how fast, and your weight.
- Alcohol first hits **judgment and self-control**, so drinkers cannot tell they are getting drunk.

Other drugs p. 2-47

- It is illegal to be under the influence of any **controlled substance**, amphetamines ("pep pills," "uppers," "bennies"), narcotics, or anything making you unsafe.
- Prescription and OTC drugs, like **cold medicines**, can cause drowsiness; a doctor's drug is OK only if the doctor says it won't affect safe driving.
- Never take a drug to hide tiredness; **rest** is the only cure for fatigue.
- **Alcohol makes other drugs much stronger**, so the safest rule is not to mix drugs and driving at all.

Illness p. 2-47

- If you are **too sick to drive safely**, do not drive.
- In an emergency, drive only as far as the **closest place where it is safe to stop**.

Hazardous materials rules for all commercial drivers p. 2-48

- Every CMV driver must be able to **spot hazardous cargo** and know whether they can haul it without an **"H" endorsement**.
- Anyone applying for an original or renewal "H" must pass a **TSA security threat assessment**, a federal background check.
- You start it **after** applying for your CDL, passing all knowledge tests, and turning in a valid medical form.
- Give a TSA agent your **fingerprints, a fee**, a copy of your CLP, and one ID.

What are hazardous materials? p. 2-48

- **HazMat** is anything that could harm people's health or safety, or damage property, while being shipped.
- There are **9 numbered hazard classes**, e.g. 1 explosives, 2 gases, 3 flammable liquids like gasoline, 7 radioactive, 8 corrosives.
- **ORM-D** (like hair spray) and **combustible liquids** (like fuel oils) have no class number.

Why are there rules? p. 2-48

- HazMat rules have **3 goals** → contain the product, communicate the risk, and ensure safe drivers and equipment.
- Shippers warn of risk with a **shipping paper** and **diamond-shaped labels**, since you may be hurt and unable to speak after a crash.
- HazMat papers must be marked or kept **on top**, in a door pouch or in clear view within reach, and on the **driver's seat** when you leave.

Placards, ID numbers, and who may drive pp. 2-48, 2-49

- Placards are outside warning signs showing the hazard class; you need **at least 4 identical** ones, front, rear, and both sides.
- Each placard is at least **9.84 inches (250 mm) square**, set on a point as a diamond; answer this, not 10¾ inches, on the test.
- ID numbers are **4 digits** after "UN" or "NA," listed in the DOT **Emergency Response Guidebook (ERG)**.
- Placards required → you need an **"H" endorsement**; not required → you may drive without one; driving placarded without an H is a crime.
- Tank endorsement → each tank over **119 gallons** and **1,000 gallons or more** total; on the test, tank only → N, tank with placarded HazMat → X.

Numbers and terms to memorize

Item	Value	Page
Accident steps (5)	Protect, Notify, Care for injured, Collect info, Report	2-44
First thing at a crash scene	Prevent another accident at the same spot	2-44
Stopping to help	Park away from the crash	2-44
Have a phone/CB	Call before getting out of the vehicle	2-44
Moving a badly injured person	Only if fire or passing traffic makes it necessary	2-44
Heavy bleeding	Direct pressure on the wound	2-44
Officer information	Name, badge number , agency	2-44
Tire fire causes	Under-inflated tires; duals that touch	2-44
En route fire check	Tires, wheels, body for heat at every stop	2-44
Where to stop a burning truck	Open area; never a service station	2-45
Engine fire	Engine off; don't open the hood ; spray through louvers, radiator, underside	2-45
Van/box cargo fire	Keep doors shut	2-45
Extinguisher position	As far away as possible, upwind	2-45
Aim	Base/source of the fire	2-45
B:C extinguisher	Electrical fires and burning liquids	2-45
A:B:C extinguisher	Also wood, paper, cloth	2-45
Water — never on	Electrical (shock) or gasoline (spreads) fires	2-45
Burning tire	Must be cooled — lots of water	2-45
Fire classes	A ordinary materials · B greasy liquids · C electrical · D metals	2-45
Deaths from drinking drivers	Over 20,000 per year	2-46
No drinking before duty	4 hours	2-46
BAC unit	mg of alcohol per 100 ml of blood	2-46
BAC .10 means	1/10 of 1% (1/1000) of blood is alcohol	2-46

Item	Value	Page
CA CMV BAC limit	0.04% or more is illegal	2-47
Liver rate	About 1/3 ounce of alcohol per hour (fixed)	2-47
Chart rule	Subtract .01% per 40 minutes of drinking	2-46
One drink	12 oz 5% beer = 5 oz 12% wine = 1½ oz 80-proof liquor	2-47
BAC depends on	Amount, speed of drinking, body weight	2-47
First brain function hit	Judgment and self-control	2-47
Amphetamine street names	"Pep pills," "uppers," "bennies"	2-47
Only cure for fatigue	Rest	2-47
Too ill to drive	Don't drive; emergency → nearest safe place to stop	2-47
HazMat rule goals (3)	Contain product, communicate risk, ensure safe drivers & equipment	2-48
Shipping paper locations	Door pouch; clear view within reach; driver's seat when out	2-48
Placards needed	At least 4 identical — front, rear, both sides	2-48
Placard size	At least 9.84 in (250 mm) square, diamond on point	2-48
ID number	4 digits , after " UN " or " NA "	2-48
ERG	Emergency Response Guidebook (lists chemicals and ID numbers)	2-48
H endorsement needed	Only when the load requires placards	2-49
Tank vehicle	Tank > 119 gal each and ≥ 1,000 gal total	2-49

Exam traps

Tempting wrong idea	Handbook answer
The first thing to do at a crash is help the injured	first protect the area so no second crash happens. Helping people feels most urgent, but a second crash can hurt everyone.
Park right next to the crash to help	park away from it; emergency vehicles need that space.
Move injured people off the road right away	don't move a badly injured person unless fire or traffic makes it necessary.
Drive to a service station for help with a fire	never — stop in an open area away from anything that can burn.
Open the hood to get at an engine fire	don't open the hood ; it feeds the fire air.
Open the trailer doors to find a cargo fire	keep the doors shut — oxygen makes it burn fast.
Get close and aim at the flames	stay as far away as possible , upwind , and aim at the base .
The fire is out once you see no smoke or flame	keep going until it has cooled ; it can restart.

Tempting wrong idea	Handbook answer
A B:C extinguisher works on everything	it is for electrical and liquid fires, not wood, paper, cloth — that needs A:B:C (or water).
Water works on any fire	never on electrical (shock) or gasoline (spreads) fires.
Coffee, fresh air, or a cold shower sobers you up	only time — the liver works at a fixed rate.
Beer is "weaker" than liquor	a 12-oz beer, 5-oz wine, and 1½-oz shot contain the same alcohol.
The CMV limit is .08	0.04% ; .08 is the general "drunk driving limit" printed on the Figure 2.23 chart, not the CMV limit. And below .04 is not automatically safe or legal.
No drinking within 8 (or 2) hours of duty	4 hours , any amount of alcohol.
Over-the-counter medicine is always fine	cold medicines can make you drowsy ; a doctor's drug is OK only if the doctor says it won't affect driving.
Pep pills help on long trips	never use drugs that hide fatigue; only rest cures it.
Any HazMat load needs an H endorsement	only loads that require placards .
Placards are 10¾ inches	handbook says at least 9.84 in (250 mm) .
Two placards (front and rear) are enough	at least 4 identical — front, rear, and both sides.
Keep HazMat papers in the glove box or under other papers	clearly identified or kept on top ; in the door pouch or in clear view within reach; on the driver's seat when you leave.

GK-14 • Transporting cargo safely

General Knowledge • test weight **High** • handbook Section 3.1–3.4

You will be able to answer

- The four cargo duties every driver has, even when someone else loaded the truck.
- When to inspect and re-check cargo during a trip (first 50 miles; every 3 hours or 150 miles; every break).
- What GVW, GCW, GVWR, GCWR, axle weight and tire load mean, and what a bridge formula does.
- How a high center of gravity or badly balanced weight affects steering, traction and rollover.
- How many tie-downs a load needs, how strong they must be, and what blocking, bracing and header boards do.
- How to handle special cargo (sealed, dry bulk, hanging meat, livestock, oversized) and **CA** California's rules for spills, wide loads and projecting loads.

Key points

Start here

- Cargo is one of the biggest General Knowledge test topics, and **weight distribution** is one of the most-missed areas.
- An **axle** connects a pair of wheels; the **steering axle** is the front one you steer, and the drive axles are turned by the engine.
- **Center of gravity (CG)** is the balance point of all the weight; a high CG means weight sits high above the road.

Section 3 introduction: why cargo safety matters p. 3-1

- Bad loading or securing is dangerous; falling cargo can hurt others, loose cargo can hurt you, and overloads can damage your vehicle.
- Even if someone else loaded it, **you** must inspect cargo and recognize **overloads and poorly balanced weight**.
- You must know cargo is secured and does not block your **view** or your access to **emergency equipment**.
- Hauling HazMat that requires **placards** also needs an **"H" endorsement**.

Inspecting cargo p. 3-1

- Check cargo in your pre-trip inspection for **overload**, balance, and proper securing.
- Re-check cargo and securing devices within the **first 50 miles**, because loads settle and straps loosen once moving.
- Inspect again after every **3 hours or 150 miles** of driving and after **every break**.
- Weight, securing, covering, and route rules **differ from place to place**, so learn them for every area you drive.

Weight terms you should know pp. 3-1, 3-2

- "W" = actual weight, "WR" = the manufacturer's rating; **GVW** is a single vehicle plus load, **GCW** a combination plus load.
- **GVWR** and **GCWR** are the manufacturer's stated loaded weights for a single vehicle and a combination.
- No maker GCWR → add power unit GVWR + towed unit weight + its load, e.g. 30,000 + 15,000 + 30,000 = **75,000 lb.**
- **Tire load** is the most weight a tire can safely carry at a stated pressure, printed on the tire's side.

Legal weight limits p. 3-2

- States set maximums for **GVWR, GCWR, and axle weights**; stay within legal limits.
- A **bridge formula** allows **less** weight on axles that are closer together, to prevent overloading bridges and roads.
- Overloading hurts steering, braking, and speed control, **increases stopping distance**, and can make brakes fail.
- Legal is not always safe; in **bad weather or mountains**, the legal maximum weight may be unsafe.

Do not be top-heavy p. 3-2

- A **high center of gravity** raises your risk of tipping over, most of all in **curves** or when swerving.
- Load cargo **as low as possible**, with the **heaviest items under the lightest** ones.

Balance the weight p. 3-2

- **Too much** weight on the steering axle → hard steering and damage to that axle and tires.
- **Too little** weight on the front axle → steering axle too light to steer safely.
- Too little weight on the **drive axles** → poor traction, wheels spin, and the truck may get stuck in bad weather.
- A **high CG** → greater chance of rollover; on a flatbed, the load may also shift sideways or fall off.

Blocking and bracing p. 3-2

- **Blocking** goes against the front, back, and/or sides of cargo, fitted tightly and fastened to the deck, so it cannot slide.
- **Bracing** runs from the cargo's **upper part** down to the floor and/or walls to stop movement.

Cargo tie-down p. 3-3

- Flatbed and no-side cargo must be tied down; tie-downs can matter in **closed vans** too, since shifting cargo changes handling.
- Aggregate working load limit must be **at least ½ the cargo weight** (handbook answer, not 1½ times); e.g., 10,000 lb needs 5,000 lb.
- Use at least **1 tie-down per 10 feet** of cargo and **never fewer than 2**, so a 20-ft load needs 2.
- Special commodities like logs, metal coils, and concrete pipe have securing rules in **49 CFR §393**.

Header boards p. 3-3

- A front-end **header board** ("headache rack") protects **you** from your cargo in a crash or emergency stop.
- Keep it in good condition so it **blocks cargo from moving forward** toward the cab.

Covering cargo p. 3-3

- There are **2 basic reasons** to cover cargo → protect people from spilled cargo, and protect the cargo from weather.
- Check covers **in your mirrors** while driving; a **flapping** cover can rip loose and block your or others' view.

Sealed and containerized loads p. 3-3

- **Containerized loads** usually travel part way by rail or ship; some lock to a special frame, others are secured like any cargo.
- You cannot inspect a **sealed load**, but you must check you are within **gross weight and axle weight** limits.

CA Spilling loads and damage to the highway p. 3-4

- **CA** It is illegal to drive a vehicle whose load spills, drops, leaks, blows, or sifts, except **clear water** and **feathers from live birds**.
- **CA** Garbage, trash, rubbish, and ashes must be kept **covered** so nothing spills onto the road.
- **CA** Aggregate material must stay **6 inches below the upper edge**, with no holes, plus fenders, seals, and **splash flaps** behind each tire.
- **CA** Anyone who damages a road on purpose or through carelessness **pays for repairs**, possibly including debris cleanup.

Dry bulk p. 3-4

- **Dry bulk tanks** have a **high CG**, and the load can shift.
- Be extremely careful and go slowly around **curves** and in **sharp turns**.

Hanging meat p. 3-4

- **Hanging meat** is a **very unstable** load with a high CG, since it can swing toward the outside of a curve.
- Take special care and go slowly on sharp curves such as **on-ramps and off-ramps**.

Livestock p. 3-4

- Live animals move around in a trailer, which makes handling unsafe.
- With a partial load, use **false bulkheads** to keep the animals **bunched together**.
- Even bunched animals can **lean in a curve**, shifting the CG and raising the chance of **rollover**.

CA Oversized loads p. 3-4

- Over-length, over-width, or overweight loads need **special transit permits**; **CA** Caltrans issues them.
- Driving is usually allowed only at **certain times**, and may need wide load signs, flashing lights, flags, a **police escort**, or pilot vehicles.

CA Special markings for wide loads p. 3-5

- **CA** Vehicle and load **over 80 inches wide** → amber combination lamp at the front of the projection and red at the rear.
- **CA** Projection **no more than 3 feet** long → at least one amber lamp if near the front, or one red lamp if near the rear.
- Memory hook → **amber faces forward, red faces back**.

CA Projecting loads p. 3-5

- **CA** Load **4 feet (48 in) or more** past the rear → **solid red or fluorescent orange** flag, at least 12 inches square, at the end.
- **CA** In darkness that rear load also needs **2 lit red lights** visible from **500 feet** to the side and rear.
- **CA** Load **1 foot or more** to the left in darkness → amber light at the far left edge, visible **300 feet** ahead and behind.
- **CA** Load extending to the left **more than 120 inches** → amber lamp front and red lamp rear, each visible 300 feet.
- **CA** Vehicle **wider than 102 inches** in daylight → red or fluorescent flag, at least **12 inches square**, left front and left rear.

CA Piggyback trailers p. 3-5

- **CA** A **piggyback** trailer carried on top of another vehicle must be **tied down securely**.
- It must not be able to **shift, tip over, or become unstable** on the highway.

Numbers and terms to memorize

Item	Value	Page
Driver's cargo duties	4: inspect; spot overloads/poor balance; know it is secured and not blocking view; know it does not block emergency equipment	3-1
HazMat needing placards	requires "H" endorsement	3-1
First cargo re-check	within first 50 miles	3-1
Later re-checks	after 3 hours or 150 miles, and after every break	3-1
GVW / GCW	actual weight of single vehicle / combination, with load	3-1
GVWR / GCWR	manufacturer's loaded-weight value for single vehicle / combination	3-1
GCWR with no maker value	power unit GVWR + towed unit weight + its load	3-1
Axle weight	weight one axle or axle set puts on the ground	3-2
Tire load	max safe weight at a stated pressure; printed on tire side	3-2
Bridge formula	less weight allowed on axles closer together	3-2
Legal max may be unsafe	bad weather, mountains	3-2
Load placement	as low as possible; heaviest under lightest	3-2
Too much on steering axle	hard steering; axle and tire damage	3-2
Too little on front axle	too light to steer safely	3-2
Too little on drive axles	poor traction; wheels spin	3-2
Blocking	front/back/sides of cargo, fastened to deck; stops sliding	3-2
Bracing	from upper part of cargo to floor and/or walls	3-2
Aggregate working load limit	at least ½ the cargo weight	3-3
Tie-down count	at least 1 per 10 ft of cargo; never fewer than 2	3-3

Item	Value	Page
Header board ("headache rack")	protects driver from cargo moving forward	3-3
Reasons to cover cargo	protect people from spills; protect cargo from weather	3-3
Sealed load check	gross weight and axle weight limits	3-3
Securement rules for special commodities	49 CFR §393	3-3
CA Allowed to escape from vehicle	clear water; feathers from live birds	3-4
CA Aggregate material level	6 inches below upper edge	3-4
CA Aggregate vehicle equipment	seals on unloading openings, splash flaps behind every tire/set, fenders	3-4
CA Oversized load permits	Caltrans	3-4
CA Wide-load lamps	over 80 in wide: amber front, red rear	3-5
CA Short projection option	projection no more than 3 ft long	3-5
CA Rear projection flag	4 ft (48 in) or more; solid red or fluorescent orange; at least 12 in square	3-5
CA Rear projection lights in darkness	2 red lights, visible 500 ft side and rear	3-5
CA Left projection	1 ft or more: amber light, visible 300 ft front and rear	3-5
CA Load extends more than 120 in	amber lamp front, red lamp rear, 300 ft	3-5
CA Vehicle wider than 102 in	red or fluorescent flag, at least 12 in square, left front and left rear (daylight)	3-5

Exam traps

Tempting wrong idea	Handbook answer
The shipper loaded and sealed it, so the cargo is not your responsibility	you are responsible whether or not you loaded it. It tempts because someone else did the work.
First re-check after 100 miles (or 25 miles)	within the first 50 miles .
Re-check every 2 hours or 100 miles	every 3 hours or 150 miles , and after every break .
GVWR is what the loaded truck weighs today	that is GVW . GVWR is the manufacturer's value. The letters look almost the same.
A bridge formula allows more weight on axles that are close together	less weight. Close axles concentrate the weight on one spot.
If the weight is legal, it is safe	in bad weather or mountains, legal maximum weight may not be safe.
Put heavy items on top to hold lighter items down	heaviest under the lightest, as low as possible.

Tempting wrong idea	Handbook answer
Too little weight on the front axle causes hard steering	too much front weight = hard steering; too little = too light to steer safely. The two are easy to mix up.
Too little weight on the drive axles causes hard steering	it causes poor traction (wheels spin).
Bracing is fastened snug against the cargo at deck level	that is blocking . Bracing reaches down from high on the cargo to the floor and/or walls.
A 20-foot load needs 3 tie-downs (one each end plus middle)	2 (1 per 10 ft, minimum 2).
A very small load needs just 1 tie-down	at least 2 , no matter how small.
Aggregate WLL must be 1½ times the cargo weight	at least ½ the weight (handbook).
Header boards protect the cargo	they protect you from the cargo in a crash or emergency stop.
You cannot check anything on a sealed load	check that you are within gross weight and axle weight limits.
Hanging meat is stable because it is fastened to rails	it is very unstable with a high CG .
With a partial livestock load, spread the animals out	use false bulkheads to keep them bunched together .
Rear flag needed at 3 feet of overhang, or flag must be 18 inches	4 feet or more; flag at least 12 inches square.
Night lights on a rear overhang must be visible 300 feet	500 feet for the 2 red rear lights; 300 feet is the left-side amber light.

GK-02 · Disqualifications, violation points, hands-free rules

General Knowledge · test weight **High** · handbook Section 1.3.1–1.3.9

You will be able to answer

- The illegal blood alcohol level for CMV drivers, and the penalty for a small amount of alcohol below it.
- How long you lose your CDL for major offenses, serious traffic violations, out-of-service order violations, railroad crossing violations and hands-free/texting violations (1st, 2nd, 3rd time).
- The six railroad crossing offenses.
- How your personal-car record affects your CDL, and who cannot hold a HazMat endorsement.
- **CA** How points are counted, how long convictions stay on your record, and when you become a "negligent operator."

Key points

Start here

- **Disqualified** ("lose your CDL") means you may not drive a CMV for a set time, or for life.
- A **conviction** means a court finds you guilty; **BAC** is blood alcohol concentration, the percent of alcohol in your blood.
- Most disqualification periods in this lesson are **minimums** ("at least").

General p. 1-14

- Disqualified for **any reason** → you may not drive a CMV.
- Most disqualification periods are **minimums** ("at least"), and some last for life.

Alcohol, leaving the scene of an accident, and felonies p. 1-14

- Driving a CMV with a BAC of **.04 percent or more** is illegal, because even a little alcohol slows your reactions.
- Any measurable alcohol below .04 → **out of service for 24 hours**.
- Major offenses (alcohol, drugs, test refusal, leaving the scene, CMV felony, driving suspended, fatal negligence) → CDL lost **at least 1 year** first time.
- First major offense in a CMV **placarded for HazMat** → at least **3 years**; a 2nd major offense → life.
- Using a CMV for a felony involving **controlled substances** → **life**, even the first time; refusing a test counts like driving drunk (implied consent).

Serious traffic violations p. 1-14

- Serious violations include speeding **15 mph or more** over the limit, reckless driving, erratic lane changes, following too closely, and driving without the correct CDL.
- 2 serious violations in a CMV within **3 years** → at least **60 days**.
- 3 or more within 3 years → at least **120 days**.
- One serious violation alone brings **no disqualification**, because it takes repeats to show a risky pattern.

Violations of out-of-service orders p. 1-14

- An **out-of-service order** means you or your vehicle may not drive for a set time or until a problem is fixed.
- Driving anyway → 1st at least **90 days**, 2nd within 10 years at least **1 year**, 3rd or more within 10 years at least 3 years.
- Repeats are counted over a **10-year** window, not the usual 3 years.
- Federal rules say 180 days / 2 years / 3 years, but answer the handbook's **90 days / 1 year / 3 years** on the test.

Railroad-highway grade crossing violations pp. 1-14, 1-15

- Railroad crossing violations → 1st at least **60 days**, 2nd within 3 years at least **120 days**, 3rd within 3 years at least 1 year.
- Unlike serious violations, the **first** railroad violation already costs 60 days, because a train cannot stop quickly.
- There are **6 offenses**; drivers not required to always stop must stop when tracks are not clear and must slow to check for trains.
- Drivers who must always stop (for example all passenger and HazMat vehicles) break the rule by **not stopping** before going onto the crossing.
- All drivers → need **room to get all the way across** without stopping, obey signals or officers, and avoid getting stuck from low clearance.

HazMat endorsement background check and disqualifications p. 1-15

- An **H** endorsement requires **fingerprints** and a background check.
- You are denied or lose it if you lack lawful permanent resident status, renounce U.S. citizenship, or are wanted, indicted or convicted for certain felonies.
- Being judged to lack mental capacity or being involuntarily committed also disqualifies you, as does the **TSA** judging you a security threat.
- The check exists because hazardous materials could be used to harm many people.

Traffic violations in your personal vehicle p. 1-15

- Under the **MCSIA of 1999**, certain moving violations in your own car can cost your CMV driving privilege.
- Car privilege suspended, revoked or cancelled for **traffic-law** violations (not parking) → your **CDL privilege is lost too**.
- Car privilege lost for **alcohol, controlled substances or a felony** → CDL lost **1 year**; a second such violation, car or CMV → life.
- You **cannot get a hardship license** to drive a CMV, because habits in a car come with you into a truck.

CA Violation point counts p. 1-15

- **CA** Record retention → major **55 years**, out-of-service **15**, accidents and serious violations **10**, railroad **4**, minor **3**.
- **CA** An accident you caused = **1 point**; reckless driving, hit-and-run or DUI = **2 points**.
- **CA** A violation in a CMV counts **1½ times** its points, because the same mistake is more dangerous in a heavy vehicle.
- **CA** **Negligent operator** → **4 points in 12 months**, 6 in 24 months, or 8 in 36 months.
- **CA** After a hearing: **6, 8 or 10 points** (not if earned in a Class C car); Class A/B without a special certificate or endorsement: **2 extra points**.

CA Violations of the hands-free or texting law p. 1-16

- CA The hands-free and texting penalties apply to your CDL **in any vehicle**, your own car included.
- CA 2nd violation within 3 years → at least **60 days + 1 point**.
- CA 3rd or later within 3 years → at least **120 days + 1 point**, because a phone takes your eyes, hands and attention off the road.

Summary: every disqualification period pp. 1-14, 1-15, 1-16

- Major offense → **1 year** (3 years if HazMat placarded), 2nd → **life**; a controlled-substance felony using a CMV → life at once.
- Serious violations within 3 years → **60 days** for 2, **120 days** for 3 or more.
- Out-of-service violations over **10 years** → **90 days / 1 year / 3 years**; railroad over 3 years → 60 days / 120 days / 1 year.
- CA Hands-free/texting in any vehicle → 2nd **60 days + 1 point**, 3rd **120 days + 1 point**.
- Car suspension for alcohol, drugs or felony → **1 year**, second → **life**; measurable alcohol under .04 → 24 hours out of service.

Numbers and terms to memorize

Item	Value	Page
Illegal BAC in a CMV	.04 percent or more	1-14
Measurable alcohol under .04	out of service 24 hours	1-14
Implied consent	driving a CMV = agreeing to alcohol testing	1-14
Major offense: 1st / placarded HazMat / 2nd	1 year / 3 years / life	1-14
CMV used in a controlled-substance felony	life	1-14
Excessive speeding	15 mph or more over the posted limit	1-14
Serious violations: 2 / 3+ within 3 years	60 days / 120 days	1-14
Out-of-service violations: 1st / 2nd / 3rd+ (10 years)	90 days / 1 year / 3 years	1-14
Railroad crossing: 1st / 2nd / 3rd (3 years)	60 days / 120 days / 1 year	1-14
Railroad crossing offenses	6	1-14
HazMat endorsement check	fingerprints + background check; TSA decides security threat	1-15
MCSIA	Motor Carrier Safety Improvement Act of 1999	1-15
Car suspension for alcohol/drugs/felony	CDL lost 1 year; second = life	1-15
Hardship license for a CMV	not allowed	1-15
CA Record retention	major 55 yrs · out-of-service 15 · accidents & serious 10 · railroad 4 · minor 3	1-15
CA 2-point convictions	reckless driving, DUI (alcohol/drugs), hit-and-run	1-15
CA Violation in a CMV	1½ times the points	1-15
CA Negligent operator	4 points/12 months · 6/24 · 8/36	1-15

Item	Value	Page
CA After a hearing	6, 8 or 10 points (if not earned in a Class C car)	1-15
CA Class A/B without special certificate or endorsement	2 extra points	1-15
CA Hands-free/texting: 2nd / 3rd+ within 3 years	60 days + 1 point / 120 days + 1 point	1-16

Exam traps

Tempting wrong idea	Handbook answer
.08 is the CMV limit	.04 or more. .08 tempts because many people know it from car driving.
Under .04 you may keep driving	any measurable alcohol = 24 hours out of service.
Refusing the test avoids a penalty	refusal = at least 1 year , same as a DUI (implied consent).
A second major offense = 3 years	life. "3 years" feels familiar because it is the HazMat number.
One serious violation = 60 days	it takes 2 in 3 years. The <i>railroad</i> rule is the one that gives 60 days for a first violation.
Excessive speeding starts at 10 mph over	15 mph or more.
First out-of-service violation = 180 days	90 days (handbook). 180 days is the federal number found in some practice banks.
Out-of-service repeats are counted over 3 years	over 10 years.
Tickets in your own car can't touch your CDL	car suspensions can cost your CDL, and hands-free/texting violations count in any vehicle.
You can get a hardship license to keep driving a truck	not for a CMV.
A CMV violation counts double	1½ times.
Accidents stay on the record 3 years	10 years; 3 years is for minor convictions.

GK-06 · Seeing, mirrors, signals, warning devices

General Knowledge · test weight **High** · handbook Section 2.4–2.5

You will be able to answer

- How far ahead a good driver looks (in seconds, and in city and highway distance), and the two main things to look for.
- The California mirror law, why convex mirrors fool your eyes, how to adjust mirrors, and the four mirror checks in a lane change.
- The three rules for turn signals, and how to warn drivers behind you when you slow down or stop.
- How to make other road users notice you: when passing, at dawn/dusk/rain/snow, and when parked at the roadside.
- Exactly where to place warning triangles — on a divided or one-way road, a two-lane road, and behind a hill or curve — and how fast you must put them out.

Key points

Seeing p. 2-12

- A safe driver knows what is happening **all around** the vehicle, ahead, beside and behind.
- Poor looking habits are a **major cause of crashes**.

Seeing ahead pp. 2-12, 2-13

- Look **12 to 15 seconds or more** ahead, because a big vehicle needs a lot of distance to stop or change lanes.
- 12–15 seconds ahead is about **one block** at lower (city) speeds and about **a quarter of a mile** at highway speeds.
- Look for the 2 main things ahead, **traffic** and **road conditions**, and move your eyes back and forth, near and far.
- A **"stale" green light** (green for a long while) will likely change before you get there → begin to slow and prepare to stop.

Seeing to the sides and rear p. 2-13

- **CA** California-registered vehicles need **2 or more mirrors**, one on the left, showing the road behind for **200 feet or more** (CVC §26709).
- Check mirrors with quick glances; don't stare. In a **convex** mirror, things look **smaller and farther away** than they really are.
- Check mirror adjustment before every trip, **only with the trailer(s) straight**, so each mirror shows part of your own vehicle as a reference point.
- Check mirrors more in 4 situations: turns, merges, tight maneuvers, and lane changes, which need **4 checks** (before, after signaling, right after starting, after finishing).

Communicating p. 2-14

- Communicating means **signaling your intentions**, telling others what you are going to do.
- It also means **communicating your presence**, making sure others know you are there.

Signal your intentions p. 2-14

- Turn signals follow **3 rules**: signal early, signal continuously until the turn is finished, and cancel the signal afterward.
- Signaling early is the **most effective way** to stop others from trying to pass you.
- To warn of slowing, **tap the brake pedal lightly** so the brake lights blink; save the **4-way flashers** for very slow driving or when stopped.
- **Do not direct traffic** by waving others past, because it could cause a crash and the blame could fall on you.

Communicating your presence pp. 2-14, 2-15

- When passing, **assume others have not seen you**; tap the horn lightly, or after dark flash headlights low → high → low.
- At dawn, dusk, rain or snow, turn on your **headlights on low beam**; clearance lights alone are not enough.
- Stopped at the roadside, use **4-way flashers** and don't rely on taillights; set out warning devices **within 10 minutes**.
- Divided or one-way road → triangles at **10, 100 and 200 feet** behind; two-lane road → within **10 feet** of a corner, 100 feet behind, 100 feet ahead.
- If a hill or curve hides you within **500 feet**, move the rear triangle back beyond it; carry triangles between yourself and oncoming traffic.

Numbers and terms to memorize

Item	Value	Page
How far ahead to look	At least 12-15 seconds	2-12
12-15 seconds at lower (city) speeds	About 1 block	2-12
12-15 seconds at highway speeds	About ¼ mile	2-12
2 main things to look for ahead	Traffic and road conditions	2-12, 2-13
Light green for a long time ("stale" green)	Begin slowing; prepare to stop	2-12
CA Minimum mirrors (CVC §26709)	2, including 1 on the left side	2-13
CA Mirror rear view distance	At least 200 feet	2-13
CA When both left and right mirrors are required	Vehicle built/loaded so it blocks the rear view, or towing something that blocks it	2-13
Convex ("fisheye," "spot," "bugeye") mirror	Wider view; things look smaller and farther away	2-13
When to check mirror adjustment	Before any trip, with trailer(s) straight	2-13
Mirror reference point	Each mirror shows part of your own vehicle	2-13
Mirror checks in a lane change	4: before, after signaling, right after starting, after finishing	2-13
Special mirror situations	4: lane changes, turns, merges, tight maneuvers	2-13
Turn-signal rules	3: signal early, signal continuously, cancel	2-14
Best way to keep others from passing you in a turn	Signal early	2-14

Item	Value	Page
CA Turn signals and 2 stop lamps	CVC §§24951 and 24600	2-14
Warning of slowing	Tap the brake pedal lightly so the brake lights blink	2-14
Driving very slowly or stopped	4-way emergency flashers (if legal)	2-14
Directing other traffic	Don't — you can be blamed	2-14
Passing at night	Flash lights low → high → low	2-14
Dawn, dusk, rain, snow	Headlights on, low beams	2-14
Parked at the roadside	4-way flashers; don't rely on taillights	2-14
Time to place warning devices	Within 10 minutes	2-14
Divided / one-way highway	10 ft, 100 ft, 200 ft toward approaching traffic	2-14
Two-lane / undivided road	Within 10 ft of front or rear corner; 100 ft behind; 100 ft ahead	2-15
Hill or curve blocks view within	500 feet → move rear triangle back (100–500 ft)	2-15
Carrying triangles	Hold them between you and oncoming traffic	2-15

Exam traps

Tempting wrong idea	Handbook answer
Objects in a convex mirror look bigger and closer	They look smaller and farther away than they really are. A wide-angle view feels like a zoom, so people guess the wrong way.
On a divided highway, put one triangle in front of the truck	All three go behind you, toward approaching traffic, at 10, 100 and 200 feet . The "one ahead" pattern belongs to two-lane roads.
On a two-lane road, put all three triangles behind the truck	One within 10 feet of a corner, one 100 feet behind , one 100 feet ahead — traffic comes from both ways.
You have 15 or 30 minutes to put out triangles	Within 10 minutes.
Near a hill or curve, keep the rear triangle in its usual spot	Move it back beyond the hill or curve so drivers get warning — Figure 2.10 shows 100 to 500 feet back.
12–15 seconds ahead at highway speed is about one block (or half a mile)	About one block is for lower speeds ; at highway speed it is about a quarter mile .
Wave a car around you when the road ahead is clear — it's courteous	Do not direct traffic. It could cause a crash, and you could be blamed.
Taillights are enough when you're parked on the shoulder	Turn on your 4-way flashers . Drivers have hit parked trucks thinking they were moving.
At dusk or in rain, clearance lights (or high beams) are enough	Turn on your headlights and use low beams .
Cancel the signal as you start turning so both hands are free	Keep it on until the turn is complete , then cancel.
A green light that has been green a long time — keep going	It will probably change before you arrive; begin slowing and prepare to stop .

Tempting wrong idea	Handbook answer
Adjust mirrors while the trailer is at an angle	Mirror adjustment can be checked accurately only when the trailer(s) are straight.
In a lane change, one mirror check before you move is enough	Check 4 times — before, after signaling, right after starting, and after finishing.

GK-10 · Night, fog, winter and hot-weather driving

General Knowledge · test weight **High** · handbook Section 2.11–2.14

You will be able to answer

- When to use high beams, when to switch to low beams, and the **500-foot** dimming rule.
- How far you can see with low and high beams, and why your speed must let you stop inside that distance.
- The warning signs of fatigue (tiredness), who is most at risk, and the only real cure.
- What to do if you must drive in fog.
- What to check before winter driving, how to spot ice, why to turn off the retarder, and how to dry wet brakes.
- How to handle hot tires, a hot radiator cap, loose belts and "bleeding tar" in very hot weather.

Key points

Why night driving is more dangerous p. 2-28

- At night you see hazards later, so you have **less time to react**, and a surprised driver is less able to avoid a crash.
- Night driving problems come from three places: **the driver, the roadway, and the vehicle**.

Driver factors pp. 2-28, 2-29, 2-30

- Wear corrective lenses **every time** you drive, keep a spare pair, and **never wear dark or tinted lenses at night** because they block light you need.
- Glare can blind you briefly, and recovery can take **several seconds**; even **2 seconds** at 55 mph covers more than half a football field.
- Fatigue risk is highest **at night**, especially after midnight, and mid-afternoon; **6 hours of sleep or less** triples your risk.
- Adults need **8–9 hours** of sleep; on long trips take a break about every **100 miles or 2 hours**.
- Caffeine and drugs don't fix drowsiness; **nothing is safe except leaving the road and sleeping** (or a 15–20 minute nap).

Roadway factors p. 2-30

- Many night crashes involve **pedestrians, bicyclists and animals**, who are hard to see on poorly lit roads.
- When lighting is poor or confusing, slow down so you can **stop within the distance you can see ahead**.
- Be extra alert for drunk drivers around **bar and tavern closing times**.

Vehicle factors pp. 2-30, 2-31

- Low beams let you see about **250 ft**; high beams about **350–500 ft**. Keep your stopping distance within your sight distance.
- At 55 mph a truck needs at least **419 ft** to stop, far beyond low-beam range; at 35 mph it needs about 222 ft.
- Dirty headlights may give only **half** their light; keep all lights, reflectors, signals and the windshield (inside and outside) clean.
- **CA** Headlights on from **½ hour after sunset to ½ hour before sunrise**, when using wipers, or when you can't clearly see a person or vehicle 1,000 ft away.
- **CA** **Never drive with only parking lights on.**

Night driving procedures p. 2-31

- Dim high beams when an oncoming vehicle **or** a vehicle you are following is within **500 ft** (answer 500 ft on the test).
- Use high beams whenever it is **safe and legal**; driving on low beams all the time badly shortens your view.
- For oncoming glare, look **slightly to the right** at the lane or edge line, and never retaliate with your own high beams.
- Keep the cab dark (dome light off), wear **no sunglasses at night**, and stop at the nearest safe place if you get sleepy.

Driving in fog p. 2-32

- Best advice is **don't drive in fog**; pull into a rest area or truck stop and wait for better visibility.
- If you must drive, **slow down before** entering fog and use **low beams and fog lights**, even in daytime.
- Turn on your **4-way flashers**, use roadside reflectors to see curves, listen for traffic, and avoid passing.
- Don't stop on the roadside unless you must, and don't assume lights ahead show where the road is.

Winter: vehicle checks pp. 2-32, 2-33

- Winter tread depth: **front tires 4/32 inch** or more in each major groove; **all other tires 2/32 inch** or more.
- Carry the right number of **tire chains plus extra cross-links**, check they fit the drive tires, and practice putting them on.
- Check coolant/antifreeze, defrosters, wipers with **washer antifreeze**, and clear ice from windows, mirrors, steps and radiator shutters.
- Don't set the winterfront too tight, or the engine can **overheat and quit**.
- Check the exhaust for leaks, because **carbon monoxide** in the cab causes sleepiness and can kill.

Driving on slippery surfaces p. 2-33

- On slippery roads be **slow and smooth**; if it is extremely slippery, pull off at the first safe spot.
- Check for ice on **bridges and overpasses**; **no spray** from other tires means ice has formed, and ice on mirrors or wipers means the road is probably icy.
- **Don't use the engine brake or speed retarder** on slippery roads, because they can make the drive wheels skid.
- Melting ice makes the road **even more slippery**; don't brake in curves and keep a longer following distance.
- Wet brakes: shift to a low gear, brake **gently**, drive through with light pressure, then dry them and make a **test stop**.

Very hot weather: vehicle checks p. 2-34

- In very hot weather inspect tires **every 2 hours or every 100 miles; never let air out** of hot tires.
- If a tire is **too hot to touch**, stay stopped until it cools, or it may blow out or catch fire.
- Keep the **radiator cap closed until the system cools**; a cap cool enough to touch bare-handed is probably safe to open.
- Without a recovery tank, turn the cap **slowly to the first stop**, step back, and remove it only after the pressure is gone.
- A **slack belt** can't drive the water pump or fan, so the engine overheats; burst hoses can wreck the engine or start a fire.

Driving in the heat p. 2-34

- Watch for **bleeding tar**, which rises to the road surface in very hot weather and is **very slippery**.
- Drive slowly enough to prevent overheating, because speed raises the risk of **tire failure, fire, and engine failure**.

Numbers and terms to memorize

Item	Value	Page
Glare blindness	even 2 seconds is dangerous; at 55 mph you cover more than half a football field	2-28
Crash-heavy hours	midnight to 6 a.m.	2-29
Sleep of 6 hours or less	triples crash risk	2-29
Working more than 60 hours/week	raises risk 40%	2-29
Heavy truck crashes involving fatigue	at least 15%	2-29
Nap when you see fatigue signs	15–20 minutes (or sleep for the night)	2-29
Sleep adults need	8–9 hours	2-30
Rest breaks on long trips	about every 100 miles or 2 hours	2-30
Caffeine	works only a few hours ; don't rely on it	2-30
Only safe cure for drowsiness	get off the road and sleep	2-30
Low-beam sight distance	about 250 ft	2-30
High-beam sight distance	about 350–500 ft	2-30
Dirty headlights	may give only half the light	2-31
CA Headlights required	½ hr after sunset to ½ hr before sunrise ; when wipers are needed; when you can't see a person/vehicle at 1,000 ft	2-31
CA Parking lights	never drive with only parking lights on	2-31
Dim for oncoming vehicle	within 500 ft	2-31
Dim when following	within 500 ft	2-31

Item	Value	Page
High beams	use whenever safe and legal	2-31
Sunglasses / tinted lenses	never at night	2-28, 2-31
Lights to use in fog	low beams + fog lights + 4-way flashers (even in daytime)	2-32
Front-tire tread	at least 4/32 inch in every major groove	2-32
Other tires tread	at least 2/32 inch	2-32
Ice clue	no spray from other vehicles; ice on mirrors/wipers	2-33
Melting ice	road gets more slippery	2-33
Engine brake / retarder on slippery roads	don't use — drive wheels can skid	2-33
Carbon monoxide (from exhaust leaks)	makes you sleepy; can kill	2-33
Winterfront too tight / shutters iced	engine may overheat and stop	2-33
Hot-weather tire checks	every 2 hours or 100 miles	2-34
Hot tires	never let air out ; if too hot to touch, stay stopped	2-34
Radiator cap	open only when cool enough to touch with bare hand ; turn slowly to first stop	2-34
Loose belts	water pump/fan won't turn right → overheating	2-34
Bleeding tar	tar on the surface; very slippery	2-34

Exam traps

Tempting wrong idea	Handbook answer
Use low beams whenever you can.	use high beams whenever it is safe and legal . Low beams feel polite, but they cut your view to about 250 ft.
Dim when following at 300 ft	500 ft for following and for oncoming. The 300-ft figure comes from the car handbook.
Flash your high beams at a driver who won't dim	never retaliate; look slightly right at the lane or edge line. Payback feels fair, but it adds glare for the oncoming driver and raises the chance of a crash.
Tinted glasses or sunglasses help with night glare	never wear them at night — they cut the light you need.
Coffee, cold air or the radio will fix drowsiness	the only safe cure is sleep (stop for the night or nap 15–20 minutes). The other tricks help a little, so they seem like answers.
In fog, high beams help you see farther	low beams and fog lights , plus 4-way flashers.
In fog, follow the lights ahead of you	that vehicle may be off the road; use roadside reflectors to find the curve.

Tempting wrong idea	Handbook answer
In fog, stop on the shoulder and wait	pull into a rest area or truck stop ; don't stop at the roadside unless absolutely necessary.
Front tires need 2/32 inch	front 4/32 , other tires 2/32 . The numbers are easy to swap.
A retarder or engine brake gives extra control on ice	turn it off on slippery roads — it can skid the drive wheels. The name sounds like a safety device.
Lots of road spray means ice	no spray means ice has formed.
Melting ice is safer	when ice starts to melt the road is even more slippery .
Dry wet brakes by pressing hard on brake and gas together	use light brake pressure; hard brake plus gas overheats drums and linings.
Let air out of hot tires to bring pressure back to normal	never — they'll be underinflated once cool. It sounds like "relieving" pressure.
The radiator cap is safe to remove if the engine isn't overheated	only when the system has cooled (cap cool enough to touch bare-handed), then turn slowly to the first stop . A "normal" hot engine is still under pressure.
You can never open any coolant cap on a hot engine	a see-through overflow/recovery container that is not part of the pressurized system can be opened hot.

GK-11 • Railroad crossings and mountain driving

General Knowledge • test weight **High** • handbook Section 2.15–2.16

You will be able to answer

- The difference between passive and active crossings, and what each railroad sign, marking and signal tells you.
- Who must stop at a crossing, where to stop (**15–50 ft**), and what to do while stopping.
- How long a tractor-trailer needs to clear one or two tracks, which trailers get stuck, and what to do if you are stuck.
- How to choose a safe downhill speed and the right gear — and why you must pick the gear before the hill.
- The proper downhill braking method (often called "snub braking"), what brake fade is, and what escape ramps do.

Key points

Railroad-highway crossings p. 2-35

- A **railroad-highway grade crossing** is where a road crosses tracks at the same level, and it is **always dangerous**.
- Approach **every** crossing as if a train is coming, because a train's distance and speed are very hard to judge.

Types of crossings p. 2-35

- A **passive** crossing has **no traffic control device**, so the decision to cross is all yours.
- An **active** crossing has flashing red lights (with or without bells), or lights with bells and **gates**, but you still must look.

Warning signs and devices pp. 2-35, 2-36

- The advance warning sign is **round, black on yellow**: slow down, look, listen, and be ready to stop.
- All vehicles carrying **passengers or HazMat** must stop at railroad crossings.
- The **crossbuck** means yield to the train; a sign under it shows the number of tracks.
- With no white stop line, stop **no closer than 15 ft** and **no farther than 50 ft** from the nearest rail.
- At gates, stop when the lights **start** flashing and stay stopped until the gates are **all the way up** and lights stop flashing.

Driving procedures p. 2-36

- **Never race a train**, and keep your speed low enough to **stop before the tracks** if needed.
- Don't expect to hear a train or rely only on signals, gates or flaggers; always look yourself.
- On **double tracks**, a train on one track can hide another, so check both ways after a train clears.
- Yard and town or city crossings are **just as dangerous** as country crossings.

Stopping safely at crossings p. 2-36

- A **full stop** is required when your cargo makes it mandatory under state or federal rules, or when the law requires it.
- When stopping, check traffic behind, stop **gradually** (use a pullout lane), and turn on your **4-way flashers**, since drivers behind may not expect it.

Crossing the tracks p. 2-36

- Never let traffic trap you on the tracks; be sure you can get **all the way** across before you start.
- A typical tractor-trailer needs **at least 14 seconds** to clear a single track and **more than 15 seconds** for a double track.
- **Don't change gears** while crossing the tracks, because a missed shift could stall you on them.

Special situations pp. 2-36, 2-37

- **Low-slung units** (lowboys, car carriers, moving vans, possum-belly livestock trailers) can hang up on raised crossings.
- A **single-axle tractor** pulling a long trailer with landing gear set for a tandem-axle tractor can also hang up.
- If stuck, **leave the vehicle and get well away** from the tracks, then call 9-1-1 and give the crossing's **DOT number** if posted.

Mountain driving p. 2-37

- In the mountains **gravity** is the big factor: it slows you going up and speeds you up going down.
- Going down, choose a **safe speed**, use a **low gear**, and use proper braking so brakes hold you without overheating.
- **Brake fade** means hot brakes need more and more pedal pressure for the same stopping power, until they may not stop you at all.

Select a "safe" speed p. 2-37

- Your safe downhill speed depends on **five factors**: total weight, grade length, grade steepness, road conditions, and weather.
- Never exceed a posted speed limit or "**Maximum Safe Speed**" sign.
- **Engine braking** is your main speed control, strongest near **governed rpm** in a lower gear; save the service brakes.

Select the right gear BEFORE starting down p. 2-37

- Shift into a low gear **before** you start down the grade.
- Don't downshift after speed builds: you may not get into **any** gear and **lose all engine braking**.
- Older trucks use the **same gear** down as up; modern trucks **may need a lower gear** going down than going up.

Brake fading or failure p. 2-37

- Too much heat from overusing the brakes instead of engine braking makes them **fade or fail**.
- Brakes **out of adjustment** stop doing their share first, so the other brakes **overheat and fade**; check adjustment often.

Proper braking technique p. 2-38

- Brakes are only a **supplement** to engine braking; get into the proper low gear first.
- Brake firmly enough for a definite slowdown to about **5 mph below** your safe speed, then release; each application takes about **3 seconds**.
- Example: safe speed **40 mph** → brake to 35 mph, release, let it return to 40, and repeat ("snub braking").
- **Escape ramps** stop runaway vehicles with a long bed of **loose, soft material**, sometimes with an upgrade; know where they are.

Numbers and terms to memorize

Item	Value	Page
Passive crossing	no traffic control device — you decide	2-35
Active crossing	flashing red lights (± bells), or lights + bells + gates	2-35
Advance warning sign	round, black on yellow ; slow down, look, listen, be ready to stop	2-35
Vehicles that must stop at crossings	all carrying passengers or HazMat ; also when cargo rules or the law require it	2-35, 2-36
Pavement marking	"X" with "RR" + no-passing marking on 2-lane roads; same meaning as advance sign	2-35
Crossbuck	marks the crossing; yield to the train ; sign below = number of tracks	2-35
Stop distance with no stop line	no closer than 15 ft , no farther than 50 ft from the nearest rail	2-35
Gates	stay stopped until gates are up AND lights stop flashing	2-36
When stopping at tracks	check behind, stop gradually, pullout lane, 4-way flashers	2-36
Clear a single track	at least 14 seconds	2-36
Clear a double track	more than 15 seconds	2-36
Shifting on the tracks	don't	2-36
Units that hang up	lowboy, car carrier, moving van, possum-belly livestock trailer ; single-axle tractor + long trailer with landing gear set for tandem tractor	2-36
Stuck on tracks	get out and away , find emergency info, call 9-1-1 , give DOT number	2-36, 2-37
Brake fade	must press harder and harder for the same stopping power	2-37
Safe-speed factors	weight, length, steepness, road conditions, weather	2-37
Main speed control downhill	engine braking (brakes are only a supplement)	2-37, 2-38
Engine braking strongest	near governed rpm , in lower gears	2-37
When to pick your gear	before starting down	2-37
Modern trucks	may need a lower gear down than up	2-37
Out-of-adjustment brakes	stop working first → other brakes overheat and fade	2-37

Item	Value	Page
Braking technique ("snub braking")	brake to about 5 mph below safe speed, release, repeat	2-38
Length of each application	about 3 seconds	2-38
Example	safe speed 40 → brake to 35 → release	2-38
Escape ramp	long bed of loose, soft material , sometimes with an upgrade	2-38

Exam traps

Tempting wrong idea	Handbook answer
It takes about 14 seconds to clear a double track	at least 14 s for a single track, more than 15 s for a double . The two numbers are close and easy to swap.
Stop as close to the tracks as possible to see better	with no stop line, stop 15–50 ft from the nearest rail.
Go when the train has passed, even if lights are still flashing	wait until the gates are up and the lights stop flashing — and check for a second train.
Signals and gates will always warn you	don't rely on signals ; always look and listen yourself.
You'll hear the train horn	some trains don't or can't sound horns, and cab noise can hide them.
City and yard crossings are safer than country ones	they are just as dangerous .
Downshift halfway across the tracks for power	don't shift gears while crossing.
If stuck on the tracks, try to rock or drive the truck free first	get out and away from the tracks, then call for help with the DOT number. Instinct is to save the truck.
Downshift once you notice speed building on the hill	choose the low gear before starting down; later you may not get into any gear.
Always use the same gear down that you used going up	that is the old-truck rule; modern trucks may need a lower gear going down.
The service brakes are your main downhill speed control	engine braking is the main control; brakes are only a supplement .
Brake down to 10 or 15 mph below the safe speed	about 5 mph below , for about 3 seconds .
Keep light, steady pressure on the brakes all the way down	use firm applications (definite slowdown), release, and repeat — steady use builds heat and causes fade .
Brake adjustment only matters for the parking brake	out-of-adjustment brakes make the others overheat and fade ; check adjustment often.

GK-04 • Vehicle inspection

General Knowledge • test weight **High** • handbook Section 2.1.1–2.1.7

You will be able to answer

- Why you inspect, what the law says, and who is responsible for a safe vehicle.
- When to inspect (before, during, after a trip), including the cargo check within the first **50 miles**, then every **150 miles or 3 hours**.
- The defects to look for in tires, wheels, brakes, steering, suspension and exhaust, with the key numbers (**4/32** and **2/32 inch, 10 degrees, ¼**).
- The **3 kinds** of required emergency equipment.
- The 7-step inspection method: gauge readings, light colors, the key, and the three brake tests.

Key points

Start here

- A **defect** is something broken, worn, missing or not working right.
- **Out-of-service (OOS)** is an official order that the vehicle may not be driven until repaired.
- The **motor carrier** is the company that runs the vehicle, usually your employer.

Why inspect p. 2-1

- **Safety** is the most important reason to inspect, for you and everyone else on the road.
- Finding defects early also prevents breakdowns, lost time and money, and crashes.
- Drivers must inspect under both **federal and state law**.
- Federal and state inspectors may inspect too; an unsafe vehicle is placed **out-of-service** until repaired.

Types of vehicle inspections pp. 2-1, 2-2

- Pre-trip → inspect every time and read the **last vehicle inspection report**; the carrier must fix and **certify** each safety item.
- Once you drive, safe operation is **your** job, **not the mechanic's**, because you are the one in control.
- During the trip → watch gauges and **look, listen, smell and feel**; check tires, brakes, lights, connections, coupling and cargo securement at every stop.
- After the trip → at the end of the trip, day or tour of duty, inspect **each vehicle you drove**.

What to look for pp. 2-2, 2-3, 2-4

- Front tires need at least **4/32 inch** tread in every major groove; all other tires at least **2/32 inch**.
- **Rust around wheel nuts** can mean they are loose; a wheel or rim **repaired by welding** is unsafe.
- Steering play of **more than 10 degrees** (about **2 inches** on a 20-inch wheel) makes the vehicle hard to steer.
- Missing **¼ or more** of a spring's leaves = **out-of-service**; a broken exhaust can let poisonous fumes into the cab.
- **Three** kinds of emergency equipment are required: fire extinguisher, spare electrical fuses (unless circuit breakers), and warning devices like **3 red reflective triangles**.

The CDL vehicle inspection test p. 2-4

- To get a CDL you must pass a hands-on **vehicle inspection test** at a real vehicle.
- For each item, **point to or touch it, name it, and explain why** you check it.
- Trucks → before every trip, check the truck is **not overloaded** and the load is **balanced and secured**; with HazMat, check papers and placards.

The 7-step inspection method pp. 2-4, 2-5, 2-6, 2-7, 2-8

- Always follow the **same order** (approach, then 7 steps); in Step 2, first make sure **parking brakes are on** and/or wheels are chocked.
- Air pressure should build from **50 to 90 psi within 3 minutes**, up to governor cut-out, usually **120-140 psi**.
- Oil pressure must be normal **within seconds**; temperatures rise gradually; the **ABS** light should go out, and staying on means a fault.
- Step 4 → engine off, take the **key** with you; front clearance lights and reflectors are **amber**, rear ones red.
- Hydraulic leak test → pump **3 times, hold 5 seconds**; service brake test at about 5 mph; check brake, turn and 4-way lights separately.

Inspection during a trip p. 2-8

- During a trip, check instruments, gauges, mirrors, tires, lights and cargo, and investigate anything you **see, hear, smell or feel**.
- Check cargo securement within the **first 50 miles**, then every **150 miles or 3 hours, whichever comes first**.
- Cargo shifts and tie-downs loosen as you drive, which is why the checks repeat.

After-trip inspection and report p. 2-9

- You may have to write a **daily report** on each vehicle you drove, listing anything affecting **safety** or that could cause a breakdown.
- The carrier uses the report to plan repairs.
- Keep a copy **in the vehicle for 1 day**, so the next driver can learn about the problems.

Numbers and terms to memorize

Item	Value	Page
Most important reason to inspect	Safety	2-1
Responsible for safe operation	You, the driver (not the mechanic)	2-1
Front tire tread	4/32 inch in every major groove	2-2
Other tire tread	2/32 inch	2-2
Bus front wheels	No regrooved, recapped or retreaded tires	2-2
Rust around wheel nuts	Nuts may be loose	2-2
Welded wheels or rims	Not safe	2-2
Steering wheel play limit	10 degrees ($\approx$ 2 inches at the rim of a 20-inch wheel)	2-2
Leaf spring out-of-service	¼ or more leaves missing	2-3
Broken exhaust danger	Poison fumes in cab or sleeper	2-3
Required emergency equipment	Fire extinguisher; spare fuses (unless circuit breakers); warning devices	2-4

Item	Value	Page
Warning devices	3 red reflective triangles, 6 fusees, or 3 liquid-burning flares	2-4
Oil pressure	Normal within seconds ; 5–20 psi idling, 35–75 psi operating	2-5
Air pressure (section 2.1)	50 → 90 psi within 3 minutes	2-5
Governor cut-out	Usually 120–140 psi	2-5
Step 4 key	Take it with you	2-6
Front / rear light color	Amber front, red rear	2-6, 2-7
Hydraulic leak test	Pump 3 , hold 5 seconds	2-8
Service brake test	About 5 mph	2-8
Cargo securement check	First 50 miles , then every 150 miles or 3 hours (whichever first)	2-8
Report copy in vehicle	1 day	2-9

Exam traps

Tempting wrong idea	Handbook answer
The main reason to inspect is that the law requires it	safety . The law and cost savings are real reasons, so they tempt.
The mechanic signed off, so the mechanic is responsible	you are responsible once you drive.
2/32 front and 4/32 others	4/32 front, 2/32 others . Easy to swap.
Rust around lug nuts is only cosmetic	the nuts may be loose ; wrench-test them.
A rim repaired by welding is fixed	welded wheels or rims are not safe .
leaf spring out-of-service only when half the leaves are gone	¼ or more ; and any defect can be dangerous.
tire chains are required emergency equipment	optional . Required: extinguisher, spare fuses, warning devices.
6 triangles	3 triangles; "6" is the number of fusees .
air pressure should build within 45 seconds	50–90 psi within 3 minutes . The 45 seconds is the dual-air test from Air Brakes, which you studied recently.
Coolant temperature should be normal right after start-up	temperatures rise gradually ; it is oil pressure that must be normal within seconds.
An ABS light that stays on is normal until you drive	it should go out ; staying on = ABS fault.
pump 5 times, hold 3 seconds	pump 3 , hold 5 .
cargo check in the first 100 miles, or 150 miles/3 hours whichever comes last	first 50 miles , then 150 miles/3 hours, whichever comes first .

GK-05 · Basic control, backing, shifting, retarders

General Knowledge · test weight **Medium** · handbook Section 2.2–2.3

You will be able to answer

- How to start moving on a hill without rolling back, and what to do if the drive wheels spin.
- The six safe-backing rules, why you back toward the driver's side, and the most important hand signal to agree on with a helper.
- Which way to turn the steering wheel when backing a trailer, and how to fix a trailer that drifts.
- The five steps of double clutching (up and down), and the two ways to know when to shift.
- The two situations where you must downshift, and how an automatic transmission helps on a downgrade.
- What a retarder does, its four types, and when you must turn it off.

Key points

Basic control of your vehicle p. 2-9

- To control a truck or bus you control its **speed** and its **direction**.
- The **4 skills** are accelerating, steering, stopping, and backing safely.
- Wear your seatbelt on the road, and set the **parking brake** whenever you leave the vehicle, because an empty parked truck can roll away.

Accelerating p. 2-9

- Don't roll back: with a manual, let the clutch **partly out** before your right foot leaves the brake.
- Release the parking brake **only after** the engine gives enough power to hold the vehicle; a tractor-trailer may also use the trailer hand valve.
- Speed up smoothly and gradually, because jerky starts can damage the vehicle and the **coupling**.
- On poor traction, speed up very gradually; if the drive wheels spin → **take your foot off the accelerator**.

Steering p. 2-9

- Grip the wheel firmly with **both hands**, on **opposite sides** of the wheel.
- Why: hitting a curb or pothole can jerk the wheel out of a loose grip.

Stopping p. 2-9

- Press the brake pedal down **gradually**; the pressure needed depends on your speed and how fast you must stop.
- With a manual, push in the clutch once the engine has slowed to **near idle**, so the engine keeps running as you stop.

Backing safely p. 2-10

- Backing is **always dangerous**, so avoid it when you can and park where you can pull forward out.
- Follow the **6 rules**: proper start position, get out and look at your path, use both mirrors, back slowly, turn toward the driver's side, use a helper.
- Back in the **lowest reverse gear** so you can fix errors and stop quickly.
- Back toward the **driver's (left) side**, even if you must drive around the block, because your view on the right (blind) side is poor.
- A helper stands near the back **where you can see them**; the most important hand signal to agree on is **"STOP."**

Backing with a trailer p. 2-10

- Backing a trailer → turn the steering wheel the **opposite** way, then follow the trailer once it begins to turn.
- Example: to move the trailer's rear left, turn the wheel **right**, then turn back left to follow it.
- Keep the path straight when possible, or curve toward the driver's side, and back **slowly**.
- Correct drift immediately by steering **toward the drift**, and use **pull-ups** to reposition whenever needed.

Shifting gears p. 2-10

- If you can't get into the right gear while driving, you have **less control** of the vehicle.
- So **correct shifting** matters.

Manual transmissions pp. 2-10, 2-11

- Most heavy vehicles with an **unsynchronized** manual require double clutching (clutch in twice per shift); synchronized ones do not.
- Double clutching has **5 steps**; in Neutral, let rpm drop before an upshift but press the accelerator to **raise** rpm before a downshift.
- Can't get into gear → don't force it: back to Neutral, release the clutch, match engine rpm to road speed, try again.
- Know when to shift by **engine speed (rpm)** or **road speed (mph)**; with progressive shifting the shift rpm rises in higher gears.
- Downshift **before** starting down a hill, usually to a gear lower than the climbing gear, and **before** entering a curve.

Multi-speed rear axles and auxiliary transmissions p. 2-11

- Multi-speed rear axles and auxiliary transmissions give the vehicle **extra gears**.
- They are usually worked by a **knob or switch** on the main shift lever; shift patterns vary, so learn the vehicle you drive.

Automatic transmissions p. 2-11

- On a downgrade, select a **low range** to get more engine braking.
- A low range keeps the transmission from shifting up past the chosen gear, unless the engine exceeds its **governor rpm**.

Retarders p. 2-11

- A retarder helps slow the vehicle, giving **less brake wear** and another way to slow down.
- The **4 types** are exhaust, engine, hydraulic, and electric; the driver can turn any retarder on or off.
- When on, it brakes **only the drive wheels**, each time your foot comes fully off the accelerator.
- Turn the retarder **OFF** on wet, icy, or snow-covered roads, because it can make the drive wheels skid.

Numbers and terms to memorize

Item	Value	Page
Basic control skills	4: accelerating, steering, stopping, backing safely	2-9
Leaving the vehicle	Set the parking brake	2-9
Starting on a hill (manual)	Partly engage clutch before taking right foot off brake	2-9
When to release parking brake on a start	Only after enough engine power is applied to prevent rollback	2-9
Drive wheels spin	Take foot off the accelerator	2-9
Hands on the wheel	Both hands, on opposite sides, firm grip	2-9
Manual transmission stop	Clutch in once the engine is near idle	2-9
Safe-backing rules	6	2-10
Reverse gear for backing	Lowest reverse gear	2-10
Safest backing direction	Toward the driver's side	2-10
Helper position	Near the back of the vehicle, where you can see them	2-10
Most important hand signal	"Stop"	2-10
Backing a trailer	Turn steering wheel the opposite way	2-10
Trailer drifts off path	Turn steering wheel in the direction of the drift	2-10
Pull-up	Drive forward to reposition while backing	2-10
Double clutching	Needed on most unsynchronized manuals; not on synchronized	2-10
Steps in a double-clutch shift	5	2-10, 2-11
Ways to know when to shift	2: engine speed (rpm) and road speed (mph)	2-11
Progressive shifting	Shift rpm gets higher in higher gears	2-11
Special times to downshift	2: before starting down a hill; before entering a curve	2-11
Downhill gear	Usually lower than the gear to climb the same hill	2-11
Automatic on a downgrade	Select a low range for engine braking	2-11
Retarder types	4: exhaust, engine, hydraulic, electric	2-11
Retarder acts on	Drive wheels only, when accelerator is fully released	2-11
Retarder OFF when	Road is wet, icy, or snow-covered	2-11

Exam traps

Tempting wrong idea	Handbook answer
Retarders help prevent skids on slippery roads.	False. A retarder can make the drive wheels skid; turn it off on wet, icy, or snowy roads. The word "retarder" sounds like a safety device, so people assume it always helps.
Back toward the right side so traffic is away from you	Back toward the driver's side . On the right (blind) side your view is poor, so it is very risky.
When backing a trailer, turn the wheel the way you want the trailer to go	Turn it the opposite way. That "toward" rule is right for cars, straight trucks and buses, which is why it tempts.
When the trailer drifts, steer away from the drift	Steer toward the drift , right away. "Away" feels natural but pushes the trailer further off.
Downshift once you feel the truck speeding up on the hill, or use the same gear you climbed with	Downshift before starting down, to a gear usually lower than the climbing gear.
Drive wheels spinning → brake, or add power to push through	Take your foot off the accelerator .
Release the parking brake, then give it gas	Give enough power first ; release the parking brake only when the engine can hold the vehicle.
The helper should stand in the middle of your path where they can see everything	Near the back of the vehicle, where you can see them . If you can't see your helper, you can't see their signals.
A retarder slows all wheels	Drive wheels only.
Can't get into gear → push harder on the shift lever	Don't force it; back to Neutral, release clutch, match engine speed to road speed, try again.
Synchronized transmissions need double clutching	Not required on synchronized; most unsynchronized heavy-vehicle transmissions need it.

GK-03 · Other CDL rules and California laws

General Knowledge · test weight **Medium** · handbook Section 1.4–1.5

You will be able to answer

- Who you must tell, and how fast, after an out-of-state ticket, a license suspension, or a crash.
- The California speed limit and lane rules for big trucks and for any vehicle that tows, and when you must use a turnout.
- California size limits: length, width, height, load overhang, and the main axle-weight numbers.
- How California (intrastate) hours of service differ from federal (interstate) hours of service.
- When you must send an SR 1 accident report, and how CHP handles an overweight truck at the scale.
- Basic clean-air (CARB) rules such as the idling limit, and what motor carrier permits, IRP, IFTA and trip records are.

Key points

CA Start here pp. 1-15, 2-13, 2-16

- This module is law, not driving skill: most questions ask for a number, so memorize the **bold** numbers.
- Rules marked **CA** are California rules; **CVC** = California Vehicle Code, CFR = federal rules, CHP = California Highway Patrol.
- **GVWR** = gross vehicle weight rating, the maker's maximum safe weight for the loaded vehicle.
- **CA** Some California rules are taught in other modules, and **triple trailers** are not legal in California.

Other CDL rules p. 1-16

- Out-of-state traffic conviction (not parking), in any vehicle → tell your licensing agency within **30 days**.
- License suspended, revoked, cancelled or disqualified → tell your employer within **2 business days** (answer this on the test for the 1.4 wording).
- H endorsement holder with a disqualifying crime, mental commitment, or giving up citizenship → notify the issuing state and surrender the H within **24 hours**.
- Hold only **1 license**; no holding a phone to talk, dialing only with **one button**, and no sending or reading texts.
- Always wear a seat belt: if you are thrown from the vehicle without one, you are **4 times** more likely to be killed.

Disqualification summary tables pp. 1-17, 1-18, 1-19

- Major offenses (alcohol, drugs, BAC **0.04%+** in a CMV, leaving the scene, felony): 1st = **1 year** (3 years with HazMat); 2nd = life.
- A felony involving a controlled substance → **life** on the 1st conviction, in any vehicle.
- Serious traffic offenses: 2nd within 3 years = **60 days**; 3rd within 3 years = **120 days**.
- Railroad crossing offenses: at least **60 days**, then 120 days, then 1 year (2nd and 3rd within 3 years).
- 1st out-of-service order violation → at least **90 days** (up to 1 year); hauling HazMat or 16+ passengers → at least **180 days** (up to 2 years).

CA State laws and rules p. 1-16

- CA Every CMV must stop at **each posted CHP inspection and testing site**, because scales and inspections find unsafe or overweight trucks.
- CA An officer who suspects an unsafe or oversize load may stop, inspect, measure or weigh you, and make you **reload or remove** part of it.
- CA Driving illegally over a highway or bridge makes you liable for all damage; the **owner and the driver** may both have to pay.

CA State air emissions rules — CARB pp. 1-20, 1-21

- CA CARB rules cover every commercial diesel vehicle in California, **even ones based in another state or country**.
- CA Diesel CMVs over **10,001 lb** GVWR may idle at most **5 minutes** when not needed, and not at all at a school.
- CA Truck and Bus Regulation (over **14,000 lb**): DPF since Jan 1, 2012; since **2020** DMV won't register a truck that doesn't comply.
- CA Smoke inspection (PSIP): fleets of 2+ diesel trucks test **every year** and keep records at least **2 years**; DPF trucks have a 5% opacity limit.
- CA Port (drayage) trucks over 26,000 lb must be in the CARB Drayage Truck Registry with a **2007** or newer engine.

CA Length — single vehicle p. 1-21

- CA A single vehicle may be at most **40 ft** long; semitrailers, buses and housecars can be exempt under certain conditions.
- CA The front bumper may stick out no more than **2 ft** ahead of the fenders, cab or radiator.
- CA An articulated bus or trolley coach may be at most **60 ft** long.
- CA Buses may add a **1 ft** safety bumper, an 18 in front wheelchair lift, and bike racks up to 36 in front or 10 ft rear.

CA Length — combination vehicles p. 1-21

- CA Truck tractor + one trailer → total length at most **65 ft**.
- CA Truck tractor + semitrailer + trailer (doubles) → at most **75 ft**, and neither trailer longer than **28 ft 6 in**.
- CA Cities and counties may post signs banning combinations longer than **60 ft** on their roads.
- CA Load length, from the farthest-forward point to the farthest-back point of vehicle or load, is at most **75 ft**.

CA Length exceptions and overhang p. 1-22

- CA Long one-piece loads (poles, timbers, pipes, structural parts) may be up to **80 ft** on a pole or pipe dolly or semitrailer.
- CA A load may extend at most **3 ft** past the front bumper or front tires; a load made only of vehicles may extend **4 ft**.
- CA Rear overhang on a single vehicle is at most **2/3 of the wheelbase** behind the last point of support, because a long tail swings wide.
- Example: wheelbase 18 ft → load may hang out at most $2/3 \times 18 = 12$ ft behind the last support.

CA Width p. 1-22

- **CA** Body of the vehicle or the load may be at most **102 in (8½ ft)** wide; wheel to wheel with air tires, **108 in (9 ft)**.
- **CA** Permitted devices (door handles, chain binders) may stick out **3 in** each side; required devices (lights, mirrors) **10 in** each side.
- **CA** Special mobile or construction equipment and loosely piled farm products (load plus racks) may be **120 in** wide.
- **CA** Buses are limited to **102 in** (city or suburban for-hire buses 104 in); local roads may ban vehicles wider than **96 in (8 ft)**.

CA Farm equipment p. 1-23

- **CA** Implements of husbandry (**farm equipment**) are generally exempt from width and length limits when moved on a highway during normal farm work.
- **CA** A **Caltrans transportation permit** may still be needed to move farm equipment.

CA Height p. 1-23

- **CA** The vehicle and load may be at most **14 ft** high, measured from the road surface.
- **CA** Exceptions: **double-deck buses** up to 14 ft 3 in, and farm equipment moved over a highway during farm work.
- Why: bridges, overpasses and wires limit how tall a vehicle can safely be.

CA Weight limits — general p. 1-23

- **CA** **Caltrans** may post bridge and highway weight limits that are higher or lower than normal CVC limits.
- **CA** Cities and counties may post higher or lower limits on their own roads and bridges and name other routes for heavy vehicles.
- **CA** A local weight sign still lets you use the street **by the direct route** to pick up or deliver goods, deliver permitted construction materials, or do utility work.

CA Axle weight limits p. 1-23

- **CA** Single vehicle, in the handbook's wording: **20,000 lb** on the wheels at one end of an axle (20,500 lb for buses) and **10,500 lb** on any one axle.
- **CA** The 10,500-lb limit does not apply to **livestock** haulers; never exceed the tire maker's load limit on the sidewall.
- **CA** In a combination, any one axle may carry at most **18,000 lb** and the wheels at one end of an axle at most **9,500 lb**.
- **CA** A front steering axle in a combination may carry up to **12,500 lb**.
- **CA** Groups of 2+ axles use CVC tables: one table if first-to-last axle is **18 ft or less**, another if it is more than 18 ft.

CA Logs and weight-to-axle ratio p. 1-24

- **CA** Log trucks have special weight limits, but that extra weight is **not allowed on interstate highways**.
- **CA** Weight-to-axle ratio: axles close together = **less weight** allowed; axles spread out = more, because roads carry only so much weight per foot.
- **CA** Two tandem sets **36 ft or more** apart may carry **34,000 lb** each, and 68,000 lb together.

CA Loading and unloading; special permit routes; penalties p. 1-24

- CA Load limits are **not enforced** while you load or unload at, or right next to, a loading or unloading area.
- CA You may not change a special permit route; the only detour is on **nonresidential streets** to avoid breaking a local city rule.
- CA Leaving the permitted route with an extralegal load without a peace officer's approval is a **misdemeanor**.

CA CHP weight enforcement at the scale p. 1-24

- CA CHP first allows a **200-lb** variation factor; if still **100 lb or more** over after that, you get a citation.
- CA After a citation you must adjust the load or get an overweight permit **before moving on**.
- Example (limit 34,000 lb): 34,250 → 50 over after allowance, no citation; 34,350 → **150** over, citation.
- CA An overweight **HazMat** load may continue as loaded if adjusting it cannot be done with reasonable safety.
- CA Livestock and field-loaded perishables are cited but may continue if a single truck is $\leq$ **1,000 lb** over per axle or group, or a combination $\leq$ **2,000 lb** gross.

CA Permits for oversize loads p. 1-24

- CA You need a permit to move any vehicle over the **width, height or weight** limits on a California street or highway.
- CA Get the permit from **Caltrans** for state highways, or from the city or county for their roads.
- CA Moving an oversize extralegal load without a permit can bring a **fine, jail, or both**, plus excess-load penalties.

CA Motor carrier permit and UCR pp. 1-24, 1-25

- CA Anyone running a CMV, **for hire or privately**, needs a Motor Carrier Permit (MCP) from the DMV.
- CA For the MCP, a CMV includes a motor truck with **2 or more axles** and a GVWR **over 10,000 lb**, or any vehicle carrying property for hire.
- CA Not counted: household goods carriers, **pickup trucks**, and 2-axle daily rental trucks under 26,001 lb used non-commercially.
- Interstate or foreign carriers hauling property must register under **UCR** (Unified Carrier Registration Act of 2005).

CA Speed limits p. 1-25

- CA **55 mph** is the top speed for trucks with 3+ axles, **any vehicle towing**, school buses with pupils, farm labor vehicles with passengers, explosives, and trailer buses.
- CA All other vehicles: **65 mph** on most highways, 55 mph on two-lane undivided highways unless posted higher, **70 mph** only where signed.
- Driving so slowly that you block normal traffic is also illegal, unless needed for safety, the law, or your vehicle's size and weight.
- Example: a pickup towing a small trailer is limited to **55 mph**, because heavy and towed loads need more distance to stop.

CA Right lane rule p. 1-25

- **CA** 55-mph-limit vehicles use the signed lanes; with no signs, drive in the **right-hand lane** or as near the right edge as possible.
- **CA** On a divided highway with **4 or more lanes** in your direction, you may also use the lane just left of the right lane.
- Why: slow, heavy vehicles stay out of the faster left lanes, so traffic flows with fewer lane-change conflicts.

CA Designated system access p. 1-25

- **CA** The lane rule does not apply when preparing to **turn**, entering or leaving a highway, or using another lane to stay on your route.
- **CA** Buses (not school or trailer buses) may use **any lane** if they are not towing.
- **CA** Larger trucks leave the designated system only at signed exits, going to a **terminal** or up to **1 mile** for food, fuel, lodging or repairs.

CA Slow vehicle rule p. 1-26

- **CA** On a two-lane highway where passing is unsafe, a slow vehicle with **5 or more vehicles** behind must pull off to let them pass.
- **CA** Use the nearest **signed turnout** or any other place with enough room, because drivers stuck behind a slow truck may try a dangerous pass.

Hours of service — which rules apply? p. 1-26

- **CA Intrastate** (inside California only) → follow California hours of service; **interstate** → follow federal hours of service.
- You are interstate if cargo came from or is going out of state, or is hazardous substances or wastes.
- You can be **interstate without leaving California**, for example hauling freight from Arizona from a Los Angeles warehouse to Fresno.

Record of duty status p. 1-26

- A record of duty status logs **all** your hours: driving, on duty not driving, off duty, and sleeper berth.
- **Each driver and each co-driver** keeps the log in duplicate.
- Show it **immediately** when asked by an authorized CHP employee, a salaried police officer, or a deputy sheriff.

CA Hours of service — federal vs. California p. 1-27

- **CA** California intrastate: drive at most **12 hours** after 10 off, and no driving after **16 hours** on duty; federal is 11 hours and the 14th hour.
- **CA** California: no driving after **80 hours** on duty in any 8 days (farm products 112); federal: 60 hours in 7 days or **70 hours in 8 days**.
- Both systems need **10 hours** off in a row before driving again; the sleeper berth split is 2 periods, one at least 8 hours.
- Bad weather adds **2 hours** of driving; in California never more than 14 hours driving or after 16 hours on duty.
- Federal carriers restart the 7/8-day period after **34+ hours** off in a row; limits exist because tired drivers react slowly.

CA Accident reporting p. 1-26

- **CA** A crash in California causing death, injury, or damage **over \$1,000** → report it to the DMV on form SR 1 within **10 days**.
- **CA** The SR 1 is required whether or not you caused the crash, **even on private property**, and in addition to police or insurance reports.
- **CA** If you do not file a required SR 1, your **driving privilege will be suspended**.
- **CA** Tell your employer within **5 days** after a crash in the employer's vehicle, or immediately if the employer requires it.

CA Financial responsibility pp. 1-27, 1-28

- **CA** General freight in vehicles of **10,001 lb** or more → at least **\$750,000** combined single limit; 10,000 lb or less → **\$300,000**.
- **CA** Oil, HazMat, or waste → **\$1,000,000** CSL; top-risk loads like explosives or poison gas → **\$5,000,000** CSL.
- **CA** Proof must stay on file while the MCP is active; if the insurance or bond lapses, the **DMV suspends the MCP**.
- **CA** A driver must be able to show financial responsibility **before operating** the vehicle.

IRP and IFTA: interstate registration and fuel tax pp. 1-28, 1-29, 1-30

- Interstate vehicles needing a CDL must register under **IRP** (apportioned plates and cab card) and **IFTA** (fuel tax) with their base jurisdiction.
- They qualify with **2 axles over 26,000 lb, 3 or more axles** at any weight, or a combination over 26,000 lb; recreational vehicles are excluded.
- IFTA uses one license and **one quarterly tax return** filed with your base jurisdiction.
- Record the odometer at the start of the day, when **leaving a state or province**, and at trip or day end; keep a receipt for every fuel purchase.
- Keep IVDR (trip report) records at least **4 years**, because fees and taxes are calculated from them and can be audited.

Numbers and terms to memorize

Item	Value	Page
Report out-of-state traffic conviction (not parking) to licensing agency	within 30 days, any vehicle	1-16
Tell employer of suspension, revocation, cancellation or disqualification	within 2 business days (p. 1-4 says by end of next business day)	1-16
H endorsement holder: disqualifying crime, mental commitment, giving up citizenship	notify issuing state and surrender H within 24 hours	1-16
Number of licenses allowed	1 (employer may not let you drive with more)	1-16
Phone while driving	no holding to talk; dial with a single button only; no texting	1-16
No seat belt and thrown from the vehicle	4 times more likely to be killed	1-16
1st OOS order violation (non-HazMat)	at least 90 days (up to 1 year)	1-19

Item	Value	Page
1st OOS violation hauling HazMat or 16+ passengers	at least 180 days (up to 2 years)	1-19
CA Diesel idling limit (GVWR over 10,001 lb)	5 minutes; none at a school	1-20
CA Single vehicle length	40 ft	1-21
Front bumper beyond fenders/cab/radiator	2 ft	1-21
Articulated bus or trolley coach	60 ft	1-21
Truck tractor + trailer	65 ft	1-21
Truck tractor + semitrailer + trailer	75 ft; each trailer max 28 ft 6 in	1-21
Local ban on long combinations (if posted)	over 60 ft	1-21
Load length (front to back)	75 ft	1-21
Long one-piece loads (poles, pipes, etc.)	up to 80 ft on approved dolly/semitrailer	1-22
Load past front bumper/tires	3 ft (load of only vehicles: 4 ft)	1-22
Load behind last point of support	max 2/3 of wheelbase	1-22
CA Width of body or load	102 in (8½ ft)	1-22
Wheel-to-wheel width (air tires)	108 in (9 ft)	1-22
Permitted devices / required devices (mirrors, lights)	3 in / 10 in each side	1-22
Local roads may ban vehicles wider than	96 in (8 ft)	1-22
Special mobile/construction equipment; loose hay loads	120 in (10 ft)	1-22
CA Height	14 ft (double-deck bus 14 ft 3 in)	1-23
Single-vehicle axle numbers	20,000 lb (buses 20,500) and 10,500 lb	1-23
Combination: any one axle / one end of axle	18,000 lb / 9,500 lb	1-23
Front steering axle (combination rule)	12,500 lb	1-23
Two tandems 36 ft or more apart	34,000 lb each; 68,000 lb total	1-24
CHP standard / variation factor	100 lb over = stop; 200-lb allowance in practice	1-24
Livestock / field-loaded perishables overweight allowance	1,000 lb over on an axle or axle group (single truck); 2,000 lb over gross (combination)	1-24
Leaving permit route without officer OK	misdemeanor	1-24
MCP needed for motor truck with	2+ axles and GVWR over 10,000 lb	1-24
CA California 55-mph list	3+ axle trucks, anything towing, school bus with pupils, farm labor vehicle with passengers, explosives, trailer bus	1-25
Other vehicles	65 mph most highways; 55 on 2-lane undivided; 70 only if posted	1-25
Lanes on 4+ lane-each-way divided highway	right lane or the lane just left of it	1-25

Item	Value	Page
Off the designated system for food, fuel, lodging, repairs	up to 1 mile	1-25
CA Turnout rule	5 or more vehicles behind you	1-26
CA California HOS	12 hr driving; 16 hr on duty; 80 hr on duty in 8 days (farm 112)	1-27
Federal HOS	11 hr driving; 14th hour, 60/7 or 70/8; 34-hr restart	1-27
Off-duty rest; sleeper split	10 hr in a row; 2 periods, one at least 8 hr	1-27
Adverse weather	+2 hr driving (CA max 14 hr driving, never after 16 hr on duty)	1-27
CA SR 1 accident report	to DMV within 10 days if death, injury, or damage over \$1,000	1-26
CA Tell employer of crash in their vehicle	within 5 days	1-26
FR: general freight, 10,001 lb or more	\$750,000 CSL	1-27
FR: oil, HazMat, waste / top-risk loads	\$1,000,000 / \$5,000,000 CSL	1-27
IRP / IFTA vehicle	2 axles over 26,000 lb, or 3+ axles any weight, or combination over 26,000 lb	1-29
IFTA tax return	one, quarterly, to base jurisdiction	1-28
Keep IVDR records	at least 4 years	1-30

Exam traps

Tempting wrong idea	Handbook answer
A tractor-trailer may go 65 mph on a freeway posted 65	55 mph for any truck with 3+ axles or any vehicle towing. The posted 65 is only for vehicles not on the 55-mph list.
Only big rigs are held to 55 mph	a pickup towing a trailer, a school bus with pupils, and a vehicle hauling explosives are also held to 55. "Pulling another vehicle" includes small trailers.
On a wide freeway, a truck may use any lane except the far left	the right lane, plus the lane just left of it only on a divided highway with 4 or more lanes in your direction (or signed lanes).
You must use a turnout when 3 vehicles are behind you	5 or more.
No SR 1 is needed if the crash was not your fault or happened in a parking lot	report to DMV within 10 days if there was death, injury or over \$1,000 damage — fault and private property don't matter.
The SR 1 goes to CHP or your insurance company	it goes to the DMV , in addition to any police or insurance report.
Tell your employer about a crash in the company truck within 10 days (mixing it with the SR 1)	within 5 days (or right away if the employer requires it).

Tempting wrong idea	Handbook answer
30 days to tell your employer your license was suspended	2 business days. The 30 days is for telling your licensing agency about an out-of-state conviction.
California intrastate drivers get 11 hours of driving and a 14-hour window	California is 12 hours driving and 16 hours on duty; 11/14 is federal.
California intrastate weekly limit is 70 hours in 8 days	80 hours in 8 days (112 for farm products); 60/70 is federal.
You never leave California, so California HOS always applies	if the cargo came from or is going out of state, or it is hazardous substances/wastes, you are interstate and use federal HOS.
The maximum width is 96 inches	102 in. 96 in is the limit a city or county may post on its own roads.
Doubles may be 75 ft with trailers up to 40 ft each	75 ft total, but neither trailer over 28 ft 6 in. A tractor with one trailer: 65 ft.
A load may stick out 4 ft in front	3 ft; 4 ft only for a load made only of vehicles.
All vehicles may be 14 ft 3 in tall	14 ft; only double-deck buses get 14 ft 3 in.
Diesel trucks may idle 10 minutes, or idle briefly at a school	5 minutes of non-essential idling, and no idling at all at a school.
An overweight HazMat load must always be unloaded at the scale	it may continue as loaded if adjusting it cannot be done with reasonable safety.
A local weight-limit sign means you can never drive that street	you may use it by the direct route to make a pickup or delivery, deliver permitted construction materials, or do utility work.
With a special permit, you may pick a shorter route	you must stay on the permit route; the only detour is on nonresidential streets to avoid breaking a local city rule. Leaving it without a peace officer's OK is a misdemeanor.
Only the owner pays for bridge damage from an overweight truck	the driver is liable, and the owner and driver may both have to pay.
You may hold your phone for a quick call	no holding the phone to talk; dialing only with a single button press; no reading or sending texts.
A 2-axle truck of exactly 26,000 lb needs IRP/IFTA	a 2-axle truck must be over 26,000 lb; a 3-axle truck qualifies at any weight.

GK-01 · Getting your CDL: classes, tests, medical rules

General Knowledge · test weight **Medium** · handbook Section 1 intro pages and 1.1-1.2

You will be able to answer

- Which vehicles need a CDL, and whether a vehicle needs a Class A, B or C license.
- The learner's permit (CLP) rules: minimum age, how long it lasts, the 14-day wait, and what a permit holder may not do.
- What each endorsement letter (H, N, P, S, T, X, F) allows, and what the restriction codes (E, K, L, M, N, O, P, V, X, Z and 88) mean.
- How many tries you get on the knowledge and skills tests, and what makes DMV mark a test as failed.
- Who you must tell, and how fast, after a conviction or a license suspension.
- The medical paperwork rules (MER, MEC, National Registry, 2-year limit) and interstate vs intrastate self-certification.

Key points

Key words first p. 1-8

- **CMV** = commercial motor vehicle, **CDL** = commercial driver license, and **CLP** = commercial learner's permit, the learning stage before the CDL.
- **GVWR** is the loaded weight the manufacturer sets for one vehicle; **GCWR** is the loaded weight set for a combination (power unit plus towed unit).
- No maker's GCWR → add the power unit's **GVWR** to the weight of the towed unit and its load.
- An **endorsement** is an extra letter allowing special driving; a **restriction** is a code that limits what you may drive.
- A **placarded load** is HazMat in an amount that requires diamond-shaped warning signs (placards) on the vehicle.

Who needs a CDL p. 1-1

- You need a CDL for a single vehicle with GVWR of **26,001 lb or more**, or a combination with GCWR of **26,001 lb or more**.
- You need a CDL for a vehicle designed, used or maintained to carry **more than 10 people**, counting the driver.
- You need a CDL to tow a vehicle or trailer with GVWR of **10,001 lb or more**, haul placarded HazMat, or tow two units at once.
- **CA** A **3-axle** vehicle that weighs **more than 6,000 lb** also needs a CDL.
- The lines sit one pound above a round number, so **26,000** or **10,000** is the trap answer.

CA Passenger vans: CDL or not? p. 1-1

- **CA** Van with GCWR **more than 26,000 lb** → CDL required; used as a **vanpool** → no CDL.
- **CA** Van carrying **more than 15** people including the driver → CDL required.
- **CA** **More than 10** people including the driver → CDL only if it carries people **for pay or profit, or a nonprofit group uses it**.
- Example: a nonprofit church van built for **12** people including the driver, not a vanpool → **CDL required**.

CA Horse trailers and Restriction 88 pp. 1-1, 1-2

- **CA** A horse trailer carries property; "living quarters" in it are only a side feature and do not change that.
- **CA** A horse trailer with living quarters and GVWR **over 10,000 lb** needs a **commercial Class A** license.
- **CA** If the truck-and-trailer GCWR is **under 26,001 lb**, DMV adds **Restriction 88**.
- **CA** Restriction 88 → only combinations with GCWR **under 26,001 lb** whose towed vehicle has GVWR **over 10,000 lb**.

Which class do you need? p. 1-8

- Combination with GCWR **26,001 lb or more** and towed unit GVWR 10,001 lb or more → **Class A**.
- **CA** Combination with GCWR under 26,001 lb but towed unit GVWR 10,001 lb or more → **Class A with Restriction 88**.
- Power unit GVWR **26,001 lb or more**, alone or towing under 10,001 lb → **Class B**; **CA** so is a single 3-axle vehicle over 6,000 lb.
- Farm labor vehicle for **10 or more**, or **more than 10** people used for profit, pay or a nonprofit → Class B (counts include driver).
- None of the above but carrying placarded HazMat → **Class C**; otherwise no CDL.

Getting a CLP p. 1-2

- You must be at least **18** and already hold a valid California license covering at least noncommercial **Class C**; carry both.
- A CLP lasts up to **180 days**, renewable for 180 more, but never past **1 year** from first application.
- Only **N, P and S** endorsements go on a CLP; N means empty tanks only (purged if it held HazMat), P or S means no passengers.
- Wait at least **14 days** after the CLP is first issued before the skills test.
- A **CDL holder** with the right class and endorsements must ride with you whenever you drive a CMV.

CA Age rules for a CDL p. 1-2

- **CA** Minimum CDL age is **18**, after first getting a California noncommercial Class C license.
- **CA** From **18** you may drive **for hire inside California**, as long as you do no interstate commerce.
- **CA** You must be **21** to drive in **interstate** commerce or haul HazMat or hazardous wastes, even on a trip inside California.

What to bring when you apply pp. 1-2, 1-3, 1-4

- Bring **Form DL 44C**; signing it means agreeing to a chemical test for alcohol or drugs when a peace officer asks, and refusing means no license.
- Bring an **identity document** as a **certified copy** proving birth date and legal presence; a photocopy is not accepted.
- **CA** Bring **one residency document** whose address matches your DL 44C, for an original and each renewal, transfer or upgrade.
- Prove your Social Security number; for an **original CDL** only the **Social Security card** is accepted.
- Also bring your approved **medical forms**, the application fee, and a **DL 170 ETP** if your employer is authorized to issue one.

Time limits and number of tries p. 1-4

- The application fee stays valid for **12 months** after the date you apply.
- Knowledge tests → **3 attempts** per application, with **no waiting period** between them.
- Skills tests → **3 attempts in total** per application, and **each failed part counts as 1**.
- Skills parts you pass count only during the **first 180 days** of the CLP; after a renewal you retake all skills parts.
- Failing basic control skills or the road test means a **retest fee** when you return.

Duties of every CDL driver p. 1-4

- Be a **California resident**, list every state that licensed you in the past **10 years**, and hand in all out-of-state licenses.
- Tell your home state DMV about an out-of-state conviction, and your employer about any conviction (form **DL 535**), within **30 days**.
- Tell your employer about a suspension, revocation, cancellation or disqualification before the **end of the next business day** (§1.4 says **2 business days**).
- Give a new employer your **10-year** commercial driving work history; one driver, one license, one record.

Endorsements pp. 1-4, 1-5

- **H** = HazMat (needs a **TSA background check**); **N** = tank, each tank **more than 119 gal** and all tanks 1,000 gal or more.
- **P** = passenger vehicles built for **10 or more** people including driver; **S** = school buses, Class A or B only.
- **T** = double or triple trailers, but **CA triple trailers are not legal in California**.
- **X** = tank vehicles with placarded HazMat, showing you passed **both** the tank and HazMat tests.
- **CA F** = firefighting equipment; required to drive it, **optional** for Class A or B holders.

Restrictions pp. 1-6, 1-7, 1-11

- Your CDL covers only what you proved on the skills test; there are **10** standard restriction codes.
- **E** = no manual transmission; **L** = no air-brake CMV; **Z** = no full air-brake CMV (tested with air-over-hydraulic brakes).
- **K** = intrastate only; **O** = no tractor-trailer (tested with a pintle hook or non-fifth-wheel hitch); **V** = medical variance.
- **M** = no Class A passenger vehicle; **N** = no Class A or B passenger vehicle.
- A CLP with P or S always gets restriction **P** (no passengers); a CLP with N always gets restriction **X** (no cargo in tank).

Who does not need a CDL p. 1-7

- No CDL is needed for someone driving in an emergency **at a peace officer's direction**, or for **vanpool** drivers.
- Towing, **not for pay**, a fifth-wheel travel trailer over **15,000 lb GVWR** or trailer coach over 10,000 lb GVWR needs only noncommercial Class A.
- Housecar drivers **over 40 ft but not over 45 ft** (with an endorsement), non-civilian military drivers, and unlicensed farm-equipment operators are also exempt.

CA Special certificates pp. 1-7, 1-9

- **CA** Some jobs need a certificate **in addition to** a CDL, such as ambulance, school bus, SPAB, transit, paratransit, farm labor and youth bus.
- **CA** Non-work cell phone use while driving a **school bus or transit vehicle** is **illegal**; emergency calls are allowed.
- **CA** HAM certificate lets a farm worker aged **21+** haul hazardous loads **without a CDL**, no more than 50 miles point to point.
- **CA** School bus drivers **65+** send MER and MEC **every year**; ambulance drivers every 2 years.

General p. 1-9

- You must pass **both** the knowledge and skills tests to get a CDL.
- The handbook prepares you for the tests but does not replace a truck-driving school; DMV calls formal training the most dependable way to learn.

CA Knowledge tests p. 1-10

- **CA** Knowledge tests come in **English, Arabic, Chinese, Punjabi, Russian and Spanish** on DMV's AKTE equipment.
- Everyone takes **General knowledge**; air brakes (including **air-over-hydraulic**), combination, passenger, HazMat, tank, doubles/triples and school bus depend on your license.
- Adding an endorsement without moving up a class → passed tests are waived if within **12 months** of your last renewal or original issue.
- Allow **2-3 hours** to take all tests; tests are not returned to you.

Testing rules: no aids, no helpers, no recording pp. 1-10, 1-11

- Knowledge test → no notes, phone or watch; any aid or stand-in test taker means the test is **failed**.
- Vehicle inspection test → the **only** aid allowed is the handbook's **Section 11 vehicle inspection guide**.
- No electronic devices, bystander help or helper marks on the vehicle or curbs → test stopped and failed.
- **No recording, including dash cams**; switch it off or block it during the road test, because the test must show what you know.

Skills tests p. 1-11

- Skills tests come only **after** passing knowledge tests, in the vehicle type you want, **all in English** with no interpreter.
- Three tests: vehicle inspection **~40 min**, basic control skills **~30 min**, road test **~45-60 min**.
- A road test is required for an **original** CDL, removing a size/equipment restriction, adding P or S, or renewing a CDL expired **more than 2 years**.
- Troops to Trucks can **waive the skills test** for qualified military drivers, but **not** for S or P.

What to study p. 1-12

- **Class A** → study Sections **1, 2, 3, 6** and 11-13, plus 5 for air brakes; Class B or C study the same without 6.
- Endorsements add: passenger → **4**, doubles/triples → **7**, tank → **8** and **9** (plus 6 for Class B or C), HazMat → **9**, school bus → **10**.

Medical documentation pp. 1-12, 1-13

- Every CDL driver gives DMV a valid **MER** and **MEC**, both dated within the last 2 years; submit only MER pages 1–4.
- Interstate CDL drivers must use a **National Registry** examiner; **CA** your employer pays if the job requires a CDL.
- **CA** California has **no variances or exemptions** from the federal physical standards.
- **Carry the MEC** whenever you drive commercially; driving after it expires can get you **cited for driving out of class**.
- **CA** School bus drivers **65 or older** need a new medical report every year; mail updates at least **4 weeks** before expiration.

Interstate or intrastate commerce p. 1-13

- Federal rule makes you **self-certify** your type of commerce as interstate or intrastate.
- Interstate commerce includes trips between two places in the same state as part of a shipment that **began or will end outside** the state.
- You can be in **interstate** commerce even if your truck never leaves California.
- **NI** is issued without limits; **NA** gets the Intrastate Only restriction 40/K; EI and EA are not issued in California.
- DMV **strongly recommends NI** if you meet the federal qualifications and might ever drive interstate.

Numbers and terms to memorize

Item	Value	Page
Single vehicle needing a CDL	GVWR 26,001 lb or more	1-1
Combination needing a CDL	GCWR 26,001 lb or more	1-1
Towed vehicle/trailer needing a CDL	GVWR 10,001 lb or more	1-1
Passenger vehicle needing a CDL	More than 10 people incl. driver	1-1
CA 3-axle vehicle needing a CDL	Weights more than 6,000 lb	1-1
Non-vanpool van always needing a CDL	More than 15 people incl. driver	1-1
Restriction 88	GCWR under 26,001 lb , towed unit GVWR over 10,000 lb	1-2
CLP minimum age	18	1-2
CLP validity	180 days , renewable 180 more, max 1 year from application	1-2
Wait before skills test	14 days after first CLP issue	1-2
CLP endorsements	N, P, S only	1-2
Age: for hire, CA only / interstate or HazMat	18 / 21	1-2
Application fee valid	12 months	1-4
Knowledge / skills test attempts	3 (no wait) / 3 total (each failed part = 1)	1-4
Out-of-state conviction → home DMV; any conviction → employer	30 days (employer: form DL 535)	1-4

Item	Value	Page
Suspension, revocation, disqualification → employer	End of next business day (§1.4: 2 business days)	1-4, 1-16
Licensing states to list / job history for employer	10 years	1-4
Tank (N) endorsement	Each tank over 119 gal , total 1,000 gal or more	1-4
P endorsement	Built for 10 or more incl. driver	1-5
X endorsement	Tank + HazMat	1-5
Travel trailer exemption (not for pay)	Fifth-wheel over 15,000 lb , trailer coach over 10,000 lb → noncommercial Class A	1-7
HAM certificate	Age 21 , max 50 miles point to point, medical every 2 years	1-7
Skills test times	Inspection ~ 40 , basic control ~ 30 , road ~ 45–60 min	1-11
Road test to renew	CDL expired more than 2 years	1-11
MER/MEC date limit	Within 2 years	1-12
School bus driver 65+	Medical every year	1-12
Mail medical update	4 weeks before expiration	1-13
NA self-certification	Restriction 40/K (intrastate only)	1-13

Exam traps

Tempting wrong idea	Handbook answer
a CDL starts at 26,000 lb	26,001 lb or more. Round numbers feel right, but the line is one pound higher.
only weight decides if you need a CDL	CA a 3-axle vehicle over 6,000 lb also needs one.
a CLP lasts 1 year	180 days , renewable for 180 more; 1 year from application is only the outer limit.
you may take the skills test right after getting a CLP	wait 14 days . "No waiting period" is the rule for retaking knowledge tests .
a CLP can carry H or T	only N, P, S .
at 18 you may haul HazMat if you stay in California	HazMat needs age 21 , even intrastate.
each skills test part has its own 3 tries	3 failures total across all parts.
tell your employer about a suspension within 30 days	by the end of the next business day (§1.4: 2 business days). 30 days is for convictions .
endorsement X means empty tanks only	endorsement X = tank + HazMat; "no cargo in tank" is restriction X on a CLP.
passing the Air Brakes knowledge test prevents any air brake restriction	testing without air brakes gives L ; testing with air-over-hydraulic brakes gives Z .
a T endorsement lets you pull triples in California	CA triples are illegal here.

Tempting wrong idea	Handbook answer
interpreters are allowed on the skills test	skills tests are English only ; only knowledge tests come in other languages.
a truck that never leaves California is always intrastate	it can be interstate if the shipment began or ends outside the state.
California can grant a medical variance	CA no variances or exemptions from federal physical standards.

GK-09 · Seeing hazards, distracted and aggressive drivers

General Knowledge · test weight **Medium** · handbook Section 2.8–2.10

You will be able to answer

- What a **hazard** is, how it differs from an **emergency** and a **conflict**, and why you plan as soon as you see one.
- What the **move-over law** asks you to do **CA**, and how to handle work zones, drop-offs, objects in the road and freeway ramps.
- Which clues show that another driver, pedestrian or vehicle may do something dangerous.
- The hand-held phone and texting rules: definitions, the **60-day / 120-day** disqualifications, the emergency exception, and the research numbers.
- How to avoid distraction, and how to spot and handle a distracted driver.
- The difference between **aggressive driving** and **road rage**, and what to do when an aggressive driver confronts you.

Key points

Importance of seeing hazards p. 2-21

- A **hazard** is any road condition or road user that could become a danger; an **emergency** is danger happening now.
- Seeing hazards early gives you time to check mirrors, signal, slow down or change lanes calmly instead of braking hard or swerving.
- Hazards often come with **clues**, and you get better at reading them with experience.

Hazardous roads pp. 2-21, 2-22

- Approaching a stopped emergency vehicle or work zone → **slow down and move to a lane not next to it**, if safe; otherwise slow down and pass carefully.
- **CA** Breaking the move-over law is an **infraction** punished by a fine (**CVC §21809**).
- In work zones drive slowly and warn drivers behind with **4-way flashers or brake lights**; keep away from pavement **drop-offs**.
- Avoid foreign objects **without sudden moves**, and on ramps be slow enough **before** the curve, since posted ramp speeds may be too fast for trucks.

Drivers who are hazards pp. 2-22, 2-23, 2-24

- Watch for people who **can't see or hear you**: blocked windows, rental trucks, delivery trucks, parked cars, bus passengers, pedestrians with headphones.
- If you see part of a vehicle but **not the driver**, assume the driver can't see you and be ready to stop.
- Confused drivers, slow vehicles and impaired drivers act unpredictably; **CA** a vehicle with the **SMV emblem** (red triangle, orange center) is not designed to exceed **25 mph**.
- Impaired-driver clues include weaving, stopping at a green light, and a **window open in cold weather**; watch especially late at night.
- A **conflict** means you must change speed, direction, or both to avoid a hit; other drivers' conflicts can become yours.

Always have a plan p. 2-24

- Hazards can turn into emergencies, so when you see one, **picture the emergency** it could cause and decide what you would do.
- Being ready to act on that plan is **defensive driving**.

Distracted driving p. 2-24

- **Driver distraction** is anything that pulls your attention off driving, risking injury, death or property damage.
- Distractions can be **inside** the vehicle (passengers, radio, eating, phone, CB, texting, telematics, daydreaming) or **outside** (crash scenes, sunlight, objects, billboards).

The crash problem p. 2-24

- In the LTCCS, **8%** of large-truck crashes involved a distraction outside the vehicle and **2%** a distraction inside.
- Distracted-driving crashes kill about **5,500** and injure about **448,000** people a year in the U.S. (NHTSA).
- A phone conversation, **even hands-free**, takes **39%** of the brain energy you normally use for driving.

Effects p. 2-25

- Distraction causes **slowed perception**, so you notice important events late or never.
- It causes **delayed decision making**, so you are slow to choose the right action.
- It causes **improper action**, such as wrong steering, gas or brake input.

Types of distractions p. 2-25

- **Physical** distraction takes your hands from the wheel or eyes from the road, like reaching for an object.
- **Mental** distraction takes your mind off the road, like talking with a passenger.
- Distractions that are **both**, like phone calls and texting, carry the **greatest** crash risk.

Cell/mobile phones p. 2-25

- CMV drivers may not use **hand-held** phones while driving: holding the phone for a call, dialing more than one button, or leaving your seat-belted position to reach it.
- 2nd conviction in 3 years → **60-day** disqualification; 3rd or more in 3 years → **120 days**; hand-held use is allowed only to contact law enforcement or emergency services.
- Dialing makes a safety-critical event **6 times** more likely and takes eyes off the road **3.8 seconds**, about 306 feet at 55 mph.
- Hands-free devices are **no less distracting** than hand-held ones; **CA** devices are legal while driving only in hands-free mode.
- If you must use a device, keep it short, **hang up** in difficult traffic, and never type or read messages.

Texting p. 2-26

- CMV drivers may not **text** while driving in interstate commerce, including reading or entering text, email, or web pages.
- Penalties: **60 days** disqualification (2nd in 3 years), 120 days (3rd+ in 3 years), and civil penalties **up to \$2,750** per violation.
- Texting makes a safety-critical event **23.2 times** more likely, with eyes off the road **4.6 seconds**, about 371 feet at 55 mph.
- The only exception is contacting **law enforcement or emergency services**.

Do not drive distracted pp. 2-26, 2-27

- Remove **all** in-vehicle distractions before driving, because reacting just **half a second** slower doubles crashes.
- Turn off devices; if you must use a phone, keep it **within reach while buckled in** and use hands-free features.
- Before driving, adjust controls and mirrors, preset radio, secure loose objects and cargo, and **set the GPS**; avoid eating, smoking and emotional conversations.

Watch out for other distracted drivers p. 2-27

- Signs of a distracted driver include **drifting** over or within lane lines, changing speeds, and being busy with maps, food, phones or passengers.
- Give them **plenty of room**, keep a safe following distance, and pass very carefully, since they may drift in front of you.

Aggressive driving and road rage p. 2-27

- **Aggressive driving** is driving in a selfish, bold or pushy way, ignoring others' rights and safety, like frequent unsignaled lane changes.
- **Road rage** is driving with intent to harm others, or physically attacking a driver or their vehicle.
- Heavy, slow traffic and tight schedules make drivers angry and quick to take others' mistakes personally.

Do not be an aggressive driver pp. 2-27, 2-28

- Your **mood before you start** strongly affects how stress hits you, so lower stress before and during the drive.
- Be realistic about travel time; if you'll be late, **accept it**, and give others the benefit of the doubt.
- Slow down, stay out of the left lane when slower than traffic, **avoid gestures**, and let others in ("Be my guest").

When an aggressive driver confronts you p. 2-28

- First and most important → **get out of their way**; put your pride aside and don't speed up or hold your lane.
- **Avoid eye contact** and ignore gestures.
- Report the vehicle's **license number**, description, location and direction; if they crash later, stop a safe distance away and wait for police.

Numbers and terms to memorize

Item	Value	Page
Hazard	Road condition or road user that is a <i>possible</i> danger	2-21
Emergency	The hazard is happening now	2-21
Move-over law	Slow down + move to a lane not next to a stopped emergency vehicle or work zone; if unsafe, slow down and pass carefully	2-21
 Move-over violation	Infraction, fine — CVC §21809	2-22
Work-zone warning to drivers behind	4-way flashers or brake lights	2-22
Slow-moving vehicle emblem	Red triangle, orange center	2-23

Item	Value	Page
CA SMV emblem vehicles	Not designed to exceed 25 mph (CVC §385.5)	2-23
Conflict	Must change speed, direction, or both to avoid hitting someone	2-24
LTCCS: outside / inside distraction	8% / 2% of large-truck crashes	2-24
Yearly U.S. distracted-driving deaths / injuries	about 5,500 / about 448,000	2-24
Brain energy used by phone talk (even hands-free)	39%	2-24
Hand-held use	Holding for a call; dialing more than one button ; reaching out of belted seat	2-25
Phone/texting disqualification	60 days = 2nd in 3 years ; 120 days = 3rd+ in 3 years	2-25, 2-26
Dialing	6x · 3.8 s · 306 ft	2-25
55 mph	80.7 ft/s	2-25
Texting	23.2x · 4.6 s · 371 ft	2-26
Texting civil penalty	Up to \$2,750 per violation	2-26
Emergency exception	Contacting law enforcement or emergency services only	2-25, 2-26
CA Electronic devices	Hands-free only	2-25
Reaction ½ s slower	Crashes double	2-26
Aggressive driving	Selfish, bold, pushy driving	2-27
Road rage	Intent to harm, or physical attack on a driver/vehicle	2-27

Exam traps

Tempting wrong idea	Handbook answer
Hands-free phones are safe.	hands-free is no less distracting; CA it is the only legal way, but still unsafe. "Free" sounds harmless.
The move-over law only means slow down.	slow down and change to a lane not next to it when safe; slowing alone is the fallback.
The posted ramp speed is safe for your truck	it may be too fast for large or heavy vehicles; slow down before the curve. A posted sign feels official.
An empty-looking box in the road is harmless	it may hold something heavy; avoid it early and smoothly.
A pedestrian looking at you will wait	stay alert — they may think they have the right of way.
SMV emblem vehicles go up to 35 or 45 mph	not designed to exceed 25 mph CA .
1st phone/texting conviction = disqualification, or 120 days for the 2nd	1st = civil penalty; 2nd in 3 years = 60 days; 3rd+ = 120 days.

Tempting wrong idea	Handbook answer
Mixing up the research numbers	dialing 6x / 3.8 s / 306 ft ; texting 23.2x / 4.6 s / 371 ft ; 39% is brain energy, not a risk multiplier.
Reaching for the phone is fine if you then use hands-free	unsafe reaching is itself a violation.
You may use a hand-held phone to tell dispatch you're late	the exception covers only law enforcement or emergency services.
Pass a distracted driver quickly	give room, keep following distance, pass with great care.
Aggressive driving = intent to harm	that is road rage . The two terms appear together, so they get swapped.
Speed up or hold your lane against an aggressive driver	get out of their way; avoid eye contact.

CV-02 · Combination air brakes and trailer ABS

Combination Vehicles · test weight **High** · handbook Section 6.2–6.3

You will be able to answer

- What the trailer hand valve is for — and the two things you must never use it for.
- What the tractor protection valve and the red trailer air supply knob do, and at what pressure the knob pops out.
- The difference between the service line and the emergency line, and what happens when each one breaks or leaks.
- How to connect glad hands, what happens if you cross the air lines, and why dummy couplers matter.
- Where the shut-off valves are, which ones must be closed, and why you chock a trailer without spring brakes.
- Which trailers must have ABS, where the trailer ABS lamp is, and how to brake with ABS.

Key points

Start here p. 6-5

- A combination adds parts that send air back to the trailer and control the **trailer brakes**, on top of the tractor's own air brakes.
- **psi** = pounds per square inch, the unit on your air pressure gauge.
- A **relay valve** on the trailer sends air from the trailer's own tank to its brakes.
- **Spring brakes** are held on by strong springs and released by air; on a trailer they work as the parking brakes.

Trailer hand valve p. 6-5

- The **trailer hand valve** (trolley valve or Johnson bar) puts on the trailer brakes only; use it **only to test** the trailer brakes.
- **Never use it while driving**, because braking only the trailer can make it skid; the foot brake works every brake at once.
- **Never use it to park**, because air can leak out and the brakes let go; park with the parking brakes, plus chocks if no spring brakes.
- To test, set the trailer brakes with the hand valve, then **tug gently** against them in a **low gear** to confirm they hold.

Tractor protection valve p. 6-5

- The **tractor protection valve** protects the tractor's own air supply if the trailer breaks away or gets a bad leak.
- You control it from the cab with the **trailer air supply** control.
- It **closes by itself** when air pressure drops to **20–45 psi**; one website says 25–40, but use 20–45 on the test.
- When it closes, no air leaves the tractor and the trailer emergency line is emptied, so the **trailer emergency brakes** come on.

Trailer air supply control p. 6-5

- On newer vehicles the trailer air supply control is a **red, 8-sided knob**; don't confuse it with the yellow diamond parking brake knob.
- **Push in** → air goes to the trailer; **pull out** → trailer air shuts off and the trailer emergency brakes come on.
- The knob **pops out by itself** when pressure drops into the **20-45 psi** range, as the tractor protection valve closes.
- Older vehicles may use a **lever** that may not work automatically, with "normal" and "emergency" positions.

Trailer air lines p. 6-5

- Every combination has **2** air lines linking each unit to the next: the service line and the emergency line.
- **Service line** (control or signal line) is **blue**; its pressure, set by foot brake or hand valve, goes to the relay valves.
- **Emergency line** (supply line) is **red**; it fills the trailer air tanks and controls the trailer emergency brakes.
- Emergency line loses air → trailer emergency brakes **come on** and the tractor protection valve **closes**.
- A major service line leak may go unnoticed **until you brake**; then tank pressure drops fast and may set the trailer emergency brakes.

Hose couplers — glad hands p. 6-6

- **Glad hands** join the tractor's service and emergency lines to the trailer's; clean the couplers and check the **rubber seals** for cracks.
- To connect, hold the couplers at 90 degrees, push the seals together, and turn to lock; match **blue = service, red = emergency**.
- **Crossed lines** on a trailer with spring brakes → spring brakes stay on; on an older trailer without them, you could drive away with **no trailer brakes**.
- **Dummy couplers** keep **water and dirt** out of unused lines.

Trailer air tanks p. 6-6

- Each trailer and each converter dolly carries **at least one** air tank.
- The tanks are filled through the **emergency (supply) line**, not the service line.
- **Relay valves** send tank air to the brakes; service line pressure tells them **how much**.
- **Drain every tank every day**, even with automatic drains, because water and oil buildup can stop the brakes working correctly.

Shut-off valves p. 6-7

- **Shut-off valves** are in the service and supply lines at the back of trailers that tow other trailers.
- They close off the air lines when no trailer is towed behind.
- Rule → all open, except at the **rear of the very last trailer**, which must be **closed** so air doesn't escape.

Trailer service, parking, and emergency brakes p. 6-7

- Newer trailers have **spring brakes**; they are not required on converter dollies and trailers built **before 1975**.
- Trailers without spring brakes have air-powered **emergency brakes** but **no parking brake**.
- Their emergency brakes come on if the emergency line loses pressure, you pull the air supply knob, or the trailer is unhooked.
- Those brakes hold only while tank air lasts, so **always chock the wheels** when parking a trailer without spring brakes.

Anti-lock brake systems — ABS p. 6-8

- Every trailer and converter dolly built on or after **March 1, 1998** must have ABS.
- Trailers have a **yellow** ABS malfunction lamp on the **left side**, front or rear corner; on older units, look underneath for the ECU and wheel speed sensor wires.
- ABS doesn't add braking power or necessarily shorten stops; it works only when a wheel is about to lock, helping you **keep control**.
- **Brake as you always have**, whether ABS is on the tractor, trailer, or both; with trailer-only ABS, ease off if a tractor jackknife begins.
- If ABS fails you still have **regular brakes**; get it serviced soon, and never drive faster or closer because of ABS.

Numbers and terms to memorize

Item	Value	Page
Trolley valve / Johnson bar	Other names for the trailer hand valve	6-5
Trailer hand valve use	Testing the trailer brakes ONLY	6-5
Tractor protection valve closes	Automatically at 20-45 psi	6-5
Red 8-sided knob	Trailer air supply control (in = air to trailer; out = air off + trailer emergency brakes)	6-5
"Normal" / "emergency"	Positions on older lever-type controls	6-5
Yellow diamond knob	Parking brakes (from Air Brakes)	5-4
Air lines per combination	2 — service and emergency	6-5
Service line	Also control line / signal line; blue ; goes to relay valves	6-5, 6-6
Emergency line	Also supply line; red ; fills trailer tanks + controls emergency brakes	6-5
Glad hands	Couplers joining tractor and trailer air lines; rubber seals	6-6
Glad hand angle	90 degrees	6-6
Dummy (dead-end) couplers	Keep water and dirt out of unused lines	6-6
Trailer tanks filled by	Emergency (supply) line	6-6
Drain trailer air tanks	Every day , even with automatic drains	6-6
Shut-off valves	Open everywhere except the rear of the last trailer (closed)	6-7
Not required to have spring brakes	Converter dollies and trailers built before 1975	6-7
ABS required	Trailers and dollies built on or after March 1, 1998	6-8
Trailer ABS lamp	Yellow, left side , front or rear corner	6-8
ECU	ABS computer; look under older units for it and wheel speed sensor wires	6-8

Exam traps

Tempting wrong idea	Handbook answer
Use the hand valve to park for a short time	Never park with it. The air can leak off and the brakes release. Use the parking brakes (and chocks if no spring brakes).
Put on the hand valve before the foot brake to prevent a jackknife	Don't. Braking the trailer alone can cause the skid. The hand valve is only for testing .
When the emergency line loses air, the tractor protection valve opens	It closes , and the trailer emergency brakes come on. "Opens" sounds like "lets air through to help".
A service line leak makes the trailer brakes come on right away	You may not notice it until you brake . That describes the emergency line, not the service line.
The trailer air tanks are filled by the service line	By the emergency (supply) line . The service line only sends the "how much" signal.
The "supply line" is another name for the service line	Supply line = emergency line. Service = control = signal.
Red 8-sided knob = parking brakes	Red 8-sided = trailer air supply . Yellow diamond = parking brakes.
Shut-off valves should all be open	All open except the back of the last trailer , which is closed .
Crossed air lines always lock the trailer brakes	On a trailer with spring brakes, yes — the brakes will not release. On an old trailer without spring brakes, you may drive away with no trailer brakes .
Emergency brakes on a trailer without spring brakes will hold it when parked	Only while tank air lasts. It leaks away — chock the wheels .
Automatic drains mean you never drain the tanks	Still open the drains daily to be sure.
ABS shortens your stopping distance / adds braking power	It does not necessarily shorten stopping and does not add power — it helps you keep control .
With ABS you should brake differently (pump or stab)	Brake as you always have , the same way whether ABS is on the tractor, trailer, or both.
Trailer ABS lamp is on the right side, or red	Yellow , on the left side.

CV-03 · Coupling and uncoupling

Combination Vehicles · test weight **High** · handbook Section 6.4–6.4.8

You will be able to answer

- The 16 coupling steps for a tractor-semitrailer, in the handbook's order, with the reason for each.
- Which step comes first, next, or just before another one — a common kind of test question.
- What goes wrong if the trailer is too high or too low, and how to prove the kingpin is really locked.
- How high the landing gear must go, and which clearances to check after coupling.
- The 10 uncoupling steps, including where to stop the tractor and why.
- The basic steps for pintle hook, drawbar and gooseneck hitches.

Key points

Key words first pp. 6-16, 6-6, 6-5

- The **fifth wheel** is the round coupling plate on the tractor; the **kingpin** points down from the trailer, and its middle part is the shank.
- The jaws close on the kingpin; the **locking lever** opens them, and a **safety latch** keeps that lever from opening by accident.
- **Landing gear** are the two front legs holding up an unhooked trailer, wound with a crank handle.
- Trailer air supply knob (red, 8-sided) → **in** = air to the trailer; **out** = trailer air off and trailer emergency brakes on.

Why the steps matter p. 6-9

- Getting coupling or uncoupling wrong is **very dangerous**.
- The handbook's steps are **general**, because rigs differ, so also learn the details of the truck you will drive.

Coupling a tractor-semitrailer — 16 steps pp. 6-9, 6-10

- Steps 1–6 → inspect **fifth wheel** (greased, **tilted down toward the rear**, jaws open), inspect area and chock, line up straight, back until it just touches, secure, check height.
- Trailer height → **low enough that the tractor lifts it slightly**; too low hits the trailer nose, too high may not couple correctly.
- Steps 7–9 before backing under → connect air lines, supply air and check for crossed lines with the **engine off**, then **pull out** the knob to lock trailer brakes.
- Steps 10–13 → back under in lowest reverse gear, tug test gently, secure and take the key, then inspect for **no space** between plates and jaws around the kingpin **shank**.
- Steps 14–16 → plug in the electrical cord, raise landing gear **all the way up** and secure the crank, check clearances, and finally **remove the chocks**.

Uncoupling a tractor-semitrailer — 10 steps p. 6-11

- Park on firm ground with the tractor **in line** with the trailer; lock trailer brakes, **back up gently** to ease jaw pressure, and set parking brakes while pushing.
- Chock the trailer wheels if it has **no spring brakes, or you are not sure**, because tank air can leak and let the trailer roll.
- Lower the landing gear; on a **loaded** trailer add a few turns in low gear, but **never lift** the trailer off the fifth wheel.
- Disconnect air lines onto dummy couplers, hang the electrical cable **plug down**, unlock the fifth wheel, keeping legs clear of the rear tractor wheels.
- Pull partly clear and stop with the **tractor frame still under the trailer**, secure, check the ground and landing gear, then pull clear.

CA Pintle hook, drawbar and gooseneck hitches pp. 6-12, 6-13, 6-14, 6-15

- **CA** These hitches are in the California handbook but rarely tested; know the **safety points** and roughly what each step does.
- **Pintle hook:** inspect the hook and mount; it is locked correctly when the lock handle sits **flush with the top of the latch**.
- Uncoupling a pintle hook → park on firm, level ground, block the tires, and check the ground can carry the tongue's weight **before** extending the jack.
- Ball-type drawbar: back the adjustment screw out at least 5 turns, torque lock screws to **350–400 Nm**, and set a **0.3–0.5 mm** gap.
- **Gooseneck:** latch the coupler clamp over a lubricated ball, and hook up **safety chains**, which every trailer must have by law; test every light.

Numbers and terms to memorize

Item	Value	Page
Coupling / uncoupling steps	16 / 10	6-9, 6-10, 6-11
First three coupling steps	Inspect fifth wheel → inspect area and chock → position tractor	6-9
Fifth wheel ready to couple	Tilted down toward rear; jaws open; safety handle on automatic lock; slider locked	6-9
Correct trailer height	Low enough that the tractor lifts it slightly	6-9
Too high / too low	May not couple correctly / tractor may hit and damage the nose	6-9
Before backing under	Air lines connected (7), air supplied and brakes checked (8), trailer brakes locked (9)	6-9, 6-10
Crossed-line check	Engine off; apply and release trailer brakes; listen; watch gauge	6-9
Backing under	Lowest reverse gear, slowly	6-10
Tug test	Landing gear slightly up; pull forward gently, trailer brakes locked	6-10
Gap between fifth-wheel plates	None	6-10
Jaws close around	Kingpin shank	6-10

Item	Value	Page
Landing gear for driving	All the way up ; crank handle secured	6-10
Clearances after raising gear	Tractor frame ↔ landing gear; tractor tires ↔ trailer nose	6-10
Last coupling step	Remove and store the chocks	6-10
Uncoupling Step 2	Lock trailer brakes → back up gently → set parking brakes while pushing	6-11
Chock when uncoupling	No spring brakes, or not sure	6-11
Loaded trailer	A few extra cranks in low gear; don't lift it off the fifth wheel	6-11
Electrical cable when unhooked	Plug down	6-11
First stop pulling out	Tractor frame still under trailer	6-11
Pintle hook locked	Lock handle flush with top of latch	6-12
Drawbar numbers	Adjustment screw out ≥5 turns ; lock screws 350–400 Nm ; clearance 0.3–0.5 mm	6-13, 6-14
Safety chains	Required by law on all trailers	6-15

Exam traps

Tempting wrong idea	Handbook answer
Position the tractor is step 1	inspect the fifth wheel first. The driving part feels like the start.
fifth wheel level	tilted down toward the rear . Some websites key "level."
trailer high enough that the tractor slides under without touching	low enough to be lifted slightly . Too high = may not couple.
too low = won't couple	too low = tractor hits the nose ; too high = may not couple. Easy to swap.
back under first, then hook up the air	Steps 7–9 (air, brake check, lock brakes) all come before Step 10.
leave the knob pushed in while backing under	pull it out to lock the trailer brakes.
keep the engine running for the crossed-line check	engine off so you can hear.
a small gap between the plates is normal	no space at all.
a good tug test means no need to look	you still get under and look at the back of the fifth wheel .
landing gear just off the pavement	all the way up .
uncoupling — pull forward to free the jaws	back up gently , then set the brakes while pushing.
crank a loaded trailer until it lifts off the fifth wheel	a few turns only; never lift it off.
drive the tractor fully out in one go	first stop with the frame still under the trailer .

CV-01 · Driving combination vehicles safely

Combination Vehicles · test weight **High** · handbook Section 6.1

You will be able to answer

- Why a loaded rig rolls over, and the two things that prevent a rollover.
- What "crack-the-whip" and "rearward amplification" mean, and which trailer is most likely to tip over.
- Why an empty rig or a bobtail tractor needs **MORE** distance to stop than a fully loaded rig.
- Which trailers get stuck on railroad crossings, and what to do if you are stuck on the tracks.
- How to spot and stop a trailer skid, and why the trailer hand valve is the wrong tool for it.
- What off-tracking is, how to make a tight turn, and which way to turn the wheel when backing a trailer.

Key points

Start here pp. 6-1, 6-2

- A **combination vehicle** (or rig) is a power unit pulling one or more trailers, such as a tractor-semitrailer, doubles, triples, or a straight truck with a trailer.
- A **bobtail** is a tractor driving with no semitrailer attached.
- A **jackknife** is when the rig folds at the hitch point, so the trailer and tractor form a "V" instead of a straight line.
- Combinations are usually heavier and longer than single vehicles, so they take **more skill** to drive.

Rollover risks p. 6-1

- Rollovers cause **more than half** of the deaths of truck drivers in crashes.
- A **fully loaded** rig is **10 times** more likely to roll over in a crash than an empty rig; some websites say 5, but answer 10 on the test.
- Two things prevent a rollover: keep cargo **as low as possible** and drive **slowly around turns**.
- Keep the load **centered** and spread out, because a load sitting to one side makes the trailer lean.
- Slow down on corners, on-ramps and off-ramps, and avoid **quick lane changes**, especially with a full load.

Steer gently pp. 6-1, 6-2

- **Crack-the-whip**: in a quick lane change the trailer swings out much harder than the tractor and can flip; in many crashes only the trailer turned over.
- **Rearward amplification** causes it; the swing grows toward the back, so the **last trailer** is the most likely to tip.
- Values: 1.0 = 5-axle tractor-semitrailer with 45-ft trailer (lowest); **2.0** = last trailer tips 2 times as easily as the tractor; triples = **3.5** (highest).
- **CA** Triple trailers are **not legal** in California, but the 3.5 number can still appear on the test.
- Following distance → at least **1 second per 10 feet** of rig length, plus **1 more second** when driving over 40 mph.

Brake early p. 6-2

- An **empty** combination takes **longer** to stop than a fully loaded one.
- Why: stiff springs and strong brakes give **poor traction**, so wheels lock easily and the trailer can swing out or the tractor jackknife.
- A **bobtail** tractor needs **more** distance to stop than a tractor-semitrailer at its maximum gross weight.
- Keep lots of space ahead and scan far down the road so you can **brake early** and never make a panic stop.

Railroad-highway crossings p. 6-2

- **Low-slung units** can hang up on raised crossings: lowboy, car carrier, moving van, and possum-belly livestock trailer.
- A **single-axle tractor** pulling a long trailer with landing gear set for a **tandem-axle tractor** can also get stuck.
- Stuck on the tracks → first **leave the vehicle** and move away from the tracks.
- Then find emergency contact info on signposts or the signal housing, call **9-1-1**, and describe the location, especially the crossing's **DOT number** if posted.

Prevent trailer skids p. 6-3

- When the trailer wheels **lock up**, the trailer swings around to the side, a **trailer jackknife**, more often when empty or lightly loaded.
- You see a trailer skid first and best in your **mirrors**, so glance at them whenever you brake hard.
- To stop the skid, **release the brakes** so the tires grip again and the trailer follows the tractor and straightens out.
- Never use the **trailer hand valve** to straighten the rig, because locked trailer brakes caused the skid; that is always the test answer.

Turn wide pp. 6-3, 6-4

- **Off-tracking** ("cheating"): in a turn the rear wheels cut toward the inside, so the rig covers a path wider than itself.
- Longer vehicles off-track more, and the **rear wheels of the last trailer** cut in the most.
- Steer the front wide enough to clear the curb, but keep the rear **close to the curb** so no one can pass on your right.
- If you must use another lane, turn wide **as you finish the turn (button hook)**; never swing left first (jug handle), which invites passing on the right.

Backing with a trailer p. 6-4

- Backing a trailer, the steering is reversed: turn the wheel **opposite** to where you want the trailer to go (top of wheel left → trailer goes right).
- Set up to back in a **straight line**; only if a curve cannot be avoided, curve toward the **driver's side**, which you can see directly.
- Correct drift immediately by turning the **top of the wheel toward the drift**.
- Get out and look at your path, use mirrors on both sides, **back slowly**, and use pull-ups to reposition.

Numbers and terms to memorize

Item	Value	Page
Truck-driver crash deaths caused by rollovers	More than half	6-1
Loaded vs empty rollover risk	Fully loaded rig is 10× more likely to roll over	6-1
Two rollover preventers	Cargo as low as possible; slow in turns	6-1
Crack-the-whip	Trailer swings hard in a quick lane change and can flip	6-1
Rearward amplification 1.0	5-axle tractor-semitrailer, 45-ft trailer (lowest value in the chart)	6-2
Rearward amplification 2.0	Rear trailer 2× as likely to turn over as the tractor	6-1, 6-2
Rearward amplification 3.5	Triples (27-ft trailers) — highest in chart	6-1, 6-2
Following distance	1 second per 10 ft of length, +1 second over 40 mph	6-1
Empty vs loaded stopping	Empty takes longer	6-2
Bobtail	Tractor with no semitrailer; takes longer to stop than a max-loaded tractor-semitrailer	6-2
Low-slung units	Lowboy, car carrier, moving van, possum-belly livestock trailer	6-2
DOT number	Crossing ID to give when you call for help	6-2
Trailer jackknife	Trailer wheels lock and the trailer swings around	6-3
First sign of a trailer skid	Seen in the mirrors	6-3
Off-tracking / "cheating"	Rear wheels take a different (inside) path than front wheels	6-3
Button hook / jug handle	Correct turn / incorrect turn	6-4
Backing a trailer	Turn the wheel opposite to where you want the trailer to go	6-4
 Triples	Not legal in California	1-5

Exam traps

Tempting wrong idea	Handbook answer
An empty rig stops faster because it is lighter.	An empty combination takes longer to stop. Lighter feels easier, but with stiff springs and strong brakes the tires grip poorly and the wheels lock.
A bobtail tractor stops more quickly than a fully loaded tractor-semitrailer.	The bobtail takes longer to stop. A small vehicle seems easier to stop.
Empty rigs roll over more easily.	Fully loaded rigs are 10× more likely to roll over. Do not mix this up with the stopping rule, where empty is worse.
In a sudden turn with doubles, the first trailer is most likely to tip	The rear (last) trailer. The first trailer is closer to the driver and feels more "involved".
Use the trailer hand valve to straighten a skidding trailer	Release the brakes. The trailer brakes caused the skid. Some drivers say they do it — the test answer is still no.

Tempting wrong idea	Handbook answer
You first notice a trailer skid by feel or sound	In your mirrors .
Swing wide to the left before a right turn	Turn wide as you complete the turn (button hook). Swinging left first invites drivers to pass on your right.
Keep the rear of the rig away from the curb in a turn to be safe	Keep the rear close to the curb so no one passes on the right — just not so close that you run over it.
When backing a trailer, turn the wheel the way you want the trailer to go	Turn it the opposite way. That is how a car works, not a trailer.
The best way to back a trailer is on a curve to the driver's side	Set up to back in a straight line . The driver's side is only for when a curve cannot be avoided.
To correct drift, turn the wheel away from the drift	Turn the top of the wheel toward the drift . It sounds backward because trailer steering is reversed.
Add the extra second of following distance at 40 mph	Only when going over 40 mph.

CV-04 · Inspecting a combination vehicle

Combination Vehicles · test weight **High** · handbook Section 6.5–6.5.2

You will be able to answer

- Which extra parts you check on a combination during the walk-around: both fifth-wheel plates, the kingpin, the air and electric lines, a sliding fifth wheel, and the landing gear.
- Where the shut-off valves must be open, and where they must be closed.
- How to test that air reaches every trailer.
- How to test the tractor protection valve, and the pressure range at which it should close.
- How to test the trailer emergency brakes and the trailer service brakes.

Key points

Key words first pp. 6-7, 6-5

- The **lower fifth wheel** is the coupling plate on the tractor; the **upper fifth wheel** (glide plate) is the flat plate under the trailer's front.
- The **kingpin** has a base, a **shank**, and a head from top to bottom.
- **Shut-off valves** sit in the service and supply lines at the rear of any trailer that pulls another trailer.
- To **charge the trailer air system**, let air pressure rise to normal, then push the red air supply knob in.

Use the same 7-step method p. 6-16

- Inspect a combination with the **same 7-step procedure** you learned in Section 2.
- A combination has **more** of the usual parts to check, such as tires, wheels, lights, and reflectors, plus some **new** parts.

Extra checks during the walk-around p. 6-16

- Lower fifth wheel → mounted firmly, greased, **no visible gap** between plates, and jaws closed around the kingpin **shank**, never its head.
- Release arm seated with the **safety latch** engaged; upper fifth wheel's glide plate firmly mounted and kingpin undamaged.
- Electrical cord secured; air lines coupled, not leaking, with **enough slack for turns** so they are not pulled apart.
- Sliding fifth wheel → greased, **every locking pin** there and locked, not so far forward that the tractor frame hits the landing gear or cab hits the trailer.
- Landing gear raised **all the way up**, undamaged, with the crank handle secured, because partly lowered gear can catch on railroad tracks.

Combination vehicle brake check p. 6-17

- Air-flow test → push knob in, apply trailer hand brake, then open the last trailer's **emergency** then **service** valves; air must come from both, then close them.
- Shut-off valves stay **open**, except at the **rear of the last trailer**, which stay closed; no air at the rear means check they are open.
- Tractor protection valve test → **engine off**, pump the brake pedal; the knob should pop out at the maker's pressure, **usually 20–45 psi** (not 25–40).
- Emergency brakes → **pull out** the knob and tug gently; they should hold. Service brakes → roll slowly and apply the **hand valve**; feel them grab.
- Leakage limits per minute, engine off → brakes released: **3 psi** (2 vehicles) or 5 psi (3 or more); foot brake held: **4 psi** or 6 psi.

Numbers and terms to memorize

Item	Value	Page
Inspection method	Same 7-step procedure as Section 2, plus extra checks	6-16
Space between upper and lower fifth wheel	None visible	6-16
Locking jaws	Around the shank , not the head, of the kingpin	6-16
Kingpin parts	Base, shank, head	6-16
Upper fifth wheel	Glide plate secure to trailer frame; kingpin not damaged	6-16
Air lines	Connected, no leaks, enough slack for turns	6-16
Sliding fifth wheel	Greased; all locking pins present and locked; not too far forward	6-16
Landing gear	Fully raised; crank handle in place and secured	6-16
Shut-off valves	Open , except closed at the rear of the last trailer	6-7
Air-flow test: air to emergency lines	Push in red trailer air supply knob	6-17
Air-flow test: air to service line	Apply the trailer hand brake	6-17
Air-flow test: valve you open first	Emergency line shut-off valve at rear of last trailer	6-17
No air at the rear	Check shut-off valves on trailers and dollies are OPEN	6-17
TPV test	Engine off, pump the brake pedal	6-17
TPV / knob pops out at	Maker's range, usually 20–45 psi	6-17
Trailer emergency brake test	Pull out knob, tug gently	6-17
Trailer service brake test	Move slowly, apply the hand valve (trolley valve)	6-17
Normal-driving trailer braking	Foot pedal (all wheels)	6-17
Static leakage, 2-vehicle combination	3 psi in 1 minute	5-10
Static leakage, 3+ vehicles	5 psi in 1 minute	5-10
Applied leakage (foot brake held), 2 / 3+ vehicles	4 psi / 6 psi in 1 minute	5-8

Exam traps

Tempting wrong idea	Handbook answer
all shut-off valves should be open	all open except the ones at the rear of the last trailer, which are closed. "All open" sounds safest because air must reach every trailer.
on doubles, the valves on the back of the first trailer are closed	only the last trailer's rear valves are closed; the first trailer tows the second one, so its rear valves stay open.
the service line has air as soon as you push in the knob	the knob fills the emergency (supply) line; the service line has air only when the trailer hand brake or brake pedal is applied.
test the tractor protection valve with the engine running	engine off , then pump the pedal.
the knob should pop out at 25–40 psi (one website's number)	usually 20–45 psi (the manufacturer's range).
to test the trailer emergency brakes, push in the knob and tug	pull out the knob (or set "emergency"), then tug gently.
test the trailer service brakes with the foot pedal	test with the hand valve (trolley valve); use the foot pedal for normal driving.
the jaws should grip the head of the kingpin	the shank .
air lines should be pulled tight so they cannot drag	secured with enough slack for turns .
slide the fifth wheel as far forward as possible	not so far forward that the tractor frame hits the landing gear or the cab hits the trailer in turns.

Practice tests

Cover the answers, circle a, b or c, then check the key at the end. Same format as the real tests: 3 choices, one correct.

General Knowledge practice test (50 questions)

Pass: 40 of 50

1. You must back into a loading dock. The easiest setup puts the dock on your right side, but you could drive around the block and approach so the dock is on your left. What should you do?

- a. Back toward the right; it is quicker
- b. Drive around the block so you can back toward the driver's side
- c. Back toward the right, but faster, so you spend less time in the blind area

2. Strong winds make it hard to stay in your lane. This problem is usually worse:

- a. For heavily loaded trucks
- b. For lighter vehicles, especially coming out of tunnels
- c. Only on bridges

3. Your engine is on fire. What should you do?

- a. Open the hood fully so you can reach the fire
- b. Keep the engine running so the fan can blow the fire out
- c. Turn off the engine and, if possible, keep the hood closed while spraying through the louvers, radiator, or underside

4. You are driving at night and catch up to a car in your lane about 400 feet ahead. What should you do?

- a. Keep your high beams on because the car is going your way
- b. Flash your high beams so the driver knows you are there
- c. Switch to low beams

5. You need a CDL to drive a single vehicle with a GVWR of:

- a. 26,000 lb or more
- b. 26,001 lb or more
- c. 10,001 lb or more

6. While backing a trailer, you see it start to drift off the correct path. You should:

- a. Turn the steering wheel in the direction of the drift
- b. Turn the steering wheel away from the drift
- c. Keep backing and correct it once you stop

7. You are driving an empty truck on a dry road. Compared with the same truck fully loaded, it will:

- a. Stop in a shorter distance, because it is lighter
- b. Need a longer distance to stop, because it has less traction
- c. Stop in the same distance, because the brakes are the same

8. CA Outside a business or residential district, a truck ahead is climbing a grade at 15 mph. You may pass it only if you:

- a. Pass at least 5 mph faster and finish within ½ mile
- b. Pass at least 10 mph faster and finish within ¼ mile
- c. Pass at least 20 mph faster and finish within 1 mile

9. According to the California handbook, a first violation of an out-of-service order costs your CDL for at least:

- a. 90 days
- b. 180 days
- c. 1 year

10. A train has passed a gated crossing. The gates are rising, but the lights are still flashing. You should:

- a. Go as soon as the train clears the road
- b. Drive carefully around the end of the gate
- c. Stay stopped until the gates are up and the lights stop flashing

11. There is a fire in the cargo of your van trailer. What should you do with the doors?

- a. Open them to find the source of the fire
- b. Keep them shut
- c. Open them only if the cargo is HazMat

12. A car behind you clearly wants to pass. You can see the road ahead is clear. You should:

- a. Not signal them — don't direct traffic
- b. Wave them past with your hand
- c. Turn on your right turn signal to tell them it is safe

13. You are given a sealed trailer that you cannot open. What must you still check?

- a. That the gross weight and axle weight limits are not exceeded
- b. That each item inside has at least 2 tie-downs
- c. Nothing, because the shipper is responsible for sealed loads

14. Which statement is TRUE?

- a. Use low beams whenever you can
- b. Use high beams whenever it is safe and legal
- c. Use high beams only where there are street lights

15. Your truck catches fire. Where should you stop?

- a. At the nearest service station
- b. In an open area, clear of buildings, brush, trees, and other vehicles
- c. Under an overpass, out of the wind

16. While turning right, you must cross into the oncoming lane and a car is coming toward you. You should:

- a. Back up to give the car more room
- b. Give the car room to pass, or stop — but do not back up
- c. Speed up to finish the turn before the car arrives

17. This morning you cleared a low bridge with a loaded van. Now you return on the same road with the van empty. You should:

- a. Expect to clear it again, because the bridge has not changed
- b. Remember that an empty van sits higher, and check your clearance
- c. Drive faster under it so the trailer does not bounce

18. CA You must turn on your headlights whenever you cannot clearly see a person or vehicle within:

- a. 350 feet
- b. 1,000 feet
- c. 500 feet

19. Your brakes fail on a long downgrade and you see a sign for a runaway truck ramp ahead. What should you do?

- a. Use the escape ramp
- b. Pass the ramp and keep trying to downshift
- c. Keep going and wait for the brakes to cool and recover

20. You are forced back onto the road before you can stop. What is the correct technique?

- a. Edge back onto the pavement slowly and gradually
- b. Wait until all wheels are on the pavement before you countersteer
- c. Turn sharply back onto the road and countersteer when both front tires are on the pavement

21. Your truck is heavily loaded. You approach an off-ramp that goes downhill and curves, with a posted ramp speed. What should you do?

- a. Keep your speed and brake firmly in the curve if needed
- b. Slow down enough before the curve, even below the posted speed if needed
- c. Drive at the posted speed, because ramp signs are set for trucks

22. You drive a 40-foot truck at 35 mph. What following distance does the handbook recommend?

- a. 3 seconds
- b. 4 seconds
- c. 5 seconds

23. A second railroad-highway grade crossing violation within 3 years costs your CDL for at least:

- a. 60 days
- b. 120 days
- c. 1 year

24. CA At night, a load extends 5 feet beyond the rear of your vehicle. Besides the flag, what lights are required at the end of the load?

- a. One amber light visible 300 feet to the front and rear
- b. 2 lit red lights visible 500 feet to the side and rear
- c. 2 red reflectors visible 200 feet to the rear

25. You get a second serious traffic violation in a CMV within 3 years. You lose your CDL for at least:

- a. 60 days
- b. 120 days
- c. 1 year

26. In the vehicle inspection section, air pressure should build from 50 to 90 psi within:

- a. 45 seconds
- b. 2 minutes
- c. 3 minutes

27. During stab braking, after you release the brakes it can take up to how long for the wheels to start rolling?

- a. 1 second
- b. 3 seconds
- c. 5 seconds

28. You are moving an oversize load on a special permit. The permit route would make you break a local city traffic rule. What may you do?

- a. Detour on nonresidential streets and return to the route as soon as possible
- b. Take any shorter route you like
- c. Continue, because the permit overrides local rules

29. According to the handbook, a HazMat placard must be at least how big?

- a. 9.84 inches (250 mm) square
- b. 10¾ inches square
- c. 12 inches square

30. Under California intrastate rules (not farm products), you may not drive after being on duty for how many hours in any 8 days in a row?

- a. 70 hours
- b. 112 hours
- c. 80 hours

31. You passed the Air Brakes knowledge test, but you take the skills test in a vehicle with air-over-hydraulic brakes. Which restriction will you get?

- a. Z – no full air-brake CMV
- b. L – no air-brake CMV
- c. None, because you passed the Air Brakes test

32. CA Which is NOT one of the basic negligent-operator point counts?

- a. 4 points in 12 months
- b. 8 points in 36 months
- c. 6 points in 12 months

33. Which is a sign that ice has formed on the road?

- a. Other vehicles' tires are throwing up no spray
- b. Other vehicles' tires are throwing up a lot of spray
- c. The road surface looks dry and dusty

34. You start the engine. The ABS light on the dash comes on and stays on. This means:

- a. It is normal; the light goes off once you drive
- b. The trailer ABS is being tested
- c. The ABS is not working properly

35. On a California freeway posted 65 mph, which of these vehicles is NOT limited to 55 mph?

- a. A 2-axle truck that is not towing anything
- b. A school bus carrying pupils
- c. A vehicle carrying explosives

36. Your "safe" speed on a downgrade is 45 mph. Using the proper braking technique, you should:

- a. Wait until you reach 50 mph, then brake down to 35 mph
- b. Wait until you reach 45 mph, then brake down to about 40 mph and release
- c. Keep light, steady pressure on the brakes all the way down

37. Compared with driving at 20 mph, driving at 60 mph makes the impact and braking distance:

- a. 9 times greater
- b. 3 times greater
- c. 6 times greater

38. There is no white stop line at a crossing. A vehicle that is required to stop should stop:

- a. No closer than 15 feet and no farther than 50 feet from the nearest rail
- b. No closer than 10 feet and no farther than 40 feet from the nearest rail
- c. No closer than 20 feet and no farther than 100 feet from the nearest rail

39. A cardboard box lies in your lane well ahead. What is the best approach?

- a. Drive over it, since boxes on the road are usually empty
- b. Wait until you are close, then swerve around it quickly
- c. Treat it as possibly heavy and steer around it early, without sudden moves

40. You are approaching an intersection. The signal ahead has stayed green for quite a while. You should:

- a. Begin slowing and prepare to stop
- b. Speed up to get through before it changes
- c. Keep your speed, because green means go

41. The road is wet, and on a dry road you would drive 55 mph. The handbook says to slow to about:

- a. 45 mph
- b. 35 mph
- c. 25 mph

42. How many red reflective triangles must you carry?

- a. 2
- b. 3
- c. 6

43. Your front wheels are skidding and the truck keeps going straight even though you turn the wheel. What is the only way to stop this skid?

- a. Turn the steering wheel more sharply
- b. Let the vehicle slow down
- c. Brake as hard as you can

44. After you start the engine, oil pressure should:

- a. Come up to normal within seconds
- b. Rise slowly over about 3 minutes
- c. Stay low until the engine is warm

45. The loaded weight that the manufacturer specifies for a single vehicle is called the:

- a. Gross vehicle weight (GVW)
- b. Gross combination weight rating (GCWR)
- c. Gross vehicle weight rating (GVWR)

46. How can brakes that are out of adjustment lead to brake fade?

- a. They lock the wheels every time you stop
- b. They only affect the parking brake
- c. They stop doing their share, so the other brakes overheat and fade

47. Compared with a flat mirror, things seen in a convex mirror appear:

- a. Larger and closer than they really are
- b. The same size, just in a wider view
- c. Smaller and farther away than they really are

48. You had to stop on the shoulder of a highway. You must put out your warning devices within:

- a. 5 minutes
- b. 10 minutes
- c. 30 minutes

49. You are 19 and hold a Class A CDL. Which job can you legally do?

- a. Hauling tomatoes grown in Fresno County to a Los Angeles market, for pay
- b. Hauling freight from Sacramento to Reno, Nevada
- c. Hauling a placarded HazMat load from Oakland to San Jose

50. CA Your load sticks out 4 feet past the rear of your truck. In daylight you must put at the end of the load:

- a. A solid red or fluorescent orange flag, 12 inches square or larger
- b. An amber light visible for 300 feet
- c. A red or orange flag at least 18 inches square

Combination practice test (20 questions)

Pass: 16 of 20

1. What is the main job of the tractor protection valve?

- a. To keep the tractor's wheels from locking in hard braking
- b. To keep air in the tractor brake system if the trailer breaks away or leaks badly
- c. To send extra air to the trailer brakes during an emergency stop

2. Where is the ABS malfunction lamp on a trailer?

- a. A yellow lamp on the left side, at the front or rear corner
- b. A red lamp on the right rear corner
- c. A yellow lamp on the right side, near the landing gear

3. To test the trailer emergency brakes, you charge the system, check that the trailer rolls freely, and stop. What next?

- a. Pull out the trailer air supply control and pull gently against the trailer
- b. Push in the trailer air supply control and back against the trailer
- c. Apply the trailer hand valve and drive forward

4. The air lines between the tractor and trailer should be:

- a. Pulled tight so they cannot drag
- b. Stored on the dummy couplers while driving
- c. Secured with enough slack for turns

5. On the static leakage test (engine off, brakes released), what is the most air pressure a 2-vehicle combination with air brakes on both vehicles may lose in 1 minute?

- a. 2 psi
- b. 3 psi
- c. 5 psi

6. Compared with an empty rig, a fully loaded rig is how much more likely to roll over in a crash?

- a. 3.5 times
- b. 5 times
- c. 10 times

7. You have backed until the fifth wheel just touches the trailer, secured the tractor, and checked the trailer height. What is the next step?

- a. Connect the air lines to the trailer
- b. Back under the trailer
- c. Raise the landing gear

8. What is the FIRST step in coupling a tractor to a semitrailer?

- a. Position the tractor directly in front of the trailer
- b. Inspect the fifth wheel
- c. Connect the emergency air line

9. You are pulling doubles and must turn the wheel suddenly to avoid a crash. Which part of the rig is most likely to turn over?

- a. The tractor
- b. The first trailer
- c. The rear trailer

10. You are driving a bobtail tractor. Compared with a tractor-semitrailer at its maximum gross weight, your bobtail:

- a. takes longer to stop
- b. stops in a shorter distance
- c. stops in about the same distance

11. The trailer hand valve should be used:

- a. only to test the trailer brakes
- b. to hold the trailer when you park for a short time
- c. to straighten out a trailer that starts to skid

12. You are uncoupling. You have locked the trailer brakes. What should you do next?

- a. Pull forward to take the pressure off the kingpin
- b. Pull the fifth-wheel release handle
- c. Back up gently to ease the pressure on the locking jaws

13. You have unlocked the fifth wheel and start to pull away. Where should you stop first?

- a. With the tractor frame still under the trailer
- b. When the tractor is completely clear of the trailer
- c. Before the fifth wheel comes out from under the trailer

14. Just before you back under the trailer, you should:

- a. Push in the trailer air supply knob to release the trailer brakes
- b. Pull out the trailer air supply knob to lock the trailer brakes
- c. Remove the trailer wheel chocks

15. You brake hard and see the trailer starting to swing out. What should you do?

- a. Apply the trailer hand valve to straighten the rig
- b. Press the brake pedal harder
- c. Release the brakes so the trailer wheels grip again

16. Which statement about ABS is TRUE?

- a. It always shortens your stopping distance
- b. It helps you keep control during hard braking
- c. It increases your normal braking power

17. What is the earliest and best way to know your trailer has started to skid?

- a. You see it in your mirrors
- b. You feel it through the steering wheel
- c. You hear the trailer tires squeal

18. As the pressure drops during the tractor protection valve test, the trailer air supply knob should pop out, usually within:

- a. 20 to 45 psi
- b. 50 to 90 psi
- c. 85 to 100 psi

19. On a set of doubles, which shut-off valves should be closed?

- a. All of them, until you are ready to drive
- b. Only the ones at the rear of the last trailer
- c. The ones at the rear of the first trailer

20. You are pulling doubles. How should the shut-off valves be set?

- a. Open everywhere except the rear of the last trailer, where they are closed
- b. Open on every trailer, including the back of the last one
- c. Closed on all trailers except at the back of the last trailer

Answer key with explanations

General Knowledge

Q	Why
1. b	Driver-side backing is worth going around the block for; right-side backing is very dangerous. p. 2-10 · GK-05
2. b	Wind pushes light vehicles more; tunnel exits are a known spot. p. 2-20 · GK-08
3. c	Turn the engine off; don't open the hood if you can avoid it; spray through louvers, radiator, or from below. p. 2-45 · GK-13
4. c	Dim when following another vehicle within 500 ft. p. 2-31 · GK-10
5. b	The single-vehicle CDL line is 26,001 lb or more. p. 1-1 · GK-01
6. a	Fix drift at once by steering toward it. p. 2-10 · GK-05
7. b	Empty trucks have less traction and need longer stopping distances p. 2-16 · GK-07
8. b	CVC §21758: at least 10 mph faster, finished within ¼ mile p. 2-18 · GK-07
9. a	The handbook says 90 days (180 days is the federal figure). p. 1-14 · GK-02
10. c	Stay stopped until the gates are fully up and the lights have stopped flashing. p. 2-36 · GK-11
11. b	Opening the doors feeds the fire oxygen; keep them shut, especially with HazMat. p. 2-45 · GK-13
12. a	Don't direct traffic; it could cause a crash and you could be blamed. p. 2-14 · GK-06
13. a	For sealed loads, check gross weight and axle weight limits. p. 3-3 · GK-14
14. b	Use high beams whenever it is safe and legal; always using low beams cuts your view. p. 2-31 · GK-10
15. b	Stop in an open area away from anything that can burn; never a service station. p. 2-45 · GK-13
16. b	Backing up could hit someone behind you. p. 2-20 · GK-08
17. b	Without cargo weight the springs lift the van higher. p. 2-20 · GK-08
18. b	CA Headlights are required when you can't clearly see a person or vehicle at 1,000 ft. p. 2-31 · GK-10
19. a	Any driver who loses brakes on a downgrade should use an available escape ramp. p. 2-40 · GK-12
20. c	Turn sharply back on, then countersteer as soon as both front tires are on the pavement. p. 2-39 · GK-12
21. b	Posted ramp speeds may be too fast for large or heavy vehicles; slow down before the curve, not in it. p. 2-22 · GK-09
22. b	$40 \text{ ft} \div 10 = 4$ seconds; below 40 mph there is no extra second. p. 2-19 · GK-08
23. b	2nd railroad violation within 3 years = at least 120 days. p. 1-14 · GK-02
24. b	In darkness, add 2 lit red lights visible 500 feet to the side and rear. p. 3-5 · GK-14
25. a	2 serious violations within 3 years = at least 60 days. p. 1-14 · GK-02
26. c	50 to 90 psi within 3 minutes. p. 2-5 · GK-04
27. a	After release, the wheels may need up to 1 second to start rolling again. p. 2-39 · GK-12

Q	Why
28. a	The only allowed detour is on nonresidential streets to avoid breaking a local city rule, returning as soon as possible. p. 1-24 · GK-03
29. a	The handbook requires at least 9.84 in (250 mm) square. p. 2-48 · GK-13
30. c	California's limit is 80 hours in 8 days; 112 is only for farm products; 70 is federal. p. 1-27 · GK-03
31. a	Testing with air-over-hydraulic brakes gives restriction Z. p. 1-7 · GK-01
32. c	The counts are 4 in 12 months, 6 in 24, 8 in 36. p. 1-15 · GK-02
33. a	A lack of spray from other vehicles means ice has formed. p. 2-33 · GK-10
34. c	The light should come on, then go off; staying on = ABS fault. p. 2-4 · GK-04
35. a	A 2-axle truck that is not towing is not on the §22406 list, so it may go the posted 65. p. 1-25 · GK-03
36. b	Brake at the safe speed down to about 5 mph below it, then release. p. 2-38 · GK-11
37. a	Tripling speed (20 → 60) makes impact and braking distance 9 times greater p. 2-16 · GK-07
38. a	With no stop line, stop 15–50 ft from the nearest rail. p. 2-35 · GK-11
39. c	Boxes may hold heavy material; see objects early and avoid them without sudden moves. p. 2-22 · GK-09
40. a	A "stale" green is likely to change before you get there, so begin slowing now. p. 2-12 · GK-06
41. b	Cut speed by about one-third on wet roads, e.g. 55 → about 35 mph p. 2-16 · GK-07
42. b	3 triangles; 6 is the number of fusees. p. 2-4 · GK-04
43. b	In a front-wheel skid, only slowing down stops the skid. p. 2-43 · GK-12
44. a	Oil pressure should be normal within seconds. p. 2-5 · GK-04
45. c	GVWR is the manufacturer's loaded-weight value for a single vehicle. p. 3-1 · GK-14
46. c	Out-of-adjustment brakes quit first, and the rest overheat and fade. p. 2-37 · GK-11
47. c	Convex mirrors make things look smaller and farther away than they are. p. 2-13 · GK-06
48. b	Warning devices must be out within 10 minutes. p. 2-14 · GK-06
49. a	At 18 you may drive for hire inside California; interstate and any HazMat need age 21. p. 1-2 · GK-01
50. a	A load 4 feet or more past the rear needs a flag of solid red or fluorescent orange, at least 12 inches square. p. 3-5 · GK-14

Combination

Q	Why
1. b	It keeps air in the tractor if the trailer breaks away or develops a bad leak. p. 6-5 · CV-02
2. a	Trailer ABS lamps are yellow, on the left side, front or rear corner. p. 6-8 · CV-02
3. a	Pull out the knob (or set "emergency") and pull gently to check the brakes hold. p. 6-17 · CV-04
4. c	Air lines need enough slack for turns. p. 6-16 · CV-04
5. b	3 psi for 2 vehicles (5 psi for 3 or more). p. 5-10 · CV-04
6. c	The handbook says a full load makes a rollover 10 times as likely as an empty rig. p. 6-1 · CV-01

Q	Why
7. a	Step 7 (air lines) follows Step 6 (height); backing under is Step 10. p. 6-9 · CV-03
8. b	Inspecting the fifth wheel is Step 1; positioning the tractor is Step 3. p. 6-9 · CV-03
9. c	Rearward amplification makes the last trailer swing the most. p. 6-1 · CV-01
10. a	A bobtail takes longer to stop than a tractor-semitrailer at maximum gross weight. p. 6-2 · CV-01
11. a	The hand valve is for testing the trailer brakes only. p. 6-5 · CV-02
12. c	Back up gently, then set the parking brakes while still pushing. p. 6-11 · CV-03
13. a	The frame catches the trailer if the landing gear collapses or sinks. p. 6-11 · CV-03
14. b	Step 9 locks the trailer brakes before Step 10. p. 6-10 · CV-03
15. c	Release the brakes; the trailer brakes caused the skid, so the hand valve makes it worse. p. 6-3 · CV-01
16. b	ABS helps you keep control; it does not necessarily shorten stops or add power. p. 6-8 · CV-02
17. a	The mirrors are the earliest and best way to see a trailer skid. p. 6-3 · CV-01
18. a	Usually 20–45 psi (the manufacturer's range). p. 6-17 · CV-04
19. b	All shut-off valves are open except those at the rear of the last trailer. p. 6-7 · CV-04
20. a	Only the valves at the rear of the last trailer are closed; all others stay open. p. 6-7 · CV-02

Plain-English glossary

"Progressive" shifting (on some newer vehicles) — the shift point climbs **higher** with each higher gear. Find out what fits the vehicle you drive.

BAC (blood alcohol concentration) — how much alcohol is in your blood, measured in **milligrams of alcohol per 100 milliliters of blood**. A BAC of **.10** means **1/10 of 1%** (1/1000) of your blood is alcohol.

Caltrans — California Department of Transportation (runs state highways).

CARB — California Air Resources Board. Its rules cover every commercial diesel vehicle that runs in California — **even ones based in another state or country**. CARB can inspect for too much smoke, tampering, and fleet-rule compliance, and can write citations with large fines. Inspections happen at **border crossings, CHP weigh stations, fleet facilities and random roadside spots**.

CDL — commercial driver license.

Center of gravity — the balance point of all the weight. The higher you stack cargo, the higher this point sits above the road, and the easier the truck tips over.

Charge the trailer air system — let the air pressure rise to normal, then push the air supply knob in.

Chocks — wedges set against the tires so the wheels can't roll.

CHP — California Highway Patrol.

CLP — commercial learner's permit, the learning stage before the CDL (it used to be called the "commercial instruction permit").

CMV — commercial motor vehicle (a truck or bus that needs a CDL).

Combination vehicle (also called a rig) — a power unit pulling one or more trailers. Examples: tractor-semitrailer, doubles, triples, a straight truck pulling a trailer.

Combined single limit (CSL) — one total amount that covers injury and property damage together.

Conflict — to avoid hitting someone, you have to change your **speed, your direction, or both**.

Conviction — a court finds you guilty.

CVC — California Vehicle Code (the state traffic law).

Disqualified ("lose your CDL") — not allowed to drive a CMV for a set time, or for life.

Drive axles — the axles the engine turns.

Driver distraction — anything that pulls your attention off the job of driving. It puts you, passengers, other drivers and pedestrians at risk of injury, death or property damage.

Dummy couplers — places behind the cab to hook the air hoses when no trailer is connected. They keep **water and dirt** out of the couplers and air lines.

Electronic device — cell phone, PDA, pager, computer, or anything used to write, send, receive or read text.

Emergency — the danger is happening *now*; you must act at once.

Emergency line — the **red** air line (supply line). It fills the trailer air tanks and controls the trailer emergency brakes.

Endorsement — an extra letter that allows a special kind of driving.

Fifth wheel — the round coupling plate on the back of the tractor (the **lower** fifth wheel). The flat plate under the trailer's nose is the **upper** fifth wheel.

Financial responsibility (FR) — proof you can pay for injuries and damage you cause (usually insurance).

GCWR (gross combination weight rating) — the loaded weight the manufacturer sets for a **combination** (power unit plus towed unit). If there is no maker's figure, add the power unit's GVWR to the weight of the towed unit and its load.

Glad hands — couplers joining the tractor air lines to the trailer: red = **emergency (supply)** line, blue = **service** line

glide plate — the flat plate under the front of the trailer.

GVWR (gross vehicle weight rating) — the loaded weight the manufacturer sets for **one** vehicle.

Hazard — any road condition, or any other road user (driver, motorcyclist, bicyclist, pedestrian), that *could* become a danger.

Hours of service (HOS) — limits on how long you may drive and work.

Implements of husbandry — farm equipment. It is generally exempt from the width and length limits when it is moved on a highway as part of normal farm work. A Caltrans transportation permit may still be needed (CVC §§36000 and 36600).

Interstate commerce — trade or transport that goes: from a place in a state to a place outside it (including outside the U.S.); between two places in the same state but **through** another state or country; between two places in the same state as **part of a shipment that began or will end outside** the state or the U.S.

Intrastate commerce — between two places in the state, as part of a shipment that begins or ends inside the state.

Jackknife — the rig folds at the hitch point, so the trailer and tractor form a "V" instead of a straight line.

Jaws — the parts inside the fifth wheel that close on the kingpin. The **locking lever** (release handle) opens them; a **safety latch** keeps that lever from opening by accident.

Kingpin — the steel pin under the trailer's upper plate. From top to bottom it has a **base**, a **shank**, and a **head** (Figure 6.16).

Landing gear — the two front legs that hold the trailer up when no tractor is under it. You wind them with a **crank handle**.

Lower fifth wheel — the coupling plate on the tractor.

MEC — Medical Examiner's Certificate (the certificate you carry).

MER — Medical Examination Report (the examiner's full report).

Motor carrier — the company that runs the vehicle (usually your employer).

National Registry — the federal list of trained, certified examiners for CMV driver physicals.

Out-of-service (OOS) — an official order that the vehicle may not be driven until repaired.

Placarded load — HazMat in an amount that requires diamond-shaped warning signs (placards) on the vehicle.

PM — particulate matter (soot).

Relay valve — a valve on the trailer that sends air from the trailer's own tank to its brakes.

Restriction — a code that limits what you may drive.

rpm — revolutions per minute = engine speed, shown on the **tachometer**.

Service line — the **blue** air line (control or signal line). Its pressure, set by the brake pedal or trailer hand valve, tells the trailer relay valves how hard to brake.

Shut-off valves — valves in the service and supply lines at the **rear of any trailer that pulls another trailer**. Closing them seals off the lines when nothing is hooked up behind

Sliding fifth wheel — a fifth wheel that can be moved forward or back on the tractor frame. **Locking pins** hold it in place.

Spring brakes — brakes held on by strong springs and released by air (you learned these for Air Brakes). On a trailer they work as the parking brakes.

Steering axle — the front axle you steer with.

Texting — entering text into an electronic device by hand, or reading text from it: text messages, email, instant messages, opening a web page, any other text entry or reading.

Traction — the friction between your tires and the road. Without traction you **cannot steer or brake**.

Tractor — the power unit that pulls a semitrailer. **Bobtail**: a tractor driving with no semitrailer attached.

Tractor protection valve (TPV) — keeps air in the tractor if the trailer breaks away or leaks badly. You control it with the **red trailer air supply knob**

Trailer air supply control — the red knob (older rigs: a lever). The handbook also calls it the **tractor protection valve control** or the **trailer emergency valve**.

Trailer air supply knob — the red 8-sided knob in the cab.

Trailer hand valve — a hand control that puts on the trailer brakes only. Also called the **trolley valve** (or Johnson bar). Use it **only for testing**

Test-day checklist

- ☐ Bring your DL 44C application, ID (certified copy), residency document, Social Security proof, medical forms (MER + MEC) and the fee.
- ☐ No notes, phone, or watch in the test — any aid or helper means the test is failed.
- ☐ Allow 2–3 hours if you take several tests. You get 3 tries per application for knowledge tests.
- ☐ Read every question **twice**. Watch for **NOT**, **EXCEPT**, "most", "first", "always".
- ☐ Two answers look right? Pick the **handbook** one — the California rule, the safer action, the exact number (26,**001**, not 26,000).

- ☐ The screen shows at once when an answer is wrong. Test-takers report a skip option that brings skipped questions back at the end (not official).
- ☐ No official time limit, but DMV won't start a knowledge test within about 30 minutes of closing.
- ☐ Last look before you go in: Part 1 of this guide.